

Psych 221 Big Ideas in Psychology Part 2

Numbered Transcript

PSYC 221 • Dr. Roy Baumeister

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Lecture 1: Past, Present, Future

At a glance — Past, present, future — the self across time.

- 1.1 Hello, everyone. As I explained at the beginning of the previous course, I'm a bit unusual as a social psychologist, more of a generalist. All the reward structures in the research fields are "stay in your lane, pick one topic, and work on that." But I figured out some ways to be more of a generalist. If you have a lot of interests or, as some people put it, intellectually promiscuous, it's fun to go and study different topics.
- 1.2 So today, we'll focus on how and why and when and so on that people think about the future. So, time is one of the essentially human traits. As I explained in part one, I'm looking for the distinctively human traits, the things that set us apart from all the other animals. We have much longer time span, can understand ourselves not just in the here and now, but in the context of the past and heading for the future. So time is a really central part of human nature, that it really isn't nearly so much for other great apes and other animals.
- 1.3 History of psychology focuses on the past. And indeed, I got on to this studying the future, talking to the great Martin Seligman, who's been my buddy over the years. He said, "Well, you know, psychology's big theories are about how the present is the result of the past." So we looked to the past to explain it. But he said, as he was looking at his own thought processes, he found himself thinking forward about the future, more than about the past.
- 1.4 Now, of course, we psychologists are weird. We can't generalize from ourselves, especially after you've spent decades studying the mind. You can't use your own. But it was an intriguing idea, and it appealed to me with my interest in self-control, which is often geared toward the future, and free will and things like that. So I got on board, and we managed to scrape together some money and do some grants.
- 1.5 But just in the context of psychology, the two big theories of the 20th century in psychology were the Freudian theory, which explains adult personality based on childhood experiences. So it's on the past. And then Skinner and the rat animal learning people, which explained behavior based on reinforcement history. Whether you were punished or rewarded for some behavior in the past determines whether you'd do it today. So, clearly heavy focus on the past.
- 1.6 Even when the cognitive revolution happened and people started studying more thought processes, memory, again, was one of the biggest things to study. So very few people looked at how people think about the future a lot more in terms of remembering the past. We have whole journals devoted to study of memory, but not much in terms of the future.

- 1.7 There were some isolated ones doing it, and a big breakthrough came when Addis and Schacter and those people figured out that the same parts of the brain that you use to remember the past are also used to project into the future. And that upended a lot of the memory theories, because they said, well, memory is pretty good, but it has all these flaws.
- 1.8 You keep revising it. If you're just supposed to remember the past, the past doesn't change, you should remember it the same way. Or your current wants and needs shape how you remember things in the past, which is a little weird. They all talk about this as flaws of memory. But if the purpose of those brain areas is to think about the future, then you should update it as your circumstances change, and it should respond to what your current wants and needs are.
- 1.9 Why so much study of the past? Well, for one thing, we're copying the causal model from the other sciences, that causality just moves forward in time in very slow steps. But psychology, again, we can think farther ahead and behind, so we shouldn't be tied to it. It's also easier to know the past than the future. You can test in the lab if somebody remembers something that you showed it to them 10 minutes ago. But to study how they think about the future, if you want to study the accuracy, then you'd have to check back with them in the future.
- 1.10 Still, it's a legitimate question, Seligman's insight, or his impression, we think more about the future than the past. So what is everyday mental life about? And I'll get to some data on that. But certainly there are advantages to thinking about the future. You can do something about the future. You can't do nearly as much about the past. You can reinterpret it and so on. But once it's happened, you can't really change it. But the future is flexible, and it would be highly adaptive to orient toward the future.
- 1.11 Obviously, if you think of evolution of very simple animals, the first thing was just to see what's going on in the immediate present, to live right here in the here and now, and then to remember things and learn lessons. Projecting into the future is more of a challenge, but being able to do it since you can do stuff about the future and not the past, that would be useful and adaptive.
- 1.12 Anyway, psychologists have finally gotten around to studying the future, and they say, "OK, what should we study? Well, I guess let's see how people make predictions." That's the obvious place to start. So, having people think what is going to happen. And there's some fine work on that, and that's useful, too. But my impression of that is not the main way people think about the future.
- 1.13 After some of the studies we did, I came around to thinking, yeah, predicting happens—who's going to win the World Series or the next election or whatever. But is that really what people need to do to guess how things will turn out? It's also plausible that they want to think not how are things going to turn out, but at what points do I have to

make choices or do I have to perform? In other words, you want to predict the choice points, where things could go in different directions, so you're ready for those.

- 1.14** So, yeah, you want certain things to happen in the future and other things that you want not to happen. You want to predict the points at which you have some control and you can steer it toward the positive rather than the negative ones, and that's to prepare for action. So the great William James introduced the theory of pragmatism, that "thinking is for doing," is the classic formulation. And so, thinking about the future is for doing stuff in the future, to prepare for action.
- 1.15** And again, that's different from predicting how things will turn out in the long run. I mean, some of that's interesting and nice, too. But where do I have to be prepared to make a decision or make a statement? I mean, god forbid, you show up in court someday and you've been accused of something, and the judge says, "Well, how do you plead?" You can't say, "Well, let me think about it. I'm not sure." You'd better have your answer; are you going to plead guilty or not guilty or whatever? You have that all worked out in advance.
- 1.16** So that's just one of many examples of which it's useful to anticipate where things can go in different directions and prepare for what you can do. The idea of this theory is that the purpose of thinking is to produce good outcomes, practical benefits in the future, especially to guide your actions, to respond to the situation, to steer things in the desired direction. Yeah, prediction is helpful, but it's secondary. As I said, you more want to predict where you come to the crossroads, rather than predict the end result.
- 1.17** In a sense, there might be two steps in thinking about the future. First, what do you want to happen? And then second, how are you going to get there? And those have different contingencies and different things you need to think about. So, you see, well, the future could have me doing this or this or this or this. And the one I want is this one. You evaluate them and choose the one you want.
- 1.18** And so that's kind of optimistic, looking for the best case, positive that you can realistically get there. And then you start to think about, how do I get there? What do I have to do? What needs to happen in order to produce that? So that's not as positive. It's more realistic, and if anything, anticipate the pitfalls and dangers and so on.
- 1.19** Let me start describing a giant study that we did in terms of people's thoughts. I did this with the great Wilhelm Hofmann, who had developed these procedures and really refined them, for looking not at thinking in everyday life—I mean, not at thinking in the laboratory where most studies of cognition are done, but for people as they go about their daily lives. We wanted to see, what are people actually thinking about?
- 1.20** And since this is when I was getting into studying time, and Seligman had helped arrange some funding for this, we thought, well, let's look at how thoughts use time. Are they thinking about the future or the past or the present or something else? And if

so, what are the emotions, what are the other thoughts, what goes on, especially with those that distinctively accompany thinking about the future?

- 1.21** So the method—we had about 500 people from the Chicago area, slightly less than half are students. They're adults, 18 to 67 years old, with the mean being a little under 30. And the way it worked is, there was a transition in this, kind of, where you used to give people beepers, but as people mostly got smartphones, you could just program the smartphone. In any case, it would go off six times a day, at least two hours apart. That was the only restriction, but spread out through the day at random points.
- 1.22** And so they were supposed to stop whatever they're doing and answer some questions about, "What was your most recent thought? What thought was in your mind, just when that happened?" So again, lots of laboratory studies that say, "Think about this, remember this, how do you evaluate this," and so on. And those are valuable and important, too. But we want to know what actually goes through people's minds when they're out, going about their daily lives.
- 1.23** Now, again, we wanted to ask them about the past, the present, and the future. So there's one other study that had done this, and they simply asked people to choose, "Was your thought about the past or the present or the future?" But I thought, oh, I don't know that that's the best way. Let's have them rate each of those separately, which turned out to be a very fortunate idea, because sometimes they said none of the above, sometimes all of the above.
- 1.24** But it did enable us, because sometimes you're thinking about the present and the future. How do I get from here to there? So you might check multiple ones. So instead of forcing people to choose one of the three, we gave them three separate ratings. And then there's a question of what is the present? Is the present this year, this week, this nanosecond, this minute? And different people interpret that in different ways.
- 1.25** So, in order to homogenize the data, I just made an executive decision that we put in the instructions. The present is 10 minutes long. It started five minutes ago, and it ends five minutes from now. So the future begins in five minutes, and the past is anything up to five minutes ago. We did some separate analyses to use a broader time frame—this week or whatever—which yielded similar results. The primary instruction that we gave them, the present is within five minutes of right this second.
- 1.26** So, how often do people think about the future? Well, it turned out, we think mostly about the present. That's the most common. But Seligman was right, thinking three, three and a half times as much about the future as the past. So the mind tends to row forward. People are thinking about what happens next. You notice those don't add up to 100%. Well, that's because they don't include the combinations. So let's add in the combinations.
- 1.27** Most of the combinations were rather unusual, like the past and the present. But the present and the future was a significant number. So a lot of times they checked both.

I'm thinking about the present and the future, and not necessarily how will I shape the future course of world events, but more, how do I get from here to lunch? Something like that. But anyway, there were significant numbers of combinations.

- 1.28** And you'll notice they still don't add up to 100% because on about a quarter of the thoughts, they didn't check any or they didn't have any time to mention. So that was interesting, too. We stumbled on discovering this whole category of human thought that apparently occurs outside of any specific time. We started calling these time zones the past, present, and future. And the first time, Hofmann said that we don't want to use that term, but then we all quickly came around to it.
- 1.29** So, reflecting on the point that psychology studies how the past causes the present—well, at least in people's minds, they're only thinking about the past and the present together, one out of every 50 thoughts. It's extremely unusual. And if you put future and past there with all the combinations, either by themselves or combinations, again, you still get three times as many thoughts about the future as in the past. So yes, we are basically forward-thinking more than retrospective thinking.
- 1.30** And this is not to say the past should be forgotten or ignored or anything. It's still important. But still, people think more about the future than the past. That's clear. Then we asked people, if they said they were thinking about the future, what was the time frame? How far into the future? Well, they're thinking about the future, but the near future. So the most common by far is later the same day. And then about at the same rate for tomorrow, a few days from now, a month from now, up to a year from now.
- 1.31** And the future then drops off. Very few thoughts about more than a year into the future. So there's kind of a shelf of the same degree of thinking into the future between tomorrow and the end of the year. But beyond that, it's quite rare. So people are thinking about the near future. That fits the pragmatic idea. That's when you have to prepare for things that aren't going to happen for another year or two. They're not really urgent to resolve.
- 1.32** With the past, same kind of thing. Earlier today was the most common if they're thinking about the past. And then there's, again, a shelf of about the same level yesterday, a few days ago, and so on that goes back. And they don't drop off till 10 years ago. So the time horizon in people's thoughts goes about a year into the future and about 10 years into the past and not that much beyond that. So there are some thoughts about what happened before you were born.
- 1.33** There was something recently, I guess, on the Internet about how often men think about the Roman Empire, which is before any of us were alive. And it does come up now and then, but most commonly up to a few years ago. Now, looking at the thoughts that did not have a time to mention, was a quarter of the thoughts. Well, in terms of how they rated these versus the time-based thoughts, they were less meaningful. So

meaning has a strong link to time. They're more involuntary, that your thought just roams there by itself.

- 1.34** But also, they were less likely to have these out-of-time thoughts when they're mentally tired or fatigued or depleted. Thoughts without time were less exciting in general than thoughts that have a time dimension. They were less happy, but less anger, less anxiety. All the emotional measures we had were lower with the thoughts that are outside of time. It's hard to know exactly what these are, since we didn't get them to record all their specific thoughts, which would be more difficult and more intrusive.
- 1.35** But some of these are, I think of it as doing your math homework or your statistics. It doesn't really have a time aspect. Some people said sexual fantasies they might have, which have a temporal sequence in them, but they're not really located in the future or the past or anything like that. Anyway, generally less emotion of all sorts. And this happens more when you're alone. So when you're with other people, your thoughts are more within time. So time is more of a social thing than we give it credit for.
- 1.36** The non-time thoughts tend to be the only category you had more likely when you were alone than when you were with other people. Everything else was more likely when you were with other people. The pleasantness, there was no difference. How much do you like the thought you're having? The thoughts outside of time were really not statistically significantly different from the thoughts in time.
- 1.37** So, anyway, the implication is, the richness of human experience is linked in time, is rooted in time. Thoughts that have some time aspect are more meaningful, have more emotion; more are associated with being with other people and so on. In particular, being with other people in person increases the focus on the present. So again, the social life focuses you on living in time, but on particularly on the here and now.
- 1.38** All right, then we asked them what we were thinking about. Initially, we prepared this long list of categories, of people thinking about the future. They could be thinking this or that or whatever. And Hofmann, an expert, said, "Well, you can't just give people pages and pages of things to check off." So we had to pair it down to a reasonable number. So they just went through and checked all that applied.
- 1.39** So, thoughts about the past. Well, surprisingly, the most common is the one you see near the bottom, which is implications of the past for the future. So even when people are thinking about the past, often it has a future aspect to it, and that was a surprise to us. But there are other kinds of things about the past. They're replaying thoughts over and over, trying to make sense of things, trying to understand what should have done, what could have happened, what did happen, deal with past events and things like that.
- 1.40** So, again there's a pragmatic focus, even when they're thinking about the past to some degree. So thoughts about the past. The biggest category was implications for the

future, and that's obviously a highly pragmatic thing. But they're also trying to understand and then sometimes replaying events voluntarily or involuntarily.

- 1.41** Also counterfactually. "I shouldn't have said that. I could have done something different," stuff like that. So, a lot of the thoughts about the past, not all, have this pragmatic coloring, consistent with William James' argument that thinking is for doing. And some ruminating thoughts just come into your mind that you can't really control and keep bothering you.
- 1.42** About the present, what are they thinking about? Well, that tends to be focused on what you're doing. I mean, the two biggest categories are doing what I'm supposed to do or what I intend to do, and paying attention, being zeroed in on what I'm doing. Those are the most common. So that, too, is very pragmatic. But there's some others—just sort of enjoying the moment, savoring the pleasant experience. There's some of that as well.
- 1.43** So, a lot of pragmatic aspects of thoughts about the present, figuring out what's going on, focused on your present activities and tasks, but also some enjoying the moment, however it is. Now, we turn to thinking about the future, by far the most common category. Three-quarters of the thoughts, they checked planning, so a lot of thinking about the future. We figured that would be one factor. We didn't think it would be that dominant, that it's twice as much as just about any others, and sometimes more than that.
- 1.44** So, far and away, planning is what people are doing when they think about the future. So planning is essentially pragmatic, obviously. It's making up a series of actions of what you're going to do to get to the results you want. And there's a variety of other things—what you hope to do, what you think will happen, what you hope will happen. There's some worrying and other things which are not necessarily all that pragmatic.
- 1.45** But anyway, in terms of thinking about the future, planning, anticipating, preparing is a very important, a large possibility. The thoughts about the future emphasize possibilities, what might happen, the uncertainty, the options. A lot of them are reflecting on situations and the structure with that. So I came to think of this as the matrix of maybe, the set of possibilities that are how we understand the future.
- 1.46** Because planning was so important, we said, well, let's look at planning versus other thoughts, to see what distinctive features go with planning. And Hofmann, who did the analysis, I thought he would compare it against all other thoughts, but he said, "No, let's compare it against just other thoughts about the future that do not involve planning." So it's statistically weaker, but more disciplined.
- 1.47** Anyway, planning is clearly more meaningful and more pleasant than, let's say, worrying or trying to guess what's going to happen or something like that. So, it got rated high on meaningfulness. It reduces stress, anger, anxiety, again compared to worrying and other sorts of things. That's a positive one. It's the controlled thinking

part of the mind, if you sort the mind into automatic or controlled; or in Kahneman's terms, System 1, System 2, planning is a System 2.

- 1.48** It takes effortful thought to think about the future. It doesn't just automatically— plans don't automatically happen. And by the same token, when you're mentally fatigued or depleted, whatever, then you don't want to engage in planning as much. And we had some lab studies making the same point. You need your mental energy to make a plan. It's highly involving, so people like planning. Again, those fit the pragmatic orientation.
- 1.49** All right, now we're going to turn to how people feel when they're thinking about the future or the past or whatever. So, remember, we asked them, if you're thinking about the future, how far into the future? Are you thinking about tomorrow and the rest of the week and the rest of the month? Because we're interested in this called mental time travel. We picked the rating farthest from the present for each thought, even if they checked multiple ones. If they checked, say, later this week and later this month, then it counted as later this month.
- 1.50** And so, classified each thought with that, and then looked at the emotion ratings that went with it. OK? So, they're going to be graphs like this. You'll see the line up and down in the middle. That's the present. So that's if they're not thinking about past or future at all, just about the present. Plus one is later today, plus two tomorrow, plus three later this week, and so on, up to after my life is over. And by the same token, minus one is earlier today, minus two is yesterday, minus three is earlier this week, and so on, going back to before I was born.
- 1.51** As you see, happiness is highest when you're focused right on the present. The farther your thoughts roam into the future or the past, the less happy you are. I think we tested both for the curvilinear effect, which was significant, and the linear effect, which is, I think, barely significant. A little bit happier thinking about the future than the past. But the big difference is, of course, focused on the present, is where happiness is maximum.
- 1.52** Now, this we had actually predicted because I thought emotions mainly exist in the present. The more you're in any emotional state, the more it should focus you right here and now. It's when you're not in emotional states that you'll think about the past and the future. So that fit that theory, but that theory turned out to be wrong. That was only for the positive feelings. Negative emotions show a different pattern.
- 1.53** There, with anxiety, that's lowest if your thoughts are right here in the immediate present. There's, as some therapists and so on point out, if things are going bad and you're struggling with whatever, it's never quite as bad right here in the present moment. If you can keep your thoughts narrowly and not worry about how your life is ruined or whatever, then the anxiety is somewhat reduced. I think the linear effect is significant in this, too.

- 1.54** There's more anxiety about the future than the past, which makes sense. You're worried about, again, is my life ruined? Will I ever be happy? Will I be able to solve this problem? Or whatever. Anger, roughly the same thing. Lowest right away in the present. Anger data were actually—I mean, this is the quadratic curve. If you actually look at the time span, there's a bump in anger about yesterday, or something that happened quite recently. So very recent things do make you more angry.
- 1.55** But when you just put it into the quadratic two-variable formula, then this is the curve of the best fit. So, again, being in the present gets rid of it. Now, emotion researchers have, for a long time, sorted emotions on two dimensions. They call it the circumplex. So obviously, there are positive and negative emotions, and then there are high arousal and low arousal ones. High arousal, for example, negative emotion might be anger or anxiety, whereas sadness is also a negative emotion, but it's usually a low arousal one. Grief will be another one like that.
- 1.56** So, we asked people to rate their arousal. The problem is, with ordinary people, we asked them, "You're feeling arousal right now?" They think it's sexual arousal. Now, that's interesting too, but that's not what we wanted to know. So we asked them how excited they were, asked for it. Anyway, this is the one pure linear effect we have. Basically, the future is exciting, the past is boring. So, the more the mind goes back in time into the past, the less physiological arousal we see.
- 1.57** Stress shows the same pattern as other negative things. It's reduced if you can get right into the immediate present. The pleasantness of the thought, which is a little different from your overall happiness, but how pleasant is it to think about that? It's an even steeper curve than the happiness one. Again, focused on the present seems to produce the highest. Actually, if you look closely at the curve, it's not even right in the present. The peak is a little bit into the future, later today, tomorrow, or later this week.
- 1.58** That's—I don't want to make too much of that, because it could just be a statistical thing, but a lot of data went into this. It reminds me, economists study about time as well. Their point is, anything departing from the present is a cost. Getting money today is worth more than getting it tomorrow or next week or next year. These questions, you could have \$1,000 a year from now, how much would you take today? Would you rather have \$800 today or \$1,000 a year from now? And some people will say, "Sure." So that's almost a law in economics.
- 1.59** But there was one exception. I found this, and my friend George Loewenstein had this study where he asked people, "How much would you pay for a kiss from your favorite movie star?" And first of all, the women weren't willing to pay, but a lot of the men would pay varying amounts to have a kiss from their favorite movie star, and they were willing to pay significantly more for a kiss three days from now than right now. I assume it's not just so they could brush their teeth or whatever, but you're going to get a kiss from your favorite movie star. It'd be fun to look forward to that for three days, right?

- 1.60 So the value of it is actually higher slightly in the future. So this kind of fits that, the pleasure of anticipation, something slightly into the future, might be that. Again, I don't want to make too strong an argument about that, but it does look like the peak is really off the immediate present, unlike just about all the other graphs. Being here now, researchers on mindfulness and so on tell you, this is a good way to cope with life, a good way to feel better, and so on. And mostly, our results have found that.
- 1.61 So, we found the more you're focused on just the present moment, the more positive emotion, then the less negative emotion, the less stress, and the thought is itself more pleasant. However, meaningfulness went very much the other way. Focusing just on the immediate present is quite low in meaningfulness. Actually, it looks like last week is the least meaningful thing to think about. But the farther it goes into the future, the more meaningful the thought is.
- 1.62 So making decisions that might affect your life for years to come, that's highly meaningful, more meaningful than deciding what you're going to do this afternoon. Whether the thought was under conscious control, this was a surprise to me, because we evolved from animals who live in the present. We have this ability to think about the future and the past. But I thought, well, that would take effort. So it would take conscious, deliberate control to pull us off in the natural thing, like all the other animals would be right in the immediate present. But the result is clearly the opposite of that.
- 1.63 I was wondering what to make of this, but I came across research on mind-wandering that Schooler and Smallwood and other people have done. It turns out, they find the same thing. The mind naturally wanders away from the present. It wanders into the future or sometimes into the past, replaying things that happen. But it takes effort to stay focused on the immediate present. So you have a difficult job or something, you have to force yourself to concentrate and stay right here in the immediate present. And when you relax or you're depleted or whatever, the mind naturally wanders into the other time zones, the past or the future.
- 1.64 So, what we have so far—the present is happy, the future is meaningful, many curvilinear relationships. So often it was the present different from the past and future, which were rather similar to each other. Negative emotion increases as you move away from the present, and the tired mind depleted and so on, not consciously controlling its thoughts, wanders away in time, perhaps especially in the past. It takes effort to stay focused on the present. And again, the future is more exciting than the present, and the present is more exciting than the past.
- 1.65 Now, what about the combinations linking the time zones? Well, the future plus the present was the most common, and that was the most meaningful. It turned out, meaningfulness increased just linearly and significantly. The least meaningful were the thoughts with no time zone. The next most meaningful were the ones that had one time zone. If they combined two, it was more meaningful. And the most meaningful were the ones where they checked all three—past and present and future.

- 1.66** So it's a big function of meaningful thought to link across time. So the thoughts that combined multiple times are not only more meaningful, there's more excitement, arousal there, more stress and frustration and anxiety, too. And a lot of them were less happy, although the present plus future combination felt reasonably happy. There's a theory in social psychology that one of the driving forces is fear of death, that it's at the root of psychological motivations.
- 1.67** So we thought, well, if it's that important, people should think about it now and then, so we included an item, death. But hardly ever were people thinking about death. I think it was about one out of every 200 thoughts, death came up. And it wasn't even their own death, because some of them they put in the past, and all our research participants were alive during this. Can't really study the thoughts of dead people. Anyway, we didn't find much aspects of death.
- 1.68** Now, they say, well, it's so terrifying that you repress the thoughts of death. That's why it drives it. It's hard to test that. But I thought somebody should do study in the future, to compare their theory versus the pragmatic theory that you can't do anything about the fact you're going to die someday. But if there's a specific threat of death, you might think a lot about that. Somebody's coming to your office tomorrow afternoon to kill you. That would probably be on your mind.
- 1.69** I've talked to people who, say, got a diagnosis of a possibly fatal disease, and so that should increase the terror of death. If that causes repression, then you shouldn't think about it at all. But I said, no, you think about it all the time. But that's, you know, you can get another opinion. You can learn what kind of treatments to look for. So, when there's a pragmatic aspect of death, I think that's what people would think about. But mortality, not so much.
- 1.70** We also looked at who thinks about the future. This is an exploratory study. We had about 20 different ways that people differ, male and female, old and young. We gave them a whole bunch of personality tests. We wanted to predict what kinds of people think more about the future than others. Basically, it just flopped. Almost nothing happened. I think the implication, though, is profound.
- 1.71** Everybody thinks about the future. It's part of life. You have to think, "What am I going to do tomorrow? How do I need to prepare for it? I'll think about that." Thinking about the past is optional. The ones that did think a little bit more about the future—optimists think a little more because, well, it's more pleasant. You think good things are going to happen, so it's more fun, people who are basically happy. And the people who want to have everything figured out, the need for closure, they think a little more about the future to get better organized and so on.
- 1.72** But mostly out of the 20, we had hardly anything be significant. They did actually predict who thought about the past, and that was somewhat impressive because we chose these traits to predict thinking about the future. They didn't do that, but they did

predict who thinks about the past, even though there are a lot fewer thoughts about the past. So if you know your statistics, it's much harder to predict when you don't—I mean, statistics is explaining variation, and if there's not much variation, it's much harder to get a significant effect.

- 1.73** But we got a bunch of these, and these are the traits doing it. When the analyses came through, I said, "OK, we have to put that down and record it." I didn't want to make a big thing of it. But one of my co-authors, who said she never thinks about the past, she made a big deal of this. She said, "Ha, look at this. Look who thinks about the past." People who are highly neurotic, grumpy complainers, disagreeable people who don't want to go along, poor conscientious, low self-esteem, ruminating, depressed—these are not the people you want to spend your weekend with, basically. Argumentative.
- 1.74** She thought, well, that was a validation of, let's not think about the past. Now, there are other reasons to think about the past, and sometimes people have to do it. So it doesn't mean you're a bad person if you think about the past. But as a general tendency, yes, I have to admit she did have a point, that these aren't really the positive desirable traits associated with dwelling a lot on the past.
- 1.75** All right, now let's move to laboratory studies. There you can study thought processes under nice, controlled conditions. So one thing is, what happens when people are thinking about the future? If your mindset is oriented toward the future, what effect does it have on your behavior? A context for this is that lots of studies, going back to 1980, had asked people to make predictions about their future life, and they're just relentlessly optimistic. So, the average person thinks good things are more likely than average to happen to them, and bad things are less likely.
- 1.76** And this is replicated over and over. There's a mountain of evidence, so we didn't need to look at that. The average person comparing themselves to other people in the same psychology class, say, predicts their future will be better than the average person in the class, so on all sorts of different domains. So the future is basically bright. OK, that's settled. We'll start from that. But how deep does this go? Let's have people put their money where their mouth is.
- 1.77** If we make them think about the future, they should be more willing to take chances, right? Because if you go for high risk, high reward, if you're optimistic, you'll expect to get the high reward, whereas the more pessimistic people focused on the present should be more cautious. So let's put people in the mindset and see if people then don't worry about risk. So high risk, high reward choices should increase after that.
- 1.78** So we had a couple of procedures for doing this. We had people come to the lab and write about what their life will be like in five years versus what it's like right now. Or there's even a grammatical one where they had to rewrite sentences, either future tense, future verb tense, or the present, and that seems to work just as well. Anyway, the findings just consistently came out the opposite of our prediction. I'm showing you

how research programs work. You have to follow the data. You have a perfectly plausible theory, but somehow, no. The more they're thinking about the future, the less they're willing to tolerate risk.

- 1.79** We thought, OK, maybe students don't understand risk. So there's a trust game that the economists have developed, where they give you a certain amount of money and they say, "Well, you can send any part of it to the other person and we'll quadruple that." And then the other person can divide it up however they want. So there's a risk the other person will just keep it, but you're both much better off if the other person splits it. So if they give you \$10, you say, OK, I'll send it all. It becomes \$40. The other person says, "Great. Here's 20 for you, 20 for me." You're all better off.
- 1.80** But you have to trust the other person to do it. And making people think about the future made them less trusting, less willing to take the chance there. Also, morality, they were blaming other people more for misdeeds as well. So thinking about the future seemed to make people, instead of being willing to take chances, they're more avoiding risk, seeking safety, and wanting to punish bad people so the future is better, and so on. So suddenly the future is dark.
- 1.81** We thought all this evidence of optimism would increase risk-taking, but we found the opposite. So how do we reconcile it? Some researchers would say, "Well, everybody else was wrong," that I've got the right answers. But we believed all the prediction answers. But this is where we came to the two-step idea, because those questionnaires where they'd ask people, optimistic, say, "What are the chances you're going to get a good job, or the chances you're going to be fired from a job, or you'll be in a car crash, or you'll have a gifted child, or whatever?"
- 1.82** And some people just make predictions off the top of their head. Well, maybe that's the immediate response. That's the step one is, what do I want to happen? But step two, I should say, well, it's not going to be that easy to get a big award in my career or whatever. What do I have to do to get there? So we came to the two steps. The first one is, what do you want? The second, and hence it's optimistic because you want good things to happen, the second is more realistic because, how are you going to get there? What are the obstacles? What can go wrong? Or to put it more succinctly, step one is dream big, step two is get real.
- 1.83** We did a series of experiments on these with one of my colleagues, and he came up with this clever procedure, which actually somebody else had developed, but he found it there, where you give people a question and there's a 10-second timer on the computer. And half of them, you say, "Be sure to respond within the first 10 seconds. You have to respond immediately, otherwise we can't use your data." And the others, you tell them, "You can't respond until the 10 seconds are up, so respond then." So we made them wait 10 seconds. In other studies, 15 or 20 seconds, it worked a little bit better, but it worked consistently, anyway.

- 1.84 And the idea was, the immediate response would be step one, and the delayed response would be step two. And that's, sure enough, what happened. I was looking online for amusement, some compilations of funny things kids have said in terms of their assignments. Here you can see, I think, the dream big and get real thing.
- 1.85 What are three things you want to do in the future? The first is to "get a girlfriend." OK, think big. Something wonderful happened. The second one is get real. Obviously, he's erased it several times, so it was a lot of thinking over what to do. He said, "Kiss her." If you have a girlfriend, that's usually part of the deal, so that's not asking too much. Then the number three, it goes back to dreaming big. "Rule the world."
- 1.86 In terms of this research, five studies, several thousand people and so on, making predictions of the future, either respond fast or later. So in study one, it was like, what are the odds you'll live past 80 years old? What are the odds in five years, you'll be in a romantic relationship, successful career, and so on? Sure enough, people were more optimistic with the fast response, the respond right away, than wait 10 seconds. And we've replicated this in two different countries multiple times and using multiple procedures and questions.
- 1.87 Anyway, fast responses show this unrealistic optimism. The slow responses are much more realistic. And the faster they respond, the more optimistic they were. And it was bigger for negative events than positive events. It goes with the idea that bad is stronger than good. It really takes a lot of time to think about bad events actually happening to you. So yes, there is optimism off the top of their head. People are optimistic, but even a few seconds' delay, that drops off sharply. So, the pragmatic prospection starts with thinking of what you want and then realistically making a plan to get there.
- 1.88 Extending this kind of work, back in the first course, I talked about research on rejection, described lab studies where we give people a rejection experience. But something that always bugged me about that is, why do people care? Why do they react so strongly to being rejected by some other people in the laboratory who they may never see again?
- 1.89 Sjøstad, my Norwegian colleague, had some nice studies to show that the answer is, OK, it's not a big deal to be rejected by a bunch of people you don't know and don't care about, but it does raise the specter of the possibility that other more important people will reject you in the future; that maybe there's something wrong with you, maybe there's something about you that's more negative, that puts people off. So that indeed explains the impact of rejection. It's not just the immediate practical consequences of being rejected, but the implication that it might happen in the future.
- 1.90 Peace of mind versus stress. I'm sure many of us have had the experience where you're ready to go to bed or something, and things keep popping into your mind. "Oh, I need to do this, or I got to make sure to do this tomorrow." The effect in psychology is

known as the Zeigarnik effect, which is that uncompleted tasks keep popping into your mind. The term, it's named for a Polish researcher who first did it, but she supposedly got the idea for it that, they went out to lunch, a bunch of psychologists.

- 1.91** When they're ordering lunch, they're very impressed. The waitress came around and asked them what they wanted. She didn't write anything down. She didn't take any notes. But she remembered everything that everyone ordered and brought it back perfectly. They were marveling at what a perfect memory this woman had. Then they went back to the laboratory and said, "Oh, I forgot my hat." He runs back to the restaurant.
- 1.92** He said, "Hi, we were here earlier." She had no idea who he was. He said, "Remember we sat over here?" "No." "Come on, I ordered the wienerschnitzel, whatever. You remembered everything so perfectly. How is that possible?" She says, "Once it's done, it leaves my memory. I can only remember it until it's unfinished." Anyway, there's this effect that unfinished tasks intrude on your mind until they get done.
- 1.93** Now, there are two main theories about why that is, in terms of understanding the conscious and the unconscious mind. One is that the unconscious is doing everything. There are a lot of people who think consciousness doesn't have any causal impact, that it's just giving you an experience of your own life, but the unconscious runs everything. And that's a strong opinion that's come up periodically. Later in the course, we'll talk about consciousness. So that's one.
- 1.94** The other is that the unconscious is trying to get you to do it. It knows that this task is there and hasn't finished. And so it's telling you, and I'm going to keep bugging you until you go and get done. So, separating those theories is a little bit difficult, but I had this really bright PhD student, Masicampo, who—so first theory is, the unconscious is doing it. The second is that it needs consciousness to get it done. He came up with a theory that it's actually a third reason, that the unconscious needs the conscious mind to make a very specific plan, so that the unconscious knows what to do.
- 1.95** These very specific plans—Peter Gollwitzer has developed a research program on these, called implementation intentions. And you need a very specific plan of exactly when and what you're going to do. And that really increases the likelihood that you're going to do something. Things like, you're going to write an essay over the Christmas break, or if you're a woman, you're going to do a breast examination or whatever.
- 1.96** So people have these general, "Yeah, that would be a good thing to do." But that doesn't help a whole lot. That doesn't really get it done. But if you have a specific plan, "I'm going to sit down Tuesday morning and start on the essay and make notes for what I'm going to do, or tomorrow when I take a shower, I'll feel my breasts and make sure there's no lump in there," then they are much more likely to do it. So the unconscious is very powerful, but it's not very flexible, and it needs the conscious mind to give it those instructions.

- 1.97** So, we did a whole series of experiments to show this. In one, run at Florida State, which is a big party school, so half of them were told to think about the upcoming party, the social event. Others think about the upcoming exam, the most important exam that you're going to do. And then some of them made a plan for when to study, and some of them didn't. So they made the implementation intention. "Wednesday morning, I'll get out my notes and go over them. Thursday, I'll get out the textbook, look at my highlight, and so on."
- 1.98** And then we wanted to see, is it intruding into the mind? So we gave people these word stems to see if they came up with words. So like S, space, space, D, space, It could be study, sandy, shade. There are a lot of different ways, but study would indicate that studying, the unconscious is focused on studying. And found, sure enough, thinking of the exam rather than the party increased those intrusive thoughts about. So, study was a more likely answer than the others for the people thinking about the exam, but not if they'd made that specific plan, not if they had worked out exactly how I'm going to prepare for the exam.
- 1.99** And notice, they hadn't made any progress on studying. They hadn't done anything. But the Zeigarnik effect went away. The task was still incomplete, but as long as they had a specific plan, then the unconscious is content with that. "I know exactly what to do. I can leave you alone until it's Wednesday morning at 10:30, when you said, OK, now get out your lecture notes and start going over them." When you have this, your mind is running on and on about things that you haven't done or need to do. Making a specific plan is a good way to quell those thoughts and make it easier.
- 1.100** And again, in multiple studies showing the same thing. So the intrusive thoughts are occurring not because the unconscious is taking care of it itself, nor because it's going to bug you until you actually finish, but rather, it's wanting the conscious mind to take the grand general intention and translate it into a very specific plan, so that the unconscious knows what to do.
- 1.101** All right. So let me sum up here with the main conclusions. First of all, as I said, the primacy of prospection. People think more about the future than about the past. I mean, yes, they think about the present a lot, too. The combination of the present plus the future, also a fairly common category of thought. Notice, too, even when they're thinking about the past, they're often thinking about the implications of it for the future. So, yes, the future has been really underappreciated in psychological thought, in terms of human mind and mental processes.
- 1.102** In pragmatic prospection, when people are thinking in terms about the future, it's, they're thinking most of all about the present, and then next about the future, but the fairly near future, which again is pragmatic. You have to decide what you're going to do for lunch today or how you're going to spend the evening. That needs to be decided. What you're going to do next year could be decided, but it's not urgent.

- 1.103** Focus a lot on the multiple possibilities, on the uncertainty that different things could happen, different possibilities. It's not so much thinking, how will things turn out in the end, but what are the choice points? Anticipate, where do I have to deal with things? Where do I have to perform well? Where do I have to make a choice or a statement? Where do I have to know what to say? Emphasis on what to do in that situation, and sometimes links to present as well. Again, a lot of the thinking about the future is planning, which is developing a series of actions that will bring you to the result you want.
- 1.104** The pragmatic prospection seems to have two steps. Dream big, think of what you want, what your ideal optimal outcomes would be. And second, get real, how are you going to get from here to there? They're both important, but there are separate biases. And lovely research by Gabriele Oettingen shows that people who do only one of those and not the other, they don't really reach their goals. They have to have, for example, it's students finishing up their university education. Do they have a clear idea of the job that they want to get? And are they taking steps to prepare for it and preparing the interviews?
- 1.105** They might be getting themselves ready for interviews, but if they don't have a clear idea what they want, they're not going to do very well there either. And likewise, if they know what they want, but they're not taking steps to prepare for it, that doesn't work either. So you need to do both. And again, the different steps have different processes and biases. As I said, the one is much more positive and optimistic.
- 1.106** The future is this matrix of maybe. It's multiple possibilities. We understand it as this or that could happen. So don't just think about the best-case scenario of what you want. There's these books on business, research on business, by Jim Collins and his colleagues, which really advanced the understanding of business procedures by looking — they got a large sample of companies, and they particularly looked for ones, the good to great ones, that had been stable at a certain level and then became much better and stayed at that level for a period of time.
- 1.107** There's another one about which companies flourish during a lot of difficult, uncertain times. We looked for then the features of, what were the executives like? What was the company culture like, and so on? One of the interesting findings was that the CEOs of the companies that flourished in the difficult times, they weren't any better than the other ones at predicting the future. Nobody knows what's going to happen.
- 1.108** I mean, obviously you'd think that would be a great advantage. If I'm pretty sure how the future is going to turn out, I can gear company policy toward that. But no, they weren't any better at that. What they were better at was preparing for multiple outcomes and multiple possibilities and contingency plans and so on. So again, mapping out all the possibilities. And well, we want this to happen, but if this happens, we have to be prepared for that, too. That's what those guys and gals were, where they really stood out.

- 1.109** In terms of the feelings, as I said, you're happier if you focus on the present, consistent with the mindfulness work, too. On the other hand, meaning links across time. So meaningfulness is highest when you're more focused on the future, or especially linking across time zones, past and present and future. Those produce the most meaningful thoughts. Focusing on the present gets rid of anxiety, and other negative feelings makes you feel better, which argues that bad events tend to spread across time. It's not just, it's a problem today, but it's going to cause problems down the road in the future. I'm going to be dealing with this for a long time.
- 1.110** So, in that sense, if you can shrink awareness to the immediate present, to the here and now, that is soothing and gets rid of some of it. Making a plan, on the other hand, reduces your anxiety over the future. The unconscious mind wants very specific plans, exactly what you're going to do. But if you're worried about things, remember, planning is inherently a positive activity in the majority of cases. So you can think about what to do. And if this happens, and even if the worst case comes up, this is what I'll do, and this is how I'll deal with it—and that seems to make people feel better.
- 1.111** Anyway, to conclude, yes, we can know the past better than the future, since it's already happened. But the future is really all we have left, and it's not surprising, therefore, that the mind orients more toward the future than on the past. Again, this is a big evolutionary step forward. Most of our animal relatives cannot do that, cannot even think more than a few seconds or at most a few minutes into the future, whereas we can calculate things, at least speculatively, years into the future.
- 1.112** And again, this is pragmatic. The main reason people think about the future is to make it better. To finish, one of my students, when I was doing this, found this news story about this basketball player who signed a \$95-million contract and took out the rearview mirror on his car, so he can't look back. All right. Thank you.
- 1.113** When thinking about the future and going farther ahead, perhaps over a year, two years, five years, does it become more troubling to plan for that because there's more unknown variables the farther you go?
- 1.114** Well, you can still plan for it. It's true, there are more unknowns, and it could be a nuclear holocaust, and whatever you planned might not come true. But yes, it's going to be more difficult, but it's a good thing to do, too. One of my wise friends, when I was starting out my career, an older colleague said, "You should always have a five-year plan. You can change it, but you should have one, at least. Where do you want to be five years from now?" I think that's very good advice. I pass that along to other people, too.
- 1.115** Certainly in research, as you saw from this, you have this nice theory, and then the data take you over this way. So, yeah, you've got to revise your plan and think, how are we going to explain this or deal with that? But it is good to have a long-term plan. The people who don't plan ahead, they end up living week to week and paycheck to

paycheck. And then they end up in their 40s and, "How did I get here? This isn't how I envisioned my life." So having a long-term plan, I think, is a good thing to do. But you're totally right about the farther ahead, the greater the uncertainty.

Lecture 2: The Roots of Evil

At a glance — The roots of evil.

- 2.1 All right. This lecture will focus on one of the great philosophical questions— why is there evil? This is part of my project. When I moved from briefly being a philosophy student into psychology, I thought we could tackle some of these big questions using social science information. And certainly there's a lot of research, a lot written about all sorts of bad things that people do to each other; there are lab studies and real-world studies and historical studies and so on.
- 2.2 So there's a lot of information about evil. It's not really all that fun to read, but you do get used to it as after a while. Now, I read a bunch of psychology books about psychology of evil. Many of them struggle with the question of, well, if we understand these people, that sort of forgives them, that sort of allows us to be on their side. And so they always hold back some. They say, "We've got to remember this person was a monster who did terrible things."
- 2.3 But I wanted to go farther than that. And so, yes, you do have to remind yourself that understanding is not forgiving, but still we want to understand what it is like to be a person like this. And a lot of these books you would cover and look at this remarkably horrible thing that this person did and be in shock with it. How could someone do that? But I wanted to know, how could someone do that? I wanted to understand more from the inside. What was it like?
- 2.4 When I wrote the book, I wanted people to read some of these stories about bad things people did and think that, "Well, I could see, if I were in that situation, maybe I would do the same thing." So that was the goal. Again, not to condone or forgive or whatever, but really to try to understand the psychological processes and experiences. Moreover, something that became clear to me is that most people who do these evil things don't think of themselves as evil, which complicates the things that much further.
- 2.5 I mean, I suppose we think of Adolf Hitler as the most evil person ever, and he has a pretty good case. He was awfully bad. But remember, he, and especially all the people working with him and so on, they were aiming for something good. They thought they were trying to take care of their country and restore the glory that had been stolen from it. And so, they didn't say, "Let's get together and perpetrate mass murder and inflict horrible suffering on all the world," including their own country, which they destroyed, probably worse than just about any other leader ever has.
- 2.6 So then it becomes that much more complicated to understand these people who don't see what they're doing as evil. Sometimes it gets to ridiculous extremes. I was working on this, there was this news item which I copied down this quotation by this guy who'd gone on a spree killing with a bunch of his friends and killed a bunch of people. He said, "Well, other than the two we killed, the two we wounded, the woman we pistol-

whipped, and the light bulbs we stuck in people's mouths, other than that, we didn't really hurt anybody." OK, come on. That's kind of absurd.

- 2.7 But more generally, again, a lot of people look at what they're doing as reasonable, as decent, sometimes as morally positive. So the question, instead of why do people do evil things, becomes why do some people do things that other people perceive as evil? And so then we have to understand both the perception of evil, how people evaluate others' acts and decide that was really bad for that person to do, and the processes that made the person perform those actions, which may be completely different from how it's perceived by others.
- 2.8 Before I get to what I got from reading the great historical and psychological things, let me describe a study that we did, that sort of got into the psychology of perpetrators to some degree. We started with a focus on anger. Anger is an interesting psychological phenomenon itself. Lots of psychologists study aggression. Why do people do bad things to the other? In the laboratory, you deliver painful electric shocks to somebody, or after they start worrying about the shocks, you put on headphones and deliver unpleasant loud blasts of noise, and so on.
- 2.9 Mostly, people won't do that in the laboratory to other people, unless you make them angry. So a lot of psychology research studies make people angry, and the usual way is have them write an essay or do something and then insult them gratuitously. OK, yeah, that makes people mad, when you're randomly insulted out of the blue, but how often does it happen in real life? I mean, how often do you go out to the mall and say, "I'm just going to say really obnoxious, offensive things to a bunch of people." So, yeah, it works, but it doesn't give us any clues to the psychology of anger, of how it arises in the first place.
- 2.10 So, what we did in this project was to get people to tell stories from their lives. Everybody told one of each, one story in which you made somebody else angry, and these are the full instructions. Most people have had these things happen to them. Choose an especially important episode and tell the full story. Or, in the other condition, an incident in which somebody made you angry, and again, the rest of the instruction was the same. And everybody did both, so these are not personality differences between them; these are the same people in the perpetrator and the victim role.
- 2.11 So we took the stories, whatever they said, didn't ask them anything else, and then coded them, looked for specific features. So here's some of the results of codings. First of all, were there long-term past events leading up to that? Did they provide background to the episode in which they made somebody angry or somebody made them angry? Well, as you see, the victims have a much longer time span. They're much more likely to give the background and this had happened or that had happened prior to that. Some of the perpetrators did, about half, but not as many.

- 2.12** Did they focus on the bad consequences of the anger event to have lasting things? Victims much more sensitive to bad consequences. Damage the relationship. Again, victim is much more likely to bring it up. "I could never trust my roommate again," or whatever. Was the victim still angry? They could do it. Well, a substantial minority of the victims said they were still angry. Hardly any of the perpetrators said they thought the person was still mad at them, even years later.
- 2.13** Did the story have a happy ending? "Yeah, we had this problem. I did something that offended someone, but we made up and we ended up being better friends than ever." About half of the perpetrators' stories ended on some positive note, less than a third of the victims' stories. And they specifically deny there being any negative consequences. Only the perpetrators brought that up. So that already provides some understanding of the context of the difference between the victim and the perpetrator mentality.
- 2.14** It resonated with a bunch of things. I grew up in Ohio, and we learned about the Civil War as this thing happened long ago and so on. But then when I lived in the South for a while, there were people were still mad about the Civil War. "We had this nice civilization going down here, and these Yankees came down and ruined everything." In fact, while I was working on the project, there was a book published, *The Crusades Through Arab Eyes*. It was a fascinating book. I'd read a number of Christian histories of the Crusades, and you don't think of the Crusades as evil, but my god, a lot of pretty bad things happened.
- 2.15** And the Christians, history talked about it, "Well, long ago, there were these problems, and pilgrims couldn't go to the Holy Land. So the Christians marched down there, and the knights and what, and there were a bunch of battles. And yeah, they did some bad things, but that was typical of the past. And then it ended, and there was some cross-fertilization and sharing of ideas. And it's all long ago, lost in the sands of time."
- 2.16** *The Crusades Through Arab Eyes* ended with a strong statement that this is something that's still vividly remembered. They still have this resentment that they had a flowering of the Islamic Arab world going on, and the Crusades were so nasty that they brought that to an end. They dwelled on the horrible things the Crusaders did, eating babies alive, roasting them, and things like that, and murdering people and so on. So, again, that kind of fits the pattern that we found in this survey of college students about incidents from their own lives. A much longer time span, still highly sensitive to the bad consequences damaging the relationship that is still on their minds today.
- 2.17** All right, another set of findings. Did they bring up—OK, the perpetrator did something bad, but there were reasons for it. There were other things like that. Most of the perpetrators had some excuses or something, and they admitted they did something bad, they did it, but it wasn't entirely their fault, or they were misunderstood, or they were meaning to do something else, or whatever. Hardly any of the victims brought up these mitigating circumstances or excuses, anything like that.

- 2.18** Was it impulsive action, just spur-of-the-moment, not really planned? Well, about half the perpetrators said, "Yeah, it just happened to occur. I don't know why I did it or whatever." Only a minority of the victims saw the perpetrators' actions that way. "Couldn't help it." Again, almost a third of the perpetrators said, "I couldn't really help it," whereas none of the victims said they couldn't help it. And was there some justification for it, even if, yes, there was something wrong? Again, about half the perpetrators, but 2% of the victims, said it was justified or justifiable.
- 2.19** So, the victim is seeing things in either extreme black-and-white terms. The perpetrators, "OK, yeah, I did something wrong, but I didn't really mean it, or there were other things going on, or I was misunderstood," whatever. Emphasizing the inconsistency, the victims' stories emphasize that, that he'd said he was going to do this, and instead he did something else. It had been an explicit process, specifically saying they were immoral, again, much bigger in the victims.
- 2.20** And was it deliberately hurtful or malicious? Only a few of the perpetrators said they were deliberately trying to do something to hurt the other person, but almost a third of the victims thought so. Again, very different perspectives on quite similar actions. Again, they're not different kinds of people. These are the same people, telling both stories; and yet, as they go from one role to the other, they interpret things quite differently.
- 2.21** This is interesting. What did the perpetrator intend to do? And particularly, was there just, can you believe what that person did? Or there was no reason. You hear this stuff even riding around in traffic, right? "Oh, this person pulled out in front of me when there was plenty of room behind me. There was no need for that. What a crazy, dangerous driver." That's how victims think, half of them depicting it as arbitrary, incoherent, contradictory, and so on.
- 2.22** The perpetrators, a few of them said that, but most of them, "I was trying to do this, or I meant to do this," or they had some reason they weren't just unbelievably doing something randomly bad for no reason. But that is how the victims will often construe it. Interesting set of findings that, I think, by themselves, none of them is surprising, but put together, can explain how some of these interpersonal conflicts arise.
- 2.23** So, did the victim not express any anger? Well, about half, almost half the victims had made me mad, but I didn't say anything; whereas perpetrators, sometimes they recognize the person didn't show it, but they sensed it. I suppose we could take this to mean that we've probably all pissed off a lot more people than we think, because a lot of times people get mad and don't say it.
- 2.24** Also, were there multiple repeated provocations? Well, the victims, a lot of them saw that. The perpetrators, significantly fewer; some of them said that, but a big difference there that is a pattern over time. Conversely, did the victim overreact? Well, a lot of the

perpetrators said, "I did something wrong, but, whoa, she really blew up at me, and that was uncalled for," whereas the victims didn't think they overreacted.

2.25 Well, if we put these together, sometimes one person does something that annoys or bothers the other person, the other person doesn't say anything. So the person does it again, does it several times, and you never mind. Finally, the other person blows up and says, "You're always late and leaving me waiting," or whatever, which, to the perpetrator, "I've done this before, you never said anything. I didn't know it was a big deal."

2.26 But the victim thinks, "Oh, I'm putting up with this over and over, and it's getting worse at it." So finally, they blow up, and they're reacting to what they see as the accumulated series of offenses; whereas to the perpetrator, it seems like an overreaction, because "I was late yesterday and you didn't seem to mind." Anyway, that sheds some light of how those things can develop.

2.27 Summarizing differences between the perpetrator and the victims—the degree of harm, the victims see a lot more harm than the perpetrators. The victims see what the perpetrator did as deliberate and intentional. "This is a person trying to do something bad," whereas the perpetrators, "That isn't what I meant, or I didn't intend to do it." The time frame, again, the victims have a longer time frame, starting way before the incident and extending a long time after it. The perpetrators tend to just say, "OK, I did this bad thing on this day, and that was it." It didn't have a past, didn't have a future.

2.28 Apology, making amends. A number of the perpetrators reported that none of the victims said that the person, or maybe one of them did something, but hardly ever. So, apologizing or trying to make it up is very salient to the perpetrator when you know you did something wrong, but not so much to the victim. Did the victim do something to provoke it? Well, a lot of the perpetrators said, "OK, I did something wrong. I shouldn't have said this or hit him, or whatever. But he had provoked me. He started it, and he'd done this."

2.29 The victims never see their role as producing this or provoking it. The perpetrator regrets it. The perpetrator is much more likely to say they regret it. The victims don't see that. It's almost like they're motivated not to acknowledge that the other person is sorry or regrets it. Again, they don't remember the apology. And both sides distort. We had some follow-up studies where people had to role-play one role or another, or read a story and identify with one character or another, tell it then from the victim's or the perpetrator's perspective.

2.30 And then he could compare their retelling of the same event with the original that we had given them. And the distortions were equal. Both perpetrators and victims were part. There's a big tendency to believe victims and be suspicious of perpetrators. And to be sure, in real life, sometimes perpetrators have reason to lie because they want to escape punishment. There's less of an objective reason for the victim to lie.

- 2.31** But in terms of the mindset that's built into the role, both victims and perpetrators, that seems to produce a distorted perception of what's going on. And that's something to keep in mind as we read news events and events in our lives that are going on now. And that's even when it gave people financial incentives to get it right, to tell the story exactly the way it happened. Again, the victims emphasize some things and perpetrators emphasize other things, and they all downplay things that are more favorable to the other side.
- 2.32** That said, let's go now with perception. What do people perceive as evil? So for this, I looked at multiple sources. I looked at religious depictions of evil. I looked at propaganda in wartime. I looked at Saturday morning cartoons. I didn't actually look at the cartoons, but I read studies that summarize the kinds of things that happen there. So they're all very different, but different depictions of bad guys. Movies have a lot of bad guys, which I became rather impatient with, since the bad guys don't correspond to what the real perpetrators are like. They're just sort of our image.
- 2.33** I started calling this the myth of pure evil. This is how we think of evil. Even on the news, on the supposedly objective news, they will assimilate stories into "this horrible person did something" to fit it. So, what are the aspects that make something appear evil to appeal to this myth? First of all, intentional harm. It has to be deliberately doing something bad to hurt another person. That's pretty widespread.
- 2.34** Usually, they enjoy hurting. You watch some of these movies, "I'm going to enjoy kicking your ass." The pleasure from actual perpetrators is often absent, and very much the opposite. But in our myth, the bad guys do bad things because it's fun to them; they enjoy hurting people. Meanwhile, the victim is totally innocent and good. It's not at all the victim's part. OK, that would fit pure evil. And there are cases like this, but in reality, it's often much more complicated than that.
- 2.35** I read studies of homicide and so on, and somebody goes into a bar and has a drink, and the other person next to him says something sarcastic, and he insults him back, and they insult each other, and the one shoves the other and takes a swing, and so on. And so, at some point, the one is dead with a knife in his heart, being led out, so that's the victim; and the other is the perpetrator being led away in handcuffs. But if you followed the event, it was very unclear who started it and so on. So there's often a substantial amount of a victim provoking it.
- 2.36** And again, the perpetrator stories in the previous study had included the victim. "Yeah, I did something wrong, but they started it or they caused it." And real life is often much more messy, but this is the myth. So the victim is totally innocent and good, and the perpetrator is really bad. Evil is also outside. I mean, the funniest thing of this is that, OK, cartoon characters on the Saturday morning cartoons, why do they speak with a foreign accent?

- 2.37** It's just harder for the kids to understand an accent, but it reinforces the stereotype that we are good, that bad people come; they speak with a German accent, Nazi-era, or an Arab accent or something like that, or whoever the current bad guys are. So this general perception of evil comes from outside. Also, it is always evil. It has been evil from the start. Whereas, actual people who get involved in these things, they didn't grow up evil. Maybe one or two are psychopaths or something like that. But mostly it's not that. But in the myth, they are just bad to the bone since birth or whatever.
- 2.38** Intentional harm is the main form of evil, but there's a second one—chaos and disorder. How many horror movies start off with the family having a picnic in a beautiful, nice setting and beautiful park, and in a couple of minutes, chaos will happen, and they'll be running, screaming, because they know monsters are showing up or something like that? An orderly, safe, peaceful society is good. Evil is a disorder, breaking into that, disrupting the normal flow of things.
- 2.39** One more thing is, the bad guys have this real high, big ego. They have a very positive view of themselves. They think they are superior. They're confident, even when the heroes are starting to defeat them; they're still sure they're going to win. So it's an inflated view of themselves, but they do have low self-control. They get carried away with rage, and a lot of the movies, the bad guys will get so frustrated and upset, they will kill their own people and do other irrational things that have proved their undoing. As we'll see, the egotism and the uncontrolled rage are real aspects of what contribute to violence, but they're also included in the myth. Those are more justifiable ones than the others.
- 2.40** Another thing I came to realize while doing this is, I called it the magnitude gap. But in a lot of evil, what the victim loses is much greater than what the perpetrator gains. And so, evil is a net negative in society. It reduces the total amount of positive value. The obvious example would be murder. The victim loses everything; you're dead. Whereas what the perpetrator gains is maybe rather small, certainly nothing on the equivalent of one's entire life or the value of a life.
- 2.41** But many other things, too. Sex crimes, the law enforcement has the expression, it's 15 minutes of pleasure for the rapist and five years of trauma and suffering for the victim. Even stealing—what the victim loses often is of less value. Stolen goods cannot usually be sold for the full value, so in general it's a reduction of that. And interesting, one of my colleagues did a follow-up some years later on helping, where you see the opposite pattern, that it's a net increase in value in society, because usually the cost to the helper is less than the benefit to the recipient.
- 2.42** Indeed, when we published that paper, one of my co-authors said he just had two experiences where he met somebody who had done him a big favor early in his career, and he said, "I wanted to thank you. What you did for me, it really meant so much." The guy said, "I don't know what you're talking about. I don't remember it at all." And then the next few days, some student had come and said, "Oh, you said this in class

once, and it really helped me feel better and took things." And he said this was just something that had come to him off the top of his head. It didn't do it, so.

- 2.43** Anyway, with positive acts, there's a net gain in value to society. But the point for today, for evil is a net loss, that the victim loses much more than the perpetrator gains. All right. So I read all kinds of historical and sociological and other accounts of violence and stuff, and I wanted to come up with, what is the basic cause of it? Narrow it down to one. There are actually four, maybe three and a half. So, let me go through these, because they have different categories of evil violence. Categories, in the sense they have different processes that produce them.
- 2.44** So the first one, fairly obvious one, is instrumental. It's a means to an end. So people do bad things. You rob somebody because you want their money. And you may use violence if they resist, but you can get hurt being robbed. But if you give them what they want, they don't really feel any need to hurt you and so on. A lot of the works on this show that it's basically not effective, that people resort to violence as one way to resolve problems or get what they want, but it doesn't really work that well.
- 2.45** A clear example being terrorism. People do commit these acts, set off bombs in public places and so on. Well, they want to bring about some political change or whatever in the world. It hardly ever works. I mean, in the short run, they do succeed in causing a lot of trouble, blowing up a town square or a building or something like that. It calls attention to their cause or whatever, but it doesn't really work. Assassination gets rid of a political leader, but you don't really end up getting the government you want. It doesn't bring about the things.
- 2.46** Stealing from others. Well, yeah, in the short run, you get some goods and so on, but people who steal a lot don't usually end up retiring rich and comfortable, and they usually end up in jail. So there's a general sense through a lot of these things that violent, evil means are not really effective. Even rape. One study asked rapists, "Well, how was the sex?" No, it wasn't very good. They didn't really enjoy it, which seems obvious to me, but apparently not to them while they're doing it. That tends to be one of the broad arguments. But anyway, yes, there's certainly a lot of it, that it becomes a means to an end.
- 2.47** Second root is threatened egotism. Now, when I started this, I'd heard the argument that low self-esteem is a cause of aggression. I'd done research on self-esteem, and I was looking forward to seeing what the evidence was for that. I was a little surprised, because people with low self-esteem tend to be easygoing, not wanting to draw attention to themselves, not wanting to take chances. None of this sounds like somebody who's ready to start a fight.
- 2.48** I looked at it and I found it repeated over and over, and everybody would say, "Yeah, low self-esteem causes aggression," and they would cite another source. So they would just claim that it was true. I'd go then look up that source, and they cited it, citing

somebody else. I would never find any original statement or any evidence that this was true, so I'm not sure how that got started.

- 2.49** There's one theory that it got started when Lenore Walker, I think, was interviewing battered women about the men who were beating them, and they were sort of making excuses for them and saying, "Well, he feels bad or he has low self-esteem." And that was somehow inferred there, but when they actually studied the guys who do this, no, they didn't have low self-esteem, if anything, higher views.
- 2.50** So, avenging humiliation was an important cause. Violence and evil occurs when you think you're not getting the respect that you deserve; others are disrespecting you, that that's what causes people to lash out. Wounded pride is another term for that. In one of the early reports that interviewed a lot of these people who'd been arrested, they didn't find any signs of low self-esteem. These are guys, I think, arrested for assault in prison. So they'd gotten into a fight and beat somebody up.
- 2.51** But it wasn't because they felt bad about themselves. It's because the other person had offended them, and they felt they had to defend their status and prestige and pride by fighting there. So, yes, the low self-esteem, as I said, widely repeated in the literature, but nobody had any evidence that it was true. The closest were, there were some surveys where low self-esteem people are a little bit more willing to admit that they've done something bad. But you score low in self-esteem by being willing to say bad things about yourself. So it's just a questionnaire thing. And in the lab studies where you look at actual aggression, which I'll have one or two in a moment, and no sign of it.
- 2.52** It was invoked to explain an expression, aggression by bullies and African-Americans, wife-beaters and so on. So, obviously they have low self-esteem, although, after a while, research came along and said, well, African-Americans actually, on average, have higher self-esteem than white people. Bullies also have high self-esteem, and wife-beaters are the same thing. So again, do aggressors seem like people with low self-esteem? What we know about low self-esteem, they're easily swayed, they go along with others. They're uncertain, shy, modest, risk-avoidant, oriented toward protecting themselves, rather than trying to look better.
- 2.53** So, the contrary theory, once we dismissed that, was that it's people who think well of themselves, but are a little bit insecure about it, and so they really need other people to validate their good opinion of themselves. And if other people don't do that—so, it's people with favorable views of themselves, but it's a subset of them. It has to be ones that are a little bit insecure about it or fluctuate or are more dependent on other people's opinions.
- 2.54** A good way to think of this is, think of the most obnoxious person you know, the most aggressive person. Does this person look like a low self-esteem person? Shy, modest, self-effacing? No. There's somebody, usually, who thought he was better than everyone else. So you have a favorable view of yourself. Somebody gives you an unfavorable

evaluation, insults you, criticizes you, whatever. Well, what can you do? You can either lower your self-esteem—"I guess they're right, I'm not as good as I thought," which people hate to do—or you can lash out and try to beat them up.

- 2.55** It's especially common, I think, among people who think they're better than they are, because life will constantly be giving them feedback that you're not really so great, but they can't stand that; and so, they'll become aggressive in response to that. Some of the evidence for this is that, when they look at levels of hostility and aggression and so on, they look at high and low self-esteem, but there is also the second measurement of, is your self-esteem stable all the time at whatever level, high or low, or does it fluctuate to some degree?
- 2.56** It doesn't usually fluctuate from "I'm Jesus Christ to I'm Satan," but it fluctuates somewhat around there, and people really dislike when it drops. So the most hostile people were the unstable, high self-esteem. So they think well of themselves, but it fluctuates, and so, they're very sensitive.
- 2.57** In contrast, the people with stable high self-esteem were among the least aggressive and the least hostile. Because why should you be? Your self-esteem, you think you're really good; no matter what happens, you're basically floating through life on a cloud of your own wonderfulness. What's to get mad about? Why get into a fight? But if your self-esteem does go up and down, and if it's high, you think you're better, and the low self-esteemers were in between those two groups.
- 2.58** When you look at group differences, we have both a difference in self-esteem and a difference in aggressiveness, it's almost always the higher self-esteem group that's more aggressive. Men are more aggressive than women, black people than white people, young people more than old people. Psychopathy, depression, even alcohol intoxication, people become more aggressive. Alcohol inflates your self-esteem. They have studies with self-esteem questionnaires, and you think you're better when you're a bit tipsy or drunk than when you're sober. So, all these things go in opposite to the hypothesis that low self-esteem causes aggression.
- 2.59** As I said, murder, assault, other things are a response to insult. There are cultures of honor, where you have to defend your positive view of self by fighting a duel or getting in a fistfight or whatever. Again, it's the positive view of self that's being defended. Rapists, there's a couple books, went into prisons and interviewed convicted rapists, and they found that they thought they were really wonderful people. She said they described themselves as multi-talented superachievers, and they're in prison. But again, showing the very high egotistical view of people who do this.
- 2.60** Domestic violence. So they had the low self-esteem theory for a while, that these guys had low self-esteem, but then that didn't pan out. They started getting more data on them, then they came around to what they called status inconsistency. So, it's like somebody who earned a PhD from long, hard studying and ends up driving a taxi. So

you have the high-status education and a low-status job. And so, what people like that were six times more likely than other people to beat their wives.

- 2.61** OK, that's true, but if you look at the inconsistency in the opposite direction, the guys who never graduated from high school but made a couple million dollars, they're six times less likely to beat their wives. It's just as inconsistent. It's not the inconsistency. It's that feeling that life isn't respecting you for what you deserve and what you do. "I worked hard for my PhD, and all I could get is a job driving a taxi. Life sucks. The world hates me." And you take it out on your loved ones or your domestic partners and things like that.
- 2.62** Bullies. Again, some of the detailed studies of bullies say, well, they say that bullies have low self-esteem. That's just a mistaken observation. The bullies have high self-esteem. They often have a couple buddies that will join in and help bully them. They also have high self-esteem. Some people will stand up to the bully and say, "Leave that person alone," and so on. They have high self-esteem, too. The victims are the ones that have the low self-esteem, the ones who are picked on a lot, but certainly not the bullies themselves.
- 2.63** Violent gangs, the same thing. Researchers who lived with them and studied them in detail, no, they're not full of low self-esteem; they usually think they're superior. Let me describe a couple of lab studies that we've done. One was measuring self-esteem and narcissism. There's questionnaire measures of each of those. Narcissism is this inflated view of being; you're entitled to special treatment, you're a superior being. Self-esteem is more your appraisal.
- 2.64** The most common self-esteem measure, which I'm not really thrilled about, just says, "I'm as good as other people." It doesn't really look for "I'm better than other people." So it's just like, "Am I inferior? I'm not inferior." But that's become the common measure. And then they wrote an essay, and then the other person insults them on the basis of it. And then they play a game where they can deliver blasts of noise to the other person if they win.
- 2.65** Self-esteem gave us nothing on this and no significant results. But the insulted narcissists, they were the aggressive ones much more than other people, and only toward the person who insulted them. It doesn't make them just generally hostile toward everyone. It's just a very specific, aggressive feeling because it's the person who impugned their good name. That's what they wanted to do, and it's mediated by the extent to which they felt their self-esteem threatened. That's what made them more aggressive.
- 2.66** If you look at the pink bars, those are the responses by the narcissists. Yes, getting insulted made them much more aggressive. If you praise them, they're not aggressive at all. Narcissists aren't naturally aggressive people. They can be perfectly lovely to you. If you have a boss or a roommate or whatever who's narcissistic, praising them a lot

will go far. We also looked at, they were insulted, and then they played the game with somebody else. No, no difference. The narcissists are only aggressive toward the person who insulted them. It's a very social communicative act. It's not stoking up the aggressive drive to attack anybody who's in sight.

- 2.67** Anyway, no evidence, once again, that low self-esteem causes aggression and lots of contrary evidence. Having a good opinion of yourself, that's a heterogeneous category. There are several different kinds of people. There are some who are just pretty competent people and know it, and they're fine with that, and they don't really care what you think. They're not aggressive at all. But again, the insecure, unstable, high self-esteem, reliant on other opinions and so on, some of them will be aggressive, mainly when criticized and mainly toward whoever is criticizing them. Threatened egotism, the second root, root of evil.
- 2.68** The third root would be idealism. This is one of the saddest ones. Certainly, some of the biggest body counts in the 20th century were driven by idealistic motives, the people who thought they were making the world a better place. The communists, I mean, that was a marvelous idealistic system, that we'll all live together in harmony and share everything; and we've just got to create this society, and have to kill all the people who are in the way in order to make that happen. And that's been true, again, going back to the religious wars, where God thinks, I'm working for God by killing all these people who don't believe in the same God I do or the same version of it. So there's violence in service of noble goals. We don't have as much of that in the United States.
- 2.69** Periodically, there are these bouts of planting bombs at abortion clinics and so on, and those are reported on the news as crimes, and they are arguably evil. But to the people doing them, "Inside the clinic, they're killing babies, and that's so wrong." And so they think it's justified to plant a bomb in order to stop this murder of innocent unborn children. So that's the third one. I'll have a little bit more to say about that in a couple of moments.
- 2.70** And the fourth, it took me a while, about this sadism. This is, do they actually enjoy hurting others? As I said, that is prominent in the myth of pure evil. They're always thinking, "Oh, they enjoyed hurting us and killing us. And they were laughing." It was a common thing at the death camps and stuff like that. But people laugh when they're nervous and they don't know what they're doing and they're very uncomfortable. To the victim, that's a fine point that you're not going to really attend to.
- 2.71** But I wanted to look for evidence from the perpetrators' side. A lot of the victims will say, "Yeah, they enjoyed it. They got pleasure out of our suffering." So it's part of the victim mentality. But do the perpetrators say it? Well, very hard to find evidence of that, but it's not nothing. A lot of these accounts would say, "I was uncomfortable taking part in this mass killing or whatever. But some of the others really seemed to get into it and seemed to enjoy it." It came up often enough that I thought, OK, maybe there is something there.

- 2.72** Now, most accounts indicate that the first time people do something bad, it's really disgusting. I think I mentioned in the other lectures, some of the work on serial killers, that they have this fantasy of, they're going to—it's often a man who's going to capture a woman and take her away and kill her, and it's going to be really satisfying. And so he does it, and it's really gross, and it bothers him deeply. He's very shaken by this, and so he doesn't do anything again for a long time. And then he starts to think, "Maybe I just didn't do it right." And then I think he tries again. After a while, they become more willing to do it.
- 2.73** There's, I hate to say, a wonderful book, it's a brilliant book called *Ordinary Men*. It was the account of these middle-aged German police officers in northern Germany in World War II, and they were sent to keep order in occupied Poland, and the German army was off invading Russia. And so, they're just keeping the peace and doing their police duty. And then one morning, the commandant says, "Well, today's job, we have to go into the Jewish village and kill everyone and shoot them."
- 2.74** And so, wow, all of a sudden they're assigned this duty, and it was extremely upsetting to them, and a lot of them couldn't bring themselves to do it. They would miss at point-blank range. But they finally got it done. And that night, they were having nightmares and vomiting, and so on. As the war went on, and they did this more and more often, they seemed to get used to it, and become more accustomed to it and more accepting of doing these things. But the first time is really bad.
- 2.75** So, I developed this theory about how sadism works. There's this opponent process theory that, when the body departs from homeostasis, there's usually an opponent process to bring it back. So, if you're not in shape, you run up a flight of stairs, you start breathing hard, and your heart starts beating faster, right? But it doesn't continue for the rest of your life. Rather, there's an opponent process that then slows it down, and gradually you catch your breath again, and your heartbeat returns to normal.
- 2.76** And so, the A process is the departure that speeds it up, the B process is the restorative. And what happens with these things over time is that the B process gets stronger. So the first time you do it, you'll be breathing hard and it'll take you a while to get back to normal; after you do it every day for a couple of months, you'll get over it really fast. Another good example, perhaps, is bungee jumping. There's this terror of falling. So you fall like that, but as soon as you fall and you realize you've survived, you have the opponent process of joy. If you do it a bunch of times, they say you start, the terror gets less and less, and the joy lasts longer. You get this euphoria.
- 2.77** So, it could be what's going on with this sadism, that your initial reaction is disgust and self-loathing and gross, yuck. But then there's an opponent process that brings you back to normal. Maybe over time, as you keep doing it, the disgust part gets weaker and weaker, and the restorative gets better. Maybe you actually start to enjoy it. Now, most people wouldn't start, wouldn't enjoy it anyway, because they think it's immoral.

But among a small part of the population who don't have a strong guilt reaction, they might allow themselves and say, "Well, I did sort of enjoy that. I did get carried away."

- 2.78** One of the revealing observations, I read studies of torturers, people who are professional torturers. We don't have that many in the United States, but in other parts of the world, it's a real job. Every so often, they get carried away and kill the victim. So I thought, well, that would be the novices, right? Because you don't really know what you're doing and you do too much. But no, it's usually the old hands. The novices are still very hesitant about what they're doing, and so they don't go that far.
- 2.79** But the old ones, "I'm really going to make them squeal and really drive it home." And sometimes they just get carried away and do too much. And that's considered bad form. It's not a plus in your job to kill them. You're supposed to pretend you're interrogating them, and maybe you're really just intimidating them or teaching them a lesson, but you're not supposed to kill them. It looks bad on the monthly report or whatever.
- 2.80** All right, a couple more points there. Talk about as aggression, social or antisocial. Well, it's antisocial in the sense that it destroys relationships, damages things within society, but it is a highly social act. As we see, the narcissists who are insulted become aggressive, just specifically toward the person who insulted them. Now, it's peculiar. Somebody insults your intelligence, beating him up doesn't really prove you're a genius, but it does discourage him from criticizing your intelligence again.
- 2.81** Maybe harking back to the animal fights for dominance, that just you beat people up to show them that you're superior, and they should do what you say. So, aggression has a long history in evolution; it's the way animals resolve disputes and so on. In terms of culture, a lot of the work has shown that life has actually gotten much more better and violence has reduced. Stephen Pinker's book really pulled this together very nicely, *The Better Angels of Our Nature*. But there was already a lot of work that he based it on.
- 2.82** So even the 20th century, which we think of as super bloody because of the two world wars, the proportion of people who died violent deaths was down from antiquity. It was much more common in the past, so things are getting better. So, as I said, we started the lecture on why is there evil, as I started the project. But after a while, I started seeing all the causes of aggression and violence and so on. Then we come to a second question, why isn't there more evil than there is?
- 2.83** Psychology has shown that people become more aggressive if they see violent movies or experience hot temperatures or are frustrated by something. Well, who hasn't been frustrated or seen a violent movie or whatever? And yet, hardly any of us have gone out and gotten into violent things or wicked things. Well, self-control is one answer, that we have self-control. People may have a lot of aggressive impulses, but we control them, which means that when self-control breaks down, aggression or violence will increase.

- 2.84** I've talked about the four root causes, idealism and threatened egotism, instrumental and so on. But those are the root causes, but what is the immediate cause? It's often a breakdown in self-control, that the person would normally restrain this. So culture usually wants to restrain aggression and teaches people to check their violent impulses.
- 2.85** Do you know who the most aggressive people are? You know who the most aggressive people in the population? They're two-year-old children. Something like 25% of their interactions, a quarter, have some physical violence, pushing or hitting somebody or whatever. So people start off—and fortunately, they don't do a lot of damage in society because they're incompetent and it's past their bedtime. But in terms of, if you just count the aggression, it's there. It's not something people learn.
- 2.86** There was this feminist argument that fathers teach their sons to be aggressive or something like that, which is wrong at multiple levels. And indeed, it's the young men who didn't have fathers who are more aggressive. But no, aggression starts off high in childhood and then gradually diminishes as people become socialized and learn to restrain it and learn to check their aggressive, violent impulses. So, when the restraints fail, that's when aggression happens.
- 2.87** Alcohol is a well-known cause of weakening self-control. It's also a very well-established cause of violence, or at least it's a multiplier. I mean, little old lady sitting at home having a glass of sherry, she's not going to turn into Arnold Schwarzenegger and demolish a gas station or something. But if people are provoked, then alcohol makes them more likely to respond aggressively. So again, it's the self-control.
- 2.88** And plenty of evidence groups with poor self-control are more aggressive, starting with children, adult self-reports, the ones who score lower as adults on self-control are more aggressive. There's a book, General Theory of Crime, that came out just before I was entering this research. And any social scientist, OK, what's the general theory of crime? Is it going to be relative deprivation or poverty or grievance or whatever? And they said, the biggest factor is low self-control, and that's held up very well since they proposed this.
- 2.89** It doesn't explain everything as usual, a new scientific theory. I've got the answer to everything. And many things have multiple causes, so it's not the only thing, but multiple signs that people with poor self-control. What persuaded me was that criminals often have low self-control, even in terms of legal activities. They're more likely, if they have a job, they're more likely to be late for work or not show up. They have more unplanned pregnancies. They're more likely to smoke cigarettes, all things that go with low self-control that are perfectly legal.
- 2.90** Again, so most crimes are impulsive. In the movies, we think of the people planning these elaborate heists and things like that. My wife's a social psychologist. When we're reading this, I said, "Well, Diane, if we were going to rob a house, how would we do it?" And so we talked about, well, you would do research to figure out where the rich

houses were, and we'd observe them to make sure when they're away. We'd have a list of the things we wanted to steal. We'd have some plan for what to do with the funds. We'd invest that to pay an annuity.

- 2.91** That's nothing like actual criminals. That's why college professors are not suited for a life of crime. Most criminals walk to the place that they rob. It's a spur-of-the-moment thing. They break into your house and they open your whiskey bottle and suck it down, and then they grab the TV or something they can carry away. And even if they make a big score, they tend to blow it right away. So there's no investing it in tax-exempt annuities or anything like that. So, it's a complete different, but a low self-control approach and, I think, make a good case for that.
- 2.92** Single parents, which is another thing associated with criminals. Now, most single parents are working hard, trying to do their best to take care of their children, but they have to do the basic jobs of getting the kid fed and clothed and off to school in time. And it's often the luxury of the second parent who can maintain discipline and enforce the rules and things like that. So they're indeed lower in self-control and will show up in the criminality. Alcohol again reduces self-control, increases aggression, emotional displaced, too.
- 2.93** Use willpower depletion, which I talked about previously, to experiment with that. Provoked or not provoked. And the people who were depleted were much more aggressive than anyone else in the experiment. So, conclusions for self-control—depleted self-control, or characteristically low self-control, produces more aggression, but again, only in response to a provocation. It doesn't create the impulse. Something in the situation makes the aggressive impulse, which you'd normally restrain with self-control; but if the self-control is weakened, then it doesn't happen.
- 2.94** Last thing I want to discuss—sometimes self-control can be used to increase aggression. This is particularly true with the idealistic form of aggression, where you feel it's your duty, it's God's will, or whatever. Now, for this, there was some remarkable book, I think it's in Robert Conquest's book about the terror famine in the Ukraine. Of course, the Ukraine is much in the news these days. But what they went through back under Stalin was just extraordinary.
- 2.95** For those of you who don't know the history, Karl Marx predicted there'd be a communist revolution in the most advanced countries in the world. And instead, it occurred in one of the most backward countries in the world. It was supposed to occur in a highly industrialized, where the factory workers would rise up against being exploited. Instead, Russia was full of peasant, dirt-poor serfs doing it. So, OK, well, we'll collectivize the farms. Slight modification to the theory.
- 2.96** So they started—instead of letting the farmers work their own farm, herded them all into working together on these giant farms. Only they weren't very productive; they didn't work. And you can think what it's like for the young man appointed manager of

one of these farms. It's one of the best days of your life, a big honor. You don't realize it's going to become a nightmare. But central planning says, "Well, you've got this big farm, and all these people, so you should grow this many bushels of wheat or whatever."

- 2.97** Only it doesn't work as well. People don't take care of the equipment. They don't work as hard if there's nothing in it for them, and so it falls short. So you're going to tell the central committee, "Well, we didn't produce as much as you said." Well, if you tell them that a couple of years in a row, you're going to lose your job, maybe be sent to prison, like your predecessor. So there's a big temptation to say, "OK, we did meet the target."
- 2.98** But then they say, "OK, then you send this much of us to Moscow, which we will redistribute here, and keep the rest to feed the farmers." So you end up stiffing the farmers who are growing, doing the work to grow the food. So they're really mad about it. But what else you're going to do? And plus then, central planning says, "OK, you met your goal last year, so next year we're going to raise it. We want you to do even better." And so, things get worse and worse.
- 2.99** And it was happening all over. There were these shortfalls, and the farmers were complaining. So, I think it was still under Lenin, he said, "Well, we're going to try something else. They don't like the collective farms. They still have to do them because that's communism. But one day a week, they can work on their own garden. They can have a garden at their house." OK, so that way you thought maybe they would feed themselves or tide over. But within a year or two, the private farms, the one day a week, were producing more farm, more food, than the collective farms.
- 2.100** So for a while, everybody was happy because there was more food, even though the collective farms were still a losing thing. But at some point, Lenin died and Stalin took over. But when he realized this was happening, and the farmers are selling their surplus and so on, but that's capitalism. That's not good. We got to have communism. So he said, "All right, stop the private gardens. Let me crack down on those."
- 2.101** So farmers are really mad about this, understandably; they're going to go back to being hungry all the time. So they resisted. And then what happened is, Stalin just got mad and said, "All right, we're going to confiscate all their food. They can't have anything other than what they do in the collective farms." It ended up being a huge famine. Millions of people starved to death. And it's just a horrible way to die. You're watching your family starve and your children die one after another.
- 2.102** Anyway, in, I think it was Conquest's book, he had an interview with somebody who had taken part. They sent these idealistic young cadres out into the peasant villages to confiscate the food from these, because that's what they had to do. This guy, many years later, talked about what it was like. I'll read you some of the excerpts from it.

- 2.103** He said, "With the rest of my generation, I firmly believe that the ends justified the means. Our great goal was the universal triumph of communism. For the sake of that goal, everything was permissible— to lie, to steal, to destroy millions of people, all those who stood in the way. To hesitate or doubt about all this was to give in to intellectual squeamishness and stupid liberalism, the attribute of people who could not see the forest for the trees.
- 2.104** I took part in this myself, scouring the countryside, searching for hidden grain, testing the earth with an iron rod for loose spots"—they buried some food in their yard to try to protect it—"testing the earth with an iron rod for loose spots that might lead to buried grain. With the others, I emptied out the old folks' storage chests, stopping my ears to the children's crying and the women's wails. It was excruciating to see and hear all this, and even worse, to take part in it. I persuaded myself, explained to myself. I must not give in to debilitating pity.
- 2.105** We were realizing historical necessity. We were performing our revolutionary duty. We were obtaining grain for the socialist fatherland. For I was convinced that I was accomplishing the great and necessary transformation of the countryside, that in the days to come, people who lived there would be better off for it, that their distress and suffering were a result of their own ignorance or the machinations of class enemy; that those who sent me, and I myself, knew better than the peasants how they should live. I saw terrible suffering and did not go out of my mind or commit suicide, nor did I lose my faith."
- 2.106** That's a real participant in one of these horrible things and his reflection on it, and I think very illuminating, what it was like, that he thought, "This is what we have to do." All right. Anyway, to finish up, perceptions of evil, as I said, are associated with this myth of pure evil that you can see, whether it's on the news or the cartoons or wartime propaganda or whatever. There are the four root causes of evil, the basic things that start thinking—they're a means to an end, threatened egotism to defend your pride, idealism to make the world a better place, and sadism.
- 2.107** Culture seeks to restrain most violence by promoting self-control, and that's working better and better over the centuries, but self-control still fails sometimes. And the best hope for lowering violence may be to focus on the proximal causes, to continue to strengthen self-control more. Anyway, thank you.

Lecture 3: Self-Destructive Behavior

At a glance — Self-destructive behavior.

- 3.1 All right. For this lecture, we're going to examine one of the perennial fascinations of psychology: why do people do self-destructive, self-defeating things? Why do they bring harm and failure upon themselves? It's certainly something that happens. It's particularly fascinating to social psychologists because we've, from the start, felt an obligation to find things that aren't obvious.
- 3.2 If you do a big research project and show that people would rather have \$10 than \$5, you'll say, well, duh. That's not a big contribution. But if you can do something to show that sometimes people will take the \$5 rather than the \$10, well, then you have something interesting. For me, it's particularly interesting because there's a widespread assumption that people are basically rational. This is even a foundational assumption in some fields.
- 3.3 In classical economics, you do what brings you the most money. Rational calculations are what gets you there. Some approaches to political science, again, that voters will support the candidate that advances their interests, whether it's financial or otherwise. So the assumption of rationality is widespread, and even philosophers and theologians and others have emphasized that.
- 3.4 I think, in psychology, we're a little bit more comfortable with people are irrational, but still, to be irrational, it's one thing to make a minor mistake or something like that, but to do something that actually brings a bad result to yourself. The essence of enlightened self-interest is to do what advances your best outcomes in the long run. And so, if you act contrary to that, well, there there's something interesting going on. And so, self-defeating behavior is interesting and fascinating because of that.
- 3.5 Now, there have been a variety of theories about that. Going back to Sigmund Freud, late in his career, he came around to thinking that this is just built into the psyche. For much of it, much of his career was, the two main motives are sex and aggression. But at the end, he sort of changed those into a life instinct and a death instinct. So part of you has this drive. Well, that kind of explains anything self-destructive that you'll do.
- 3.6 And some Freudian disciples followed this up. Menninger said people smoke cigarettes because of their death instinct. It's part of their wanting to bring their life to an end, and they're indulging that. I think that theory is pretty far gone by this time; not many people really buy it, but it was one theory that's possible.
- 3.7 Another is, it's a form of pathology, especially one that might occur among ordinary people or not the severely mentally ill. You would do things like pulling out their hair or some people cut themselves a little bit. Maybe there's something wrong there that produces that. There's another view that's based on Freud's ideas that if you've done

something wrong, then you feel guilty, and then you want to be punished. So it's part of the script that if you're bad, you should suffer.

- 3.8 The data don't really back these things up very well. In particular, if you look at people who are actually guilty, they don't really seek punishment. I mean, the police's job would be much easier if guilty people wanted to be punished. They'd be driving over to the police station, say, "Hey, I was speeding on the freeway this afternoon. Could you write me a ticket and charge me a fine?" That would make things a lot easier. Or, "I robbed a liquor store the other day, and please lock me up."
- 3.9 But no, guilty people, even though they know they're guilty, they hire a lawyer and make excuses and do everything they can to avoid punishment. So trying to think what would motivate people to do self-defeating things came up with at least three possible theories. One is that, for some reason or another, they have this intention to harm themselves. They come up with a desire, an impulse, to do something that brings bad outcomes on themselves, possibly feeling guilty, possibly something else.
- 3.10 Second, there'll be trade-offs. Maybe something good and something bad are connected. And so they're going for the good thing and they accept the bad thing or they sort of disregard it. If it's a small bad thing and a big good thing, well, then that's not self-defeating. But if in the long run, the harm outweighs the benefit, then that would qualify as self-defeating. So that would be plausible.
- 3.11 Third, maybe people are going about something positive, but they go about it in a way that backfires or produces the opposite result. They don't understand themselves or the world. They misregulate themselves, and that produces the effect. And these could operate in combination or separately.
- 3.12 The first category of intentional, deliberate self-destruction, there's really no evidence of that among normal populations. Again, some categories of mental illness will show this to some degree, and that is worth exploring, but clinical and abnormal psychology are outside our realm here. And in terms of social psychology, normal adult people—I can't really find any evidence that people do that. So, we can cross the first one off, but there's plenty of evidence of the other two. So, let's look at some of those.
- 3.13 So, trade-offs are an example where the good thing and the bad thing are attached, are connected, and you're really reaching for the good thing, and you get the bad thing along with it. Now, one of the most interesting ones that come out of the social psych labs is called self-handicapping, which is defined as creating an obstacle to your own success, largely done so that you have an excuse for failure, so that a failure can be blamed on it.
- 3.14 This was initially proposed as a theory of why people became alcoholics or otherwise abusers of alcohol. In particular, they're focused on people who might have a great start to their career, but then get carried away into alcohol abuse. And some of the

same applies to drugs. You have maybe a rock-and-roll band that has a great first album and so on, and then they develop a drug problem or whatever.

- 3.15** Well, the idea was, they did so well at first, everybody expects them to be great. They have a real high reputation. They have to sustain it, and they're not sure that they can. They're not confident that the second album can be as good as the first. And if it's not, people will say, "Well, I guess they just got lucky one time. They're not, they're really kind of mediocre," and they hate that. But they can say, "Well, the first album was great, but then they got into the booze and drugs. And so the second one wasn't as good, but they're a troubled genius."
- 3.16** And a lot of people would rather be a troubled genius than exposed as a mediocrity after all. The theory was, people want to avoid information, and by taking alcohol or drugs, which are known to impair your performance, that damages the information. Mostly, people seek information, but sometimes they transform into avoiding it.
- 3.17** There are other examples of self-handicapping—underachieving, not putting forth full effort. Here's an example of students who don't study for the exam, or even maybe make a show of going out to a ball game the night before the big exam. Well, then they have an excuse, they didn't do that well. But, well, "I didn't really study, so it doesn't really show I'm bad."
- 3.18** And perhaps, when the kids are in school, then the parents might cooperate for this. A lot of parents would rather say, "Oh, my Johnny, he's really bright, but he doesn't want to do all the business work. He doesn't apply himself, so he doesn't get as good grades." And the other parents will nod along and say, "That's OK." They'd much rather say that than, "Oh, my Johnny, he works really hard, but he's just dumb as the floor, and he just can't do it." So, by not applying yourself, you have an excuse to do badly.
- 3.19** It turns out, it's something people do much more in public situations than private ones. So, it's when other people are looking, it's not so much just to defend your own concept of yourself, but your public reputation, which is really more important to people, how others think of them. And so, again, being the troubled genius is better than, again, having been exposed as a mediocrity.
- 3.20** Now, the term self-handicapping, some people don't necessarily manufacture the excuse. They'll claim it. You show up for a tennis match with your buddy and say, "Oh, my knee, I hurt it yesterday. I'm probably not going to be able to do very well." So then they've got an excuse. If you win, they're, "Well, I would have beaten you if my knee were OK." And maybe their knee is just fine.
- 3.21** But what is more interesting is where the people actually do things to sabotage themselves, like, again, being drunk before an important performance or developing an alcohol problem that prevents you from working. And the beauty of self-handicapping is that, not only does it protect you from the implications of failure, it improves the

implications of success, because you still have a drinking problem, but you do a great job.

- 3.22** If the second album is really good, you think, "Wow, these people are really spectacular because, despite all their problems or whatever, or despite going to the ball game the night before the exam, he got an A," it increases, so it enhances success and protects against failure. So what's not to love? Of course, objectively, it does make your performance worse. So in that sense, it is purely self-defeating.
- 3.23** I'll describe for you one of the first experiments to show this self-handicapping, how it worked. It was two people I knew well. They told the subject coming to the experiment that we have two new drugs that are ready for the market. They're metabolic regulators, blah, blah, blah. And he said, "We just want to check and make sure—there have been some signs they have temporary side effects. They won't last, they won't ruin anything, but just for the first few minutes after you take them, there'll be a difference."
- 3.24** Essentially, one makes you smarter and one makes you stupider. And so I said, "We're going to give you an intelligence test twice, and we'll see if there's any effect." And so, the first test, they gave it to some people. They gave them hard questions and gave them accurate feedback, whether they got it right or wrong. They told them they did pretty well. In the other condition, though, there were questions. Often there was no right answer. It was multiple choice. But the person would guess and they kept saying, "You got it right. You got it right."
- 3.25** And so they knew that, "Wait a minute, I'm not really solving these questions. I'm having a lot of lucky guesses." And then at the end, they made a big fuss, and the research assistant came in, and the others said, "Oh, this is the highest score we've seen so far. Oh, congratulations." They make a big fuss over the guy for being super smart. And then he says, OK, and then they explain the drugs.
- 3.26** "And so, why don't you pick which one you want to take? Either the upper or the downer." Well, who wouldn't want to be smarter? Well, the people who think they've just established a reputation as a genius, and "I know I'm not really going to be able to do that well the second time," they chose the downer, the drug that would make them stupider. They had to suspect, at some level, they were going to do worse on the second one, but that would just make the data look really good.
- 3.27** "See, your drug really does have a big effect. It's not that I'm not really as smart as you thought I am." And so that worked. And multiple other studies have shown that kind of effect with different procedures. But you see, so it's objectively choosing the downer drug, akin to alcohol that makes you stupider, temporarily, so that you have an excuse. And if you did happen to do great on the second one, it would even double your credit. Choosing the drug that would supposedly make you smarter, if you then got stupider, that would be a real black eye.

- 3.28** OK, so that's self-handicapping. Why do people do alcohol, drugs, cigarettes? As I said, there was that Freudian theory that people smoke cigarettes in order to kill themselves. No, they don't. They smoke cigarettes because it's pleasant, and they feel good, and they have positive reactions to it. It's the same with alcohol and other drugs. These all bring positive feelings. They can be destructive if you do them in excess. But people basically accept the bad attached to it, in exchange for getting the good.
- 3.29** And in particular, this is something our theme will come back to, the pleasure, the positive effect is immediate, whereas the cost, the harm is far down the road. Long-term cigarette smoking, maybe in 30 or 40 years, you'll get cancer. But the pleasure is immediate. And who worries about 30 or 40 years from now?
- 3.30** Another sign of self-defeating behavior is that people don't follow through in their medical care. They don't follow the doctor's instructions. He says, "Take these pills until they're gone," but the pills make you feel bad. So you start to feel better after a couple of days and you stop taking them or showing up for follow-up appointments. And it's especially, if the procedure either feels bad or it's embarrassing, then people are less likely to follow through.
- 3.31** Estimates of compliance with medical instructions are remarkably low. Some are as low as 20%, some are as high as, I think, 70 or 80% , and this is astonishing to upper middle-class people like us, who probably do everything, or almost everything the physician says. But a lot of people don't do that. So again, that's self-destructive.
- 3.32** If you have an illness problem and you don't take your medicine or follow through on the therapy, you're setting yourself up for problems. But there will be the short-term gain that you don't have whatever is unpleasant about it. You don't have the side effects from the pills that make you feel sick or the procedures that embarrass you or whatever that is.
- 3.33** Shyness. Shy people want to make friends. They want to be liked. They want to be popular, but they're afraid of being rejected. And so they don't go to parties or they turn down invitations, or if they go, they stand in the background at the wall, rather than getting into conversation with people. It saves them from the embarrassment of possibly being rejected. But in the long run, it's destructive because they end up being lonely, and that's not what they wanted. They would like to have friends.
- 3.34** So, there's that aspect. But again, it's a trade-off, to go and approach someone and talk to someone who would be dangerous or risky or bring anxiety. So they avoid that, but at a long-term cost of lack of enlightened self-interest. Violence, revenge, things like that— people seek revenge, even at cost to themselves. There's moderate research on revenge that people, especially if their pride is insulted, they will go do things to get even, but often risky or costly to themselves.
- 3.35** People will sometimes pay. There are lab experiments where you can retaliate against someone who's offended you, but you have to pay some or sacrifice some of the

money you earned or whatever. People will still do that, even though they end up leaving the experiment. Why did I give up some of my pay, just to get even with this person who had offended me or beaten me or something like that? I'll never see that person again. I don't care. I wish I had the money. And that's in the lab.

- 3.36** In life, of course, going out to seek revenge can really get people in much more trouble. Most murders are impulsive. Most murderers regret the action. A lot of criminals are recidivists; they'll repeat, do the crime multiple times. Murder is not really one of them, because it's usually a very specific thing. And the person is immediately sorry and realizes they've just ruined their own life, as well as ending the person that they killed. But they aren't thinking in those terms. So it's, again, short-term focus with severe long-term cost. You're going to spend the rest of your life in prison.
- 3.37** Procrastination is an interesting one. This is one we did some research on. My wife was teaching a course in health psychology, and she got people to report on their symptoms every week, and did they go to the student clinic, and other sorts of things. But we said, well, we can see how the procrastinators fare. A lot of people claim that procrastination is actually good. "I do my best work under pressure." You'll hear them say that, so leave it to the last minute; that's when I come through.
- 3.38** Others say, no, you're not going to do your best work. Some say, well, it doesn't really matter. Well, let's see. She set up this study. There was a paper assignment, a term paper. And so, she set the due date at a date that the class did not meet. She said, "It's due on this Thursday." But she said, "If you can't get it done then, you can turn it in at the next class meeting, which is the final class meeting." So you can automatically give yourself an extension.
- 3.39** She said that so they don't call her up at night, saying, "I need an extension." And she said, "And if you can't make that, you can have one more extension, to the very last day that there are any classes." So several more extensions. The procrastinators, which we identified by question, some of them just wrote down the third date as the due date. They didn't even write down the first one. And sure enough, they did turn it in late.
- 3.40** She had tracked people all semester and looked at the procrastinators versus not, based on a trait scale, and then looked at the grades on the midterm, on the final, and on the term paper, as well as their health reports. Well, their grades were worse on everything. The more they were identified as procrastinators, the worse they did on the term paper. I think they also got lower grades on both exams. But they had fewer symptoms of ill health. They reported being healthier.
- 3.41** Now, the health reports only went up till late in the semester. It didn't go all the way to the end, but that fit the pattern. OK, there's a short-term gain, because the procrastinators are enjoying life during much of the semester, going out and playing frisbee or having a glass of wine or whatever, while others are buckling down and stressing themselves out a bit to keep up and get their work done on time.

- 3.42 The problem with this, I said, well, it didn't follow through all the way to the end of the semester on health because, yeah, the procrastinators are feeling great when they're not having to do anything, but then they have to do the work at the end, and we should have gotten the health effects there, just to see what there is. And she said, "Yeah, you're right." So we did the whole thing again the next semester.
- 3.43 So the second one, we followed the health all the way through. First one, once again, they got lower grades on both exams and the term paper. So procrastinators, don't believe that they do their best work under pressure; they got lower grades on everything. And by the way, when I give people career advice, just get in the habit of doing everything a week or a month before it's due, then life is so much nicer. You don't have all the deadline stress. I know people who live that way. I try to do it, too, and it is really much nicer.
- 3.44 Anyway, once again, the health was better and the stress was lower early in the semester. But at the end of the semester, that reversed, and it reversed so extremely that, when you added it all together, the procrastinators were sicker overall, even though they'd been healthier for much of the semester. They had a lot more stress at the end of the semester because they haven't kept up with their assignments, and they have to write that paper, and same thing for their other courses, presumably.
- 3.45 It's totally self-defeating. The only benefit was the early benefit in low stress and better physical health. So again, extreme case of short-term gain, but it's severe long-term cost to both health, well-being, stress, and to scholastic performance. Conclusions from the procrastination study and subsequent work is, they confirmed this, found similar things. It's clearly self-defeating in total. Worse academic performance, worse health, worse stress.
- 3.46 The sickness is, again, mainly late in the semester when you're having to do all the work that you haven't been keeping up with. So again, a strong pattern of short-term gain, long-term loss. So, conclusions from the trade-offs. You're looking for something good and positive. You get something bad that's attached to it. Again, the short-term gain, long-term cost turns out to be a clear formula for self-destructive trade-offs.
- 3.47 And remember, we evolved from animals who lived in the present. So doing what's best in the long term, at sacrifice of the present, probably just goes against the evolved nature. It's what our higher rational mental powers are supposed to be good for, but if we don't use them to full extent, then we suffer the consequences. And culture is certainly set up so that you have to pursue good benefits over a long period of time.
- 3.48 We're going to be talking about addiction in one of the next couple lectures. They also focus much more on the present and discount the future. I've talked about cigarette smokers here, and they think a lot less about the future. They focus much more on the present because, with addiction, there's the pleasure you get in the immediate present

with the costs down the road, delayed costs. So, don't think about those; just focus on the present that facilitates continuing to addiction.

- 3.49** Now, there's one more aspect to these trade-offs that's worth mentioning, which is a definite gain versus merely probable cost. We should go back to the example of cigarette smoking. I think the data suggests, if you're a long-term heavy smoker, your odds of getting cancer are something like 9 or 10%. People think it's 80%, but no, it's something like 9 or 10%. But that's compared to 1 % if you're a non-smoker. Some non-smokers do get lung cancer.
- 3.50** So that's a huge increase in the risk, but it's not definite. And the majority of smokers will not get lung cancer. So a definite gain and a possible cost, that's another thing that is a recipe for taking a self-defeating trade-off. A lot of people will take a slight chance of something bad happening, whereas the pleasure, the positive, whatever, is definite.
- 3.51** Those two, short-term gain, long-term cost, definite gain, merely possible cost, both of those enable people to discount the negative side, the downside. And then, when those happen, that qualifies as a self-destructive behavior, and they're sorry that it happened. But at the time, that's how they were thinking about it and focusing on the immediate and the definite.
- 3.52** All right, let's turn to the third category, and the second category of real ones. This is where things backfire, so you misregulate. And the important thing here is, again, you're pursuing something good. Those Freudian-type theories that people have an impulse to seek death or seek failure or destruction, no, there's very little evidence of that, at least in the normal population. So, here, people are seeking some positive outcome, but going about it in a way that backfires.
- 3.53** So, for example, some people feel depressed and think, "Well, maybe I'll have some alcohol and that will cheer me up." But often—it's called self-medication— it often produces the opposite result, that you sit at home alone and pour yourself a couple glasses of whiskey, and you end up feeling worse than you did before. So that would count as self-destructive, as one of these misregulating. You intended to produce a positive result. You thought it would make you feel good, but instead it produces the opposite result.
- 3.54** Persevering. The colloquial expression is throwing good money after bad. But often, when things don't go well initially, people think, "Well, we've got to try harder. I've got to do the same thing that much more." So especially if they're somewhat invested in a certain strategy and it isn't working, they do it more intensely. They push harder at it, and in the process, cause themselves that much more failure or cost or other negative outcomes.
- 3.55** So, again, ego is sometimes involved in this. If it's a private decision, you can sometimes say, "OK, the heck with that. That's not working," and go try something else. If you've argued for some procedure, and other people were against it, and you got your way,

implemented it, and then it doesn't work, then you're much more reluctant to give up and say, "Oh, I'm sorry, you guys were right all along." So you say, "Oh, we're just not doing it right. We've got to try harder and put in more effort there."

3.56 And it's like the sunk-cost fallacy, when people have invested something in an outcome, and then they keep trying and pursue that. It also goes into abusive relationships. Often the abuser is sorry afterwards and promises not to do it again. You think, "Well, we've already put in this much or come this far," even though then he turns abusive or she turns abusive again. And so, it goes back and forth, and people stay in these things for a long time. But again, multiple things can keep you persevering in strategies that basically are not successful, not effective.

3.57 Choking under pressure. It's something I studied. It's become now, a lot of different researchers have picked up on that. I mean, that's a paradoxical effect because pressure means that it's important to do well, and choking means that's exactly when you don't do well. So, why is it? The theory I came around to was that it mainly operates with skill tasks. There's skill and effort in athletic performance.

3.58 And skill tasks, as I've said, you initially learn it with paying a lot of conscious attention for what to do, but then it becomes automatic, and so your unconscious knows how to do it. So how to serve a tennis ball, you don't have to think about it. If you've done it thousands of times and practiced it, but then you're in the tournament and it's the key moment, and when it's important, you start to think, "Oh, I should pay more attention to it." And the conscious mind then intrudes into the process and messes it up.

3.59 Consciousness produces uncertainty or indeterminacy, so things can go in different ways, so it's good for learning. But when you're serving for the championship of the tennis tournament, you want it to go exactly the way it always went. I approach this using sports examples, but musical performers and others say, "Oh, I know exactly what you're talking about."

3.60 At home, you practice the sonata on the piano, you do it perfectly every time, and then you get up on stage in front of a thousand people, and you're really nervous, and that's important. You want to do it perfectly then, but you start paying attention to how your hands are moving, but you don't really know consciously how to do it; and so you mess up, and it backfires, and you end up doing worse. So it's a mental response. It's understandable.

3.61 Walter Mischel talked about this in terms of the marshmallow experiments that he did, where the kids are told, "If you can not eat this marshmallow for 10 minutes, till I come back, then you can have two of them." And he said, well, the kids think, "OK, it's important, I got to pay attention to it." So they sit there and they stare at the marshmallow, and that makes them want more to eat it.

3.62 I think it doesn't affect effort aspects of performance, because effort is controllable and you can do it. It's my pet theory, the saying that, in sports, offense sells tickets and

defense wins championships. You've heard that expression? Well, offense tends to be more skill and defense tends to be more effort, because you're reacting to the other and so on. So people come, they buy tickets to see the offense in these dazzling displays of skill.

- 3.63** A quarterback throws the ball 40 yards down field and it lands right in the hands of somebody running full speed. That's really a lovely thing to watch. But in the championship game, you start being aware of exactly how you're holding the ball, and the conscious mind intrudes, and you tend to throw it too far or miss or something like that. The defense depends more on effort, and that doesn't break down. And so, in the championship game, all the eyes are on you and everything's at stake, increased effort will get you through.
- 3.64** We're even watching some of those basketball championships and the three-point shots won't fall anymore and so on. But they just contest every pass on defense and put in as much effort, and that's what gets them through to win the title. Anyway, choking under pressure, another example of a backfiring strategy. It's not something you do deliberately. It's, again, an automatic; if it's important, pay more attention to it. But with a skill, you don't want to pay more attention. The increased attention will just mess it up.
- 3.65** Learned helplessness. This is something that came out of the animal literature. It's, again, I mentioned Martin Seligman. This is what he studied at the start of his career. The original finding with animals is that, if they're given electric shocks and there's no way to escape it, they just learn to submit and do nothing, so that when they're put in another situation and there is an escape option, they could get out of it, they don't learn. They just somehow figure out, there's nothing I can do about it, and they don't discover the escape.
- 3.66** In humans, it looks like it's almost a kind of self-handicapping. The human data on learned helplessness are much more complicated, and a lot of people try harder at first when they're learning their helplessness; when they're experiencing helplessness, they try harder, so it takes them longer to get to the place where they really do finally give up. But some of it, there's a tendency to, if I keep trying and failing, it's just awful, so I'm not going to try, and you withhold effort. Again, that's a kind of self-handicapping.
- 3.67** Well, OK, of course you failed, but you didn't really try. That's not as devastating as if you were trying your hardest and still failing. Something I also remember, too, there was a memoir of somebody who had been tortured in South America, one of those politically repressive regimes, and he talked about his experiences there. This is something I read when I was researching the book on evil.
- 3.68** And it stuck in mind, he said, people said, "Well, why do you comply? Why don't you at least fight with them when they tell you to, when they bring you down to the torture chamber and they're going to do this and take off your clothes and put them here. Why

don't you fight, at least make it more difficult for them?" But he said, "It just makes it so much harder and so much worse somehow."

- 3.69** So he said he'd pile up your clothes here. You just follow the instructions. Lie down here. You do it. You know it's going to be bad, but somehow, fighting against it and trying makes you suffer that much more. Partly, they might actually hurt you more because they're mad because you're causing trouble. But people like us, who've never been through that, it's hard to intuitively imagine, but understanding some of the psychology— maybe appreciate, yeah, maybe that is the best way to deal with that and to go through it. So, learned helplessness may fall in this category.
- 3.70** There are interpersonal strategies that don't really work with people who study praise and flattery and so on. Low-status people are constantly praising high-status people. High-status people aren't really impressed by this. It doesn't really work or help that much, but somehow they can't resist doing that. Sometimes it makes them just seem pathetic. One or two more things, setting unrealistic goals. Sometimes people do that. I think trying to achieve more, that that must be a good thing, but you set yourself up for failure.
- 3.71** Perfectionism, I think I've mentioned before. Perfectionism is positively correlated with a whole slew of mental illnesses. It's really not healthy. It was kind of a shock to me to do this because, being a scientist, of course, you want to try to be perfect, to do the perfect research and prevent any mistakes or problems or flaws or anything like that. So maybe it's nice ideal professionally, but in your personal life, to be perfectionistic is just setting yourself up for disappointment, because we don't achieve perfection very much.
- 3.72** OK, a few general patterns. One is threats to esteem, that people do more self-defeating things when there's some threat to their ego. And this is, again, parallel to the evil and violence reaction to it, which, again, engaging in violent actions is often self-defeating. For whether you win or lose the fight, you often end up in trouble. So it's not a low self-esteem, but threats to your ego make people more willing to do things like seeking revenge, even at cost to the self.
- 3.73** If somebody privately offends you or blocks you, you can let it go. But if it's public, other people will think badly of you, then you have to fight. Self-handicapping, obviously, is based on protecting your positive reputation. Other people are thinking well of you, and so you develop the alcohol problem or the drug problem or the underachievement pattern or whatever, to protect this image of yourself as a really talented high achiever. Revenge at cost to yourself, I already mentioned.
- 3.74** Procrastination is another one. A lot of people procrastinate out of a misguided perfectionism. They think they're not quite ready to start or they don't know how to get going on it. They'd better have a really good idea before they start writing the paper, so that's one element that they do it. And then procrastination can be a kind of

self-handicapping strategy because you wait till the last minute and then you go through this real crisis, and you get it done and you turn it in. So you feel proud of yourself.

- 3.75** And of course, it's not a great paper, but considering you just did it in the last 48 hours, you feel like that was good, so you have an excuse for failure. And if you get an A on it after all, wow, that enhances your positive image of self. Choking under pressure is another one. You want to succeed, but you go about it in a way that backfires and causes you to fail.
- 3.76** Shyness, again, is a threat to esteem. You're afraid to look bad. You want to make friends, but if you go and say something stupid, and then you'll embarrass yourself and look really badly and feel ashamed. Some of the medical things it's involved in, not at all, but ones that either other people knowing you have this problem will reflect badly on you or you're embarrassed during the procedure or whatever. Again, people don't want to do that.
- 3.77** Escape can be a motive for some of these things as well. The self is burdensome, and they want to forget themselves, to lose self-awareness. And so when bad things have happened, people will shut out meaningful thoughts and try to go down to a low level of awareness. There's disaster, but if you can just stay focused on the immediate here and now, it isn't as bad as thinking about future aspects, future outcomes.
- 3.78** So, it applies in some cases to alcohol, which reduces your awareness of self, so you forget your problems and troubles. You don't really think about yourself, so whatever happens doesn't seem to reflect badly on you. This is why people do stupid and embarrassing things when they're drunk, because they aren't thinking, "Well, at the office party, I took off my shirt and put a lampshade on my head and danced on the table. I'm not really thinking that Monday, when I come back to work, I'm going to be embarrassed by this," and so on.
- 3.79** Suicide, that's another case. They're basically trying to escape. Obviously, suicide is literally self-destructive, in that you kill yourself. But it's often an escape and all the implications about yourself and your identity being spoiled. Binge-eating is another thing; often happens, triggered by something embarrassing about the self. Binge-eating mostly occurs among people who are doing fairly well. It's more common among women than men, who have high expectations and so on, but feel they're not living up to them.
- 3.80** And so they're feeling bad about themselves, and focusing just on the putting food in your mouth and chewing and swallowing is a very low-level activity that temporarily blots out those other thoughts. Masochism has traditionally been understood as self-destructive, although I looked into the research on sexual masochism; it doesn't really seem to be. Some of them say it's good for you. I don't see how it's either good for you

either. It's just a weird form of sexual pleasure. But it does, again, seem to have this escaping from ordinary identity.

- 3.81** Emotional distress. We spend a bit of time on this. A lot of the self-defeating behaviors seem to have some element of unpleasant emotion, that people are more likely to do a variety of these things. Self-handicapping, shyness, revenge, drug, alcohol abuse, and so on, often associated—we find unpleasant emotion there. Maybe they want to escape that or handle that, or maybe it changes the way you make decisions, so it's not clear that the distress leads directly to it.
- 3.82** The same with suicide. The suicide, in the pre-suicidal state, they're usually not feeling all that bad. They're easy, kind of numbed out. They're feeling nothing, and they're very narrowly focused on very mechanical sorts of things. There's some studies of suicide notes, for example. The ones in the movies, they're really grand. "I've decided life isn't worth living, and teach my son to be a good man," and all these grand things like that.
- 3.83** And the actual suicide notes are, "I paid the electric bill, don't forget to feed the cat." Very humdrum, mundane things, indicating this very low-level thinking in that state. But that's what they're doing, basically, to keep the distress at bay, which is, by being very concrete in here and now, you can shut out some of the emotional distress. So, given that distress doesn't always lead to self-defeating behavior, we need a more complex theory.
- 3.84** So, one thing I got actually from reading the suicide literature is maybe there's foolish risk-taking. This came up—there are several theories of suicide, and you're looking for data to support this or that theory. But there's a lot of ambiguity about whether certain things are suicides, and so a whole category of them, if you include them or don't include them, they really change the data and change which theory might be supported.
- 3.85** So, there's one on single-car crashes. So somebody's dead in a car, whether it ran into a tree or ran off a bridge or something like that. Was that a suicide or not? They didn't leave a note, so it's very hard to tell. And the researchers argue back and forth, "We should count those, we shouldn't count those." And then there was this great paper that said, "Maybe it's not clear to them either. It's not clear to us. But maybe they weren't really setting out that I'm going to drive my car off the bridge and kill myself. It's just a rainy night, and I'm just going to drive too fast and not wear a seatbelt, and we'll see what happens."
- 3.86** And I've seen people in a very upset state, fighting with suicidal thoughts, just doing this. They're just, "I don't wear a seatbelt anymore." So they're not really trying to kill themselves; they're just taking a bigger chance, and sometimes that chance will come true. And so, could this be one of the ways that self-defeating behavior arises from emotional distress? Again, it's not the impulse, "I'm going to do something bad to

myself," because there's very little evidence of that happening. But people do take risks, and you could be more heedless about the risks that you take.

- 3.87** We looked into this in a research project. For one, we had people who did something, and it had good or bad consequences. We coded their stories. Did they include making a risky decision? Well, most of the ones with bad consequences did. Bad moods were there in both cases. I forget exactly the prompt for the story. But the risky decision, I mean, there were some with the positive outcomes, but a lot of the ones, almost all of them with the negative outcomes, they mentioned doing something risky.
- 3.88** So we said, "Let's try this in the laboratory. Let's see if we can get people to do self-defeating things in the laboratory." Now, what counts as self-defeating in the laboratory? Well, we came up with this procedure that we would tell them to choose. You get to choose one of two lotteries, and you'll be in one or the other. I will draw a card or draw a marble or whatever, and you'll win or lose. So the lotteries, there's one safe one, a big chance of a small reward, and the other is a small chance of a big reward.
- 3.89** Now, statistically, you can make these the same by calculating the expected gain, but we didn't want to have them be the same. We wanted one to be clearly better than the other, especially the small chance of the large reward would be the worst choice. If you brought your accountant or your statistician along with you to the lab, they would say, "Don't choose that one. The other one is definitely better."
- 3.90** Here's an example of the matrix we used. The one lottery had a 2% chance of winning \$25. This was run back in the '90s, so given inflation, probably double the amount of money now to relate to it. That's a lot of money to a student, versus \$2, which was still something then, with a 70% chance of winning. So if you calculate the expected gain, the one was 50 cents, the other was \$1.40, almost three times as good.
- 3.91** So, again, your statistics professor would always say, "Choose the play-it-safe one. That's the objectively better choice." And we also added, if you lose, you have to go to the other lab and go through a noise stress procedure that will be unpleasant, because we wanted it to be not just the absence of a good, but actually something bad. And so that makes the difference even bigger, a 98% chance of having an unpleasant experience. OK.
- 3.92** First study, we manipulated embarrassment. We had people have to sing My Way. We played it for them and we told them we were studying how voice relates to personality. "We want to get some recordings of your voice, and talking isn't good enough, so we need you to sing. Here's the song we want you to sing." We played it, we gave them the lyrics. It's Frank Sinatra singing. I think it's a terribly corny song; I can barely listen to it. Maybe for Frank Sinatra, a great man at the end of his life, if that made sense, but for a 19-year-old student to sing, "I did it my way," it was hard.

- 3.93** Plus, he sings it in kind of a difficult key, as his voice was different from average. So when they tried to copy it themselves, it gets to go too low, so they try to jump up an octave; and if you're not a trained musician, when you jump up, you don't jump by an octave, you jump by a fifth, but that changes the key. So you're just wandering through, trying to rediscover the memory and singing these.
- 3.94** Anyway, it's an embarrassing thing. Meanwhile, the experimenter is sitting there, looking at them and recording. In fact, this is a middle-aged woman who ran it, and she said, "On my recorder, the microphone isn't that good, so I'm going to have to sit in front of you and hold it up so you can sing right into it." So, she's looking right in. Anyway, these poor people, they're very embarrassed.
- 3.95** They had two control conditions. One was a nature video, which is neutral, and then the others watched the Saturday night video, which was funny. The measures— which one of them chose the foolish risk, the 2% chance of \$25, coupled with a 98% chance of having an unpleasant noise stress experience? And as you see, the embarrassed ones were much more likely than the others to choose it.
- 3.96** So next, we went on and tried anger. The procedure for this was the same woman who did it, and she told them, "I want you to think about somebody that you were mad at and tell the story, only not just tell it neutrally, but we want you to get angry and tell them off and say the things that you should have said, and what they did that was wrong, and go ahead and curse and swear and really let your feelings get in there."
- 3.97** Again, she holds up her—she starts the tape recorder, and so the person starts doing it. He starts getting mad, and just as he's getting into it, then she hits stop and says, "Wait a minute, wait a minute, it's not recording right. Back up and start over." So she rewinds and starts it over. All right, this is awkward. And then, so buddy does it and he starts getting—"You promised me this, you damn, whatever." And then she stops again and says, "I don't know. I think this tape is not right. Let me go get another tape and then we'll try it again."
- 3.98** So by now, he's mad from this memory. He's frustrated with this woman and her lousy equipment and so on. And that's when we had them choose which lot are you going to do. And as you see, the nature video, the comedy video, much lower rates of choosing the long shot. I think we had a better matrix in the second one than the first. But again, the people who are angry and frustrated are much more likely to grab for the big reward.
- 3.99** Now, why is it? Why would they choose it? We had a bunch of theories. Well, maybe \$25 would be enough to make you not feel bad or something like that. But we thought, well, maybe they're just not thinking it through. It's sort of impulsive, that you're in a bad emotional state and you're up \$25. Good, grab for that, and not really consider the odds. So we ran it again with a study where, before you decide, take a moment to list the pros and cons of each, and that wiped out the effect.

- 3.100** You see the bottom line, when they had to make a thoughtful decision, it was less than a minute to think it over on average. But that got rid of the effect, and the other conditions were the same as before. So, again, the key reason seems to be that, when you're upset, you're not thinking things through; and if you could just make yourself pause and consider the ups and downs, that's enough to counteract the self-destructive pattern.
- 3.101** So, there are a couple emotions. We looked at anger, frustration, and embarrassment, but is it all negative emotions? Well, we tried others. We had sadness. There was a famous movie, which the hero dies at the end, and his kid is crying and all that, so it was very moving, but that didn't work. We had another where we looked at arousal. Remember the circumplex, there's high and low arousal. We had people run in place for five minutes or so. It turned out, some people thought that was cool and they felt good. Some people felt nothing, and some people didn't like it at all.
- 3.102** So it's kind of like, we had one procedure but produced three different conditions. We had them rate how they felt from minus 10 to plus 10. So we did minus 2 to minus 10. You're feeling bad, minus 1, to plus 1 was nothing. And then 2 to 10 was feeling good. The neutral arousal and the high arousal did not produce the self-defeating thing, but the ones who didn't like it, who felt bad after it, they did go for the stupid risk-taking aspect. So it's only the high arousal, bad emotions.
- 3.103** So again, if you sort emotions by the two dimensions, high arousal, low arousal, good versus bad emotion, it's these, the high arousal, bad emotions— that's the quadrant that seems to lead to this self-defeating behavior. Restating conclusions. So high arousal distress causes this shift. It favors high-risk, high-payoff outcomes. They're usually grabbing for the positive thing. It's not intentional self-destruction by any means.
- 3.104** Our original theory that, well, you had more to gain, less to lose, if you feel bad, we couldn't find any support for that. Revised theory is failed self-regulation. You don't really think through the options to make a rational choice that will be good for you in the long run. When is immediate gratification especially welcome? Well, the worse you're feeling, the more you want to feel better right away. And so, the short-term, long-term trade-off will be more and more resolved toward the short term, the worse you feel.
- 3.105** People who have quit drinking for a while, have an alcohol problem and managed to quit. But when something really bad happens, they break up with their boyfriend, girlfriend or something like that, that's occasion when they'll go back and indulge again and resume and possibly relapse in terms of their addiction. And a lot of these things, too, when you're feeling bad about life in general, you just want to feel nothing, that works as well.

- 3.106** All right, we're in the final two minutes. I've got one more theory to cover here. Immediate gratification becomes more appealing. So essentially, it's a shift in priorities, that when you feel intensely bad, especially this high arousal, negative emotion, that makes you put more priority on feeling good right away, not worrying about the cost downstream, not worrying about the delayed aspects.
- 3.107** There are a whole bunch of studies that did that. I talked about this procedure developed here at Arizona State, where they tell people there's this pill that will freeze your mood, so you'll be in the same mood for the next hour, no matter what you do. Well, that removes any incentive to try to make yourself feel better, because you can't feel better. There's no such pill. It's just a placebo. But people are told that's what's going to happen.
- 3.108** And so, if behavior changes that way, it shows that what they were doing was done and intends to feel better. So let me just quickly get to the results. I'll describe one of the studies. You imagine causing a car accident that was your fault, and you killed a child, versus a good condition, maybe where you saved a child's life. OK, so you feel bad. It's not real, but a guided imagery, you can make people feel pretty bad, as if that would happen.
- 3.109** Do bad moods cause people to overeat fatty junk food snacks? Well, it turns out, yes, we measured how much junk people ate. And so, sure enough, in the basic condition, when they were happy, they didn't eat very much junk. But you see the 79, that's the highest positive number, people ate a lot of junk when they felt bad, when they were in a negative emotional state. But if they were told their mood was frozen, they didn't eat. That was the lowest number. So there's no point in eating junk food. It won't make you feel better.
- 3.110** All right, so sadness, distress led to more eating of junk food, but not if you didn't think it would cheer you up. Foolish risk-taking is one pathway for causing people to do self-destructive things. They're not trying to harm themselves, but they're just being more heedless and careless, taking more chances; and that comes in particular response to high arousal, negative emotional states.
- 3.111** And it happens particularly because it cuts short thinking. People don't process it, don't think through, "What are the implications? What are the consequences? Will I be sorry down the road?" In particular, grabbing for a long-shot positive outcome, disregarding the downside of the risk of things that could go wrong and happen negatively. Second, shift in priorities. The worse you feel, the more you want to escape; especially if you're feeling bad about yourself, the more you want to escape from self-awareness, escape from the emotional state.
- 3.112** And so, again, you're looking, especially for something that you think will make you feel good in the short run, even if you know it's bad for you in the long run. The basic conflict between restraining these impulses and seeking immediate pleasure, when

you're in this bad state, the more intense it is, the more you want to feel better, and so the more you work to get out of this, even if the downstream consequences are negative. And that takes care of it. Thank you.

- 3.113** What would you say the most practical approach or method for maybe limiting these self-destructing behaviors would be?
- 3.114** Probably the single most effective would be to consider the long term. Because again, a lot are disregarding the long-term consequences and focusing on the immediate gain. And so, pausing to think through it. We had a couple of studies that just made people stop and think, and that eliminated some of the self-defeating effect. But other things, like relapsing on drugs or alcohol just because you feel really bad, it turns out that's doubly difficult because people who are alcoholic, they have a certain tolerance, and then they go off for a period of time.
- 3.115** It happened to my brother-in-law, and broke up with his girlfriend. And so he said, "OK, I'm going to get drunk just once," and so on. But he drank like he used to drink. He ended up in the hospital because he didn't have the tolerance. He almost had alcohol poisoning. The same thing with heroin. I've heard heroin addicts who reform for a while, and they're off it. Their tolerance goes down, they don't realize it, and they think something bad happens. They think, "Well, I need another fix." They go take what they used to take. And again, it's way more than their system is able to tolerate then.
- 3.116** But pausing to reflect on the long-term consequences, that is the definition of enlightened self-interest. So you won't go as wrong if you can discipline yourself to do that.

Lecture 4: Addiction

At a glance — Addiction.

- 4.1 All right. Today's lecture, I'm going to grapple with the interesting phenomenon of addiction. My interest in it, as you probably know by now, I'm not an addiction researcher, or certainly not a therapist. I've never sat in a room with an addict, trying to help them with their problems. But I read extensively in the research literature, so I know a lot about it second- and third-hand. For me, the interest is that it clues into the general issues about how does the mind work and how are people psychologically put together?
- 4.2 Obviously, there are practical reasons to be interested in addiction. It's a problem that many individuals struggle with and creates problems for society as well. That's all well and good. But also just for understanding—it's a challenge to a variety of our theories. Is it a special case of motivation? People have wants and needs. Addiction, they acquire as a new set of them. Is it a different kind? Is it just a stronger version? Does it resemble the others?
- 4.3 But also issues of volition. Volition is voluntary control of behavior. Is that where the problem lies? And addiction challenges are some of our notions of self-control and free will. People say they want to quit, and yet don't seem able to quit. Is there something deficient there? Certainly, the inner conflict is an aspect. A philosopher, Frankfurt, put it, "The addict wants the drug, but at a higher level, wants to not want the drug." It's the wanting about wanting.
- 4.4 Of course, self-destructive behavior, which we've already talked about in a previous lecture, addiction is often literally quite self-destructive, causes all sorts of problems. We know self-defeating behavior is real. And as I said, trade-offs are one major category where you're pursuing something good, but something bad comes with it. Many addictions have this feature, that they bring joy and pleasure and so on. But down the road, they cause problems and health and interpersonal problems and money problems and so on.
- 4.5 Subjectively, the good seems to outweigh the bad, but objectively, or in the long run, at least the bad becomes stronger and outweighs the good. One of the patterns of trade-offs that we talked about, short-term gain, long-term cost, addiction certainly has that aspect to it. Likewise, the definite gain, you're going to get high or have your drink or cigarette or whatever, the gain is pretty much guaranteed. The cost down the road is merely possible, not guaranteed. So both of those are familiar patterns in self-defeating trade-offs.
- 4.6 Indeed, Ainslie, who's written about this, had the interesting point that, you know when you decide to quit, it's always rational to have one more cigarette. One more cigarette is not going to kill you. But the problem is, these rational choices to always

have one more, accumulating over time, produce the destructive result, and you don't end up quitting, and you may end up with some of the health problems associated with it.

- 4.7** So, short-term gain with long-term cost. Why do addicts resolve this differently than other people? Well, the first point is, addiction seems to change your relationship to time. I've shown in previous lectures that people rate their thoughts more meaningful. Meaning really connects across time. That's one of the functions of it. And indeed, we evolved as part of our free will to base present actions on the future and the past, to import other time periods into our decision-making here in the present.
- 4.8** And so, again, addiction changes your relation to the future; that could upset the rational decision-making process. Now, economists have ways of looking at how people relate to time and, as they generally show, people discount the future. In a standard research paradigm, they'll say, well, you could have \$1,000 a year from now. You can't really give them that, but imagine you could have \$1,000 a year from now, or you could have some money today, but a lesser amount. How much less would you take?
- 4.9** If it's \$100 today versus \$1,000 a year, most people would say, "Well, sure, I'll wait a year for the other \$900." If it's \$995 today versus \$1,000 in a year, they'd say, "Well, I'll take the money today. I'm not going to wait a year for five bucks. Put it in the bank and make more than that." So people will both make the immediate and the delayed choice, depending on how much they lose. And the key point is, somewhere in the middle is where you switch from saying, "I'll take it now," to, "No, it's worth waiting the year for that." So what is that point?
- 4.10** And this researcher, Warren Bickel, has done this with cigarette smokers and found a striking difference. The average for nonsmokers, these are people who never smoked, 825 is about the break-even point, where "I don't care whether I have it today or wait a year for it." In contrast, smokers, wow, 575. Much bigger discounting of the future. I think they also surveyed people who had smoked and quit, and they looked like the never-smokers, which—it's not random assignment, so you can't be completely sure. And maybe people with a better sense of the future are better able to quit.
- 4.11** But it looks to me like it's a sort of defensive thing, that while you're smoking, you don't want to think about the future. Focus on the present, when it's lovely and fun, and don't think about, down the road, you might get illnesses and suffer and die. And so, when you quit, it comes right back to being able to contemplate the future. Anyway, that's one sign of discounting the future.
- 4.12** Another of his studies that I really like, that was widely remarked, he had people write a short story, basically a one-paragraph story, starting with, "One day, Bill woke up and began to think about the future." Now, I ran this with college students, and the average, you looked at how far into the future the story extended. What is Bill

thinking? It's an imaginative, projective thing, but this is what the future means to them. They weren't aware that this was the key point of the experiment.

- 4.13** It was about five years; a little bit under five years was the average, which, for your average college student, that's probably a reasonable horizon. He ran it again with heroin addicts, and it was two weeks. So it was just a shocking difference. But the heroin addicts really live in a much shorter time frame. And the future, the future, the week after next, my goodness, that's not very far into the future. So again, addiction changing your relationship to time.
- 4.14** And a variety of things also change people's relationship to time and make people focus on the present and discount the future, a number of things that will contribute to drug abuse and possibly incipient addiction. This would be emotion, stress; being intoxicated also gives you a more short-term focus on the present. So, addiction does fit one pattern of self-defeating behavior by seeking immediate pleasure and downplaying or ignoring, just not thinking about the long-term cost that might come with it.
- 4.15** All right, next. One of the fundamental issues is, what is motivation? Is it more about motivation or volition? Both are important in psychology. And is it another change of wanting and needing and craving and so forth? Or is it a change in how you control your behavior? If it's a motivation issue, is it some kind of special case that people acquire this desire, or is it similar indeed to other forms of motivation? Perhaps it's a more extreme form of wanting.
- 4.16** On the other hand, maybe it has something to do with losing control of your actions, or the control of action maybe remains intact but serves a different master. It serves the addiction, and you exercise it in order to figure out how to get the heroin or get the cigarettes and where you can ingest and so forth. Let's look at both of those. They both have a case.
- 4.17** In terms of the change of motivation, when I started in it, this is what I thought it was all. It's just, you develop these cravings, and certainly some people have them. It's not the definitive answer. And motivation, you can sort it into wanting but also liking, and researchers on this started to separate. Even in the brain, there are slightly different areas for wanting and liking. I mean, you mostly want stuff that you like, so there's some overlap.
- 4.18** But the way you can look at it is, you discover a new source of pleasure, which is then reinforcing, and then, "Oh, that's really good. Let's do that again." And so, you acquire the addiction. And people pushing for the motivation argument will say, "Well, look, it's not really like a deficit in self-control, necessarily." Often, you have to use self-regulation to get here to do it in the first place. People usually don't like their first sip of beer or their first cigarette. They're doing it with peers or whatever, and they have to push themselves to try it.

- 4.19 But maybe when you get it, then it brings pleasure, and then it makes you start to want it that much more. In a sense, something you initially didn't want, when you get it and enjoy it, then you start to want it; and the wanting becomes reinforced, so then you feel it more often. And in that way, perhaps the addiction develops. You start wanting something more and more, because every time you want and then indulge, it makes you feel really good. And that, again, a basic reinforcement.
- 4.20 On the other hand, look at it as change in volition. There's the brain disease theory that some people have put forward, that addiction is basically some kind of disease of the brain. They even extend it that it's a disease of free will, that you lose your ability to control your behavior. Nora Volkow, one of the top officials in the National Institutes of Health, has argued, yeah, it's a disease of free will, which is sort of an extension, for her, of the brain disease idea.
- 4.21 The public tends to perceive it that way. They focus more on what you're doing than what you're wanting. They say, well, even when you want to quit, you keep doing it. That's what addiction means. You're not somehow succeeding at doing what you want, which is to stop the addiction. There's also a great line of research contending that it's not a free will issue, it's a habit issue, that for many addicts, it just becomes a habit often activated by cues.
- 4.22 It's five o'clock, time to have a cocktail. Many people report doing that. It's not that they suddenly have a craving at five o'clock. It's just, at the end of the day, they enjoy that. And so they automatically make themselves a cocktail without really having any sort of a craving aspect. And the cues, addicts supply themselves with cues. If you're a smoker, you buy ashtrays and lighters and things like that, that all remind you.
- 4.23 I had a friend in the Netherlands. I used to go over there every year and teach a two-week intensive seminar. One of the guys there had told me he had spent much of his early career sitting at his computer, reading, writing, thinking, and smoking. And he said it was just lovely. It was wonderful to think big thoughts and write things around or analyzing data and figuring out what they mean. This is part of the fun of being a professor, and having a cigarette while he did. It just made it pleasant.
- 4.24 At some point, he decided he had to quit smoking, but of course, he couldn't quit writing on his computer. That's part of the job, is to do that. He said it was just really hard to go and sit in that same chair, at the same desk, and type, and he would just have this, this habitual impulse to light up the cigarette, which he had to deny himself. I went back the next year, he'd actually moved to a new apartment, I think for that reason.
- 4.25 Indeed, they sometimes say that, if you really want to quit smoking, that's one thing to do. Just go to a new place, so there's no odor of cigarette smoke. You have never smoked there, and so you can accept that, no, I don't smoke in the place. Some people also succeed when they're on vacation—again, away from home and away from all the

cues that sustain the habit. Quitting, when you decide to do it, requires overriding it, and that becomes difficult. In order to quit, that has to be the default, that you just say no to yourself.

- 4.26 There was an interesting paper some time ago that asked people who were heroin addicts and who smoked cigarettes and were alcoholics—OK, triple threat—they asked them, "What would be the hardest of the three to quit?" I thought it's heroin, from what I've seen in the movies and things. I don't really have any first-hand knowledge. But no, they said cigarette smoking would be the hardest of the three to quit. That puzzled me. It's obviously a much lesser intense drug than heroin. Why can that be so much more difficult?
- 4.27 But one thing that occurred to me is, you take heroin, even if you're a really serious addict, you do it once a day. That's it, so seven times a week. Whereas heavy cigarette smokers will do it 40 or 50 times. So the habit is much stronger. Have a cigarette, have a cigarette, have a cigarette. And so when you quit, you have—those impulses are still there, those habitual, time for a cigarette. But you have to say no to yourself, over and over and over again. Maybe that's one of the reasons why they report it's more difficult.
- 4.28 These self-reports by addicts, admittedly, we can't trust them fully as much. We know in psychology, self-reports in general are somewhat suspect. But addicts in particular rationalize things and so on. Some policemen and some treatment providers have the expression, "All addicts are liars." Yeah, we want to go ask and see what they say and so on. But again, take whatever they say with a big grain of salt. But still, that was the most common answer. Quitting cigarettes would be the hardest of the three.
- 4.29 In terms of free will, well, I gave a talk on that in the first set of lectures. Free will, people argue about it, but they're arguing based on different definitions. As I said, I tend to agree with both the ones for and against because they're not talking about the same concept. Of course, free will is not all or nothing, but as I said it's on a continuum. Some actions are freer than others.
- 4.30 Essentially, everyone agrees, even the people who don't want to call it free will, say, "Yeah, the humans have an advanced form of controlling their action." It's beyond what other species can do—being responsible and autonomous, important parts of it, conscious choice, and so on. Is it lost in free will? As I said, there have been at least writings saying it's a free will disease. It's evolved from the brain disease people.
- 4.31 The brain disease model was very popular for a while, especially since the big shots at National Institutes of Health were promoting it, and people tended to go along with it, because that's the way to get federal money. But a lot of people are very skeptical of the notion of a brain disease and so on. I always had a sneaking suspicion.
- 4.32 The National Institutes of Health people used that because it was useful, because they depend on Congress for funding, and especially when the Republicans are in charge,

they don't want to have a lot of taxpayer money go to help drug addicts. And so they're skeptical. But understanding it as a disease, well, OK, they're in favor of health. So it might be, I just wondered, this is my personal speculation. I don't have any evidence, but I didn't know if they really believed the brain disease thing or if it was just a useful way to keep the money flowing.

- 4.33 The loss of free will, again, is, some people suggest that, if you quit and then relapse, that's a sign that your free will was damaged. I don't find that very compelling. I mean, if free will means anything, it means you can change your mind, right? So you decide to quit and then you decide to resume. There's no argument against free will. And we might ask, is free will damaged in general among cigarette smokers? As one case, a large number of addicts, but obviously not. Lots of people, smokers, have been very successful in life.
- 4.34 We've had presidents of companies and presidents of countries. We've had scientists and artists and air traffic controllers, all being regular addicted smokers, and were able to perform their job quite well. So it's not that the free will, per se, is damaged. Maybe it isn't working right in this one particular sphere of your life, the addictive behavior, but it's certainly, by and large, intact for dealing with the rest of life.
- 4.35 One way to put this together is to say, well, it's mainly about motivation. Maybe your urges are so strong, you can't resist them. Maybe your cravings, or if you try to deny them, they get stronger and stronger until you just are powerless to resist. There's some studies on this. They look at people's cravings after they quit. This is not true. Your desire for cigarettes when you decide to quit goes down; it fluctuates.
- 4.36 Now, people think it's going to get stronger and stronger, and "eventually I'll give in, so I might as well give in now." But this looks like a rationalization because, when they actually track your cravings over at least several days after quitting, they go down and they do fluctuate. And occasionally, they get a little bit strong. But in general, no, and they don't become super strong. The idea that it's such a strong desire that you can't possibly resist it, "I can't help it," addicts like that because it gives them an excuse. But there's precious little evidence that it's true.
- 4.37 As I say, people believe that. It's the stereotype of addiction, and the addicts encourage it themselves because it's nice for them, but it doesn't appear to be true. I can present results from a study we did on something else. It was on desires, as people went about their everyday lives, with Wilhelm Hofmann again. He made this graph, which actually won an award. It was one of the best science graphs of the year. It has a lot of information about what people want, and the larger it is, the more frequently; and the red pie slot is whether they indulged it and so on.
- 4.38 But you don't need to go into all that. It's just for present purposes, the important thing is look at the strength of the desire, which is along the bottom. The weakest desires people report, over the end, are desires for tobacco and alcohol. So these are people,

presumably regular smokers, I mean, they only counted people who wanted tobacco. You can say, well, it's a low score because lots of people don't smoke. Well, no, they weren't reporting their desires. If they reported a desire for tobacco, they reported it as a weak one.

- 4.39 Again, suggesting the issue it's not an overwhelming craving. It is just for tobacco or maybe a lot of them, if you have a 40- or 50-cigarette-a-day habit. And alcohol, the same thing. Yeah, people want it, but it's not, "I need alcohol, please." It's, in general, quite a weak one. We looked, maybe some of them are super strong, but no, it had a lower frequency of super high intensity ratings as well.
- 4.40 That made me start thinking about the process of taking substances. Sometimes that may have some appeal. I'm going to compare the process versus the outcome. The outcome is the state, like being high on cocaine. The process is what you do to ingest the drug. And if you compare different substances, the process varies widely in the appeal.
- 4.41 I mean, injecting heroin into your arm—I always thought, out of scientific curiosity, if I ever had the chance to do heroin somewhat legally and safely, of course I would try it, just to find out what it's like. I'm not going to get addicted. But sticking a needle in my arm, it would put me off a bit. I would hesitate at that. That's dangerous. In contrast, drinking beer, it's fun to do. Just the process is fun, regardless of whether you do it. We see lots of people have, now like the alcohol-free beers because they can have the experience and the taste and all that, without any outcome at all.
- 4.42 Peyote is a drug. I don't think it's addictive, but people do consume it, and you have to chew this very awful-tasting stuff and swallow it, and you usually vomit at some point, so that's a little bit off-putting. There are other bad-tasting drugs. Ecstasy is one of them, as well. Snorting cocaine, I'm not sure whether the process is all that appealing or not; we certainly don't have the equivalent of alcohol-free beer with cocaine, so you can have the pleasure of snorting powder up your nose without actually getting high. So that suggests to me, they don't really enjoy that.
- 4.43 Drinking champagne, that's something I like, and that's lovely. There's all the ritual of popping the cork and clinking the glasses, and then it's delicious. Smoking marijuana, that's neither good nor bad. It's something people do. It's not usually the smoking part that appeals to them, the process. Anyway, these various drugs do get you into altered states, but the process has high variability in the appeal.
- 4.44 Putting all this together, maybe addiction has several different processes, several different effects. As you're attaching this, you can take the liking, wanting, and doing aspects separately, and the process and outcome, and make sort of a table. Addiction can change any or all of these. Different addictions might be different combinations of those six things. That's going to make treatment difficult, because if they figure out

what they think the main problem is, and even if they treat that completely successfully, you might get pulled back in by one of the other things.

- 4.45 I want the taste of a beer or champagne or whatever, even though I've cured my craving for alcohol. I think of it this way, a three-by-two table. So there's six possible ingredients, and I want to talk about them a little bit. But again, any given addiction may be a mixture of these. And again, that may be why they often are so difficult to treat and cure. They're not mutually exclusive. I'm not calling them six different kinds of addiction, but any one addiction can have two or three or only one, or could conceivably have all six and in different amounts, but not necessarily all of them.
- 4.46 So that's what makes some of the different types, is the different processes, and treatment may have to focus on these. Just to spend a little time on each of these, first, the liking. You might like the outcome of being the buzz that you get. You might like the process and not care about the buzz. People who drink wine, a lot of people have had this experience or with beer or whatever, you're with your friends, you're having a good time talking and laughing and so on. Then you realize the bottle's empty. So he says, "Well, let's go have another bottle."
- 4.47 It's not that you want to be more drunk. It's just you're enjoying sitting there and sipping the wine with your friends and having a conversation and laughing and so on. So you have another one. And it's, again, the attachment to the process, not necessarily driven by the outcome. People who argue in favor of alcohol note that it does promote interpersonal harmony and enjoyment and probably sex. So it has a variety of prosocial consequences. So liking, again, part of addiction is discovering a new pleasure, something that feels really good. People don't get addicted to stuff that feels bad.
- 4.48 There is this question of the joyless addiction, and I saw, through the literature, people would talk about it and say, "Well, some of my clients say they're addicted, but they don't get any pleasure out of it at all." This, to me, is a question for research. Is that real or is this another one of those addictive rationalizations? They say, "Oh, no, I'm not indulging. I'm not just a selfish pleasure-seeker. I'm a helpless person with a disease. I have to do it. It doesn't give me any pleasure at all."
- 4.49 On the other hand, it occurs to me there is a way that could be correct, that they might sort of sneakily say, "Yeah, it doesn't give me any pleasure," and that maybe it just gets rid of feeling bad. Again, you can look at subjective feeling on a, say, 11-point scale here, from plus five to total bliss, to minus five, total misery. Well, some drugs cure the misery and block it out. You don't necessarily have pleasure.
- 4.50 But if your life is mostly, you're feeling at minus four, to get to zero, that would be very appealing. You could still say, "Oh, no, I don't get any pleasure out of it, but it does kill the pain," and that's part of the appeal. There could be joyless addictions in that sense. Anyway, I don't know. I didn't see any convincing data that there are addicts who don't

get any pleasure about it. It's mostly anecdotes and therapists saying it, but it is something they do hear and say repeatedly.

- 4.51** Turning to wanting. To me, again, that was the central change when I started, that you start to want something that you didn't know existed before and you didn't really have any pattern for, but you start to want to do it. Basically, you acquire the desire, and then the desire gets reinforced; as I said, getting begets wanting. We did do a whole series of studies on, which was just when methods were changing and so on, so we never published these, but we tried getting people hooked on a pleasure.
- 4.52** So we would assign people to spend a week playing a handheld video game. And they did it, and then they'd come back to the lab, and we'd say, "Well, you can do that one, or you can do a different one, a new one." Well, a lot of people wanted to stick with the same one, whichever one we had assigned them. We had some people addicted to taking naps. We had them take a nap every day in the middle of the afternoon for a week, and then, oh, they kind of started to want to have naps. I mean, it's not a life-changing addiction or anything like that, but they did show increased desire, consistent with the idea that you do something and you enjoy it, then you start to want it more.
- 4.53** Another was getting people hooked on the news, students who never watched the news. They said, "Well, watch the news every night and take a couple of notes on what you found." And by the second week, the experiment's over, but they're still wanting to follow the news. So this depicts this aspect, at least; addiction is consistent with other motivations, that you do something, you enjoy it, and then you start to want it more regularly, and it shows up, and you don't have to do it anymore or you have a choice. No, you kind of like that and you go back to it.
- 4.54** Again, though, the impulses are weak, not strong. It's not craving. It's just, "I've got to have a nap. I've got to watch the news." These are weak desires, but enough to guide behavior and establish the pattern. Also, there's some interesting research that desires adjust to opportunity, which is something that blew my mind when I first thought it, because I thought the desires are rooted in your psyche or something. But this fellow Dar did some really fascinating studies.
- 4.55** So he took heavy smokers, addicted smokers; and in one set, they were Orthodox Jews in Israel, and in the other, they were flight attendants. And well, what happens? They're not allowed to smoke, the Orthodox Jews, on the Sabbath. So you think they're going nuts by Saturday afternoon? No. They have their last cigarette before sunset on Friday, and then the desire goes away, and they don't really have it again. And then when the sun starts to go down the next day, they say, "Oh, I feel like having a cigarette again."
- 4.56** And they're able to do this. And you had them also pick a day of the week and abstain during it, and they were pretty good at that, pretty good at that, too. Same with the flight attendants, because you can't smoke on the plane, right? So you have your last cigarette before the plane takes off. So a flight attendant, in the middle of a nine-hour

flight, hasn't had a cigarette for five hours, still doesn't have much in the way of cravings, less cravings than, say, if it's in a 90-minute flight, when they're only one hour into it, because they're coming toward the end and that they're already having more desire for it.

- 4.57 Some people can relate to this idea, talking about urination. When there's no chance, you don't really feel like it. But when suddenly, oh, there is a chance, you need to go to the bathroom, now is it, suddenly people have the urge, and they feel it much more strongly. So again, the desire adjusts to the opportunity.
- 4.58 I suppose another example is baseball fans. What happens to them? Some people, their interest in baseball gets to the point where it really looks like an addiction. What happens when the season's over? No, they're not suffering, climbing the walls or anything. It's not there. But spring training starts, so then they start to want it and get interested in it again. So people can quit addictions when it's just not possible. When it's there, somehow that activates the desire more.
- 4.59 And last, doing, you form the habit of using. As I said, a lot of people will have their cocktail at five o'clock every day, whether they feel like it or not. It's just part of the habit, the craving. We don't really ever report having a strong craving for either the process or the outcome. It's just something you do. And as I said, users accumulate cues. The drinkers have all their fine glasses and fancy bottles and things like that.
- 4.60 So it's not loss of free will in this case, because volition is the wrong place to look. It's really more in the automatic system. A habit is something rooted in the unconscious that you do automatically, and so that may be what changes. It's not that your free will is somehow damaged, but you get more of these automatic habitual responses—time to have a drink, time to have a cigarette, time to shoot up, whatever. So you have the same conscious control. It does, however, have to contend with these automatic forces, either the habit or the frequency of desire.
- 4.61 In terms of the process and outcome, something that struck me as I read the literature, back in the '70s, the surgeon general had started to say that smoking is bad for you. So they decided, well, people are going to smoke anyway, so let's change cigarettes to be low-tar, low-nicotine. Well, that didn't work at all. People were really mad and grumpy, and they smoked more, and they would put tape around the cigarette to block the filtering holes and things like that. So, that was a complete failure, in terms of helping people quit smoking.
- 4.62 And that's when they started to realize that nicotine is an addictive substance, and so people want to keep a certain level in their bloodstream. When that goes down, then the body starts to want another fix of nicotine. So they said, OK, how about the patch? They put something on your shoulder that delivers nicotine directly into your bloodstream. That should have solved the problem, and it was surprisingly

unsuccessful. I'm still kind of stunned why that didn't work very well. A lot of people are different, and it probably worked for some, but in general, it didn't do it.

- 4.63 But one thing about it is, there's no doing. Whereas what does seem to work was vaping. A lot of people I know have quit cigarettes, and then they just vape, and they get the nicotine, but there's also the doing aspect, and it's fun. And part of the smoking is doing stuff with your hands and lighting up and all that stuff. I've tried a few cigarettes over my life. They never did anything for me, but cigars were more fun. I occasionally smoke cigars, but it doesn't produce much of an outcome. You don't get high from a cigar, but it's fun lighting and doing all the fussing with it, so that part is kind of cool.
- 4.64 And so vaping may work better, precisely because it addresses both the process and the outcome. It's not the only possible interpretation, but I think it's a very plausible one. I gave this talk at an addiction conference, and one guy had a really interesting comment, that the other thing you miss from the patch is the transition. It keeps you at a steady state, but they want the whoosh of getting high. So it's where they get the experience all at once. You smoke a cigarette and you go from, to being livelier again.
- 4.65 So that is another interesting point, and I think that calls for more research on the whoosh. I heard people talk about that, too, with marijuana, which is not generally considered addictive, except in a few cases. But as they switched to edibles, the edible comes on so slowly that you don't have the whoosh of suddenly you're high, and they sort of miss that aspect. So that may be a factor, and we notice changes more than steady states.
- 4.66 So again, look at this—six possible ingredients. And again, it's not six types of addiction, but six things, and any particular addiction may be a mix of those. I think that's potentially a more powerful way to understand the complexity of addiction, and why different treatments work and don't work, and why, as I said, if you pick the main one, even if you successfully cure that, they might get sucked back in from one of the others.
- 4.67 That raises the question, why do addicts relapse? It could be any of these that could contribute to this, and therapists who help with one might cause a lapse, even if you just want to have one cigarette or one drink or something when you haven't had any for a while. But lapse does lead to relapse. There's Alcoholics Anonymous, says, "If you're an alcoholic, it's a lifetime condition. You can never have even one drink. If you have one drink, bingo, it's gone."
- 4.68 Alcoholics Anonymous works about as well as all the other therapies, which is not that well. If you know people in AA, they do periodically relapse. It's social support for abstaining, and the strategy is questionable. Again, good for them. It works. It helps a lot of people. But it insists not a single drop. But then when they do happen to have one drop, then it's all over. It's like an army putting all its troops in the front line. No

reserve. So you have the best defense right there against any breakthrough. But if they break through, no reserves.

- 4.69 Their contention, that a lapse will inevitably cause a full relapse—a lapse is a one-time thing; a relapse is going back to using as full habit, as you were before—well, empirically, that's not true. I mean, people do lapse here and there without a full relapse, but it does certainly increase it. Now, is that a sign or whatever? There's a great experiment that I'm not even sure the ethics committee would let people run today, but it was really valuable information.
- 4.70 They got people to quit smoking. I forget whether they were wanting to quit or not, but they got smokers to quit. And then on, I think, day three, after they had quit, they brought them back into the lab, and half of them were randomly assigned to smoke one cigarette. So they reported their cravings and so on. The third day of quitting smoking, you have some wishes for it and so on. Well, the smoking the one cigarette took care of those, but they relapsed at a much higher rate than the ones who didn't smoke in the lab.
- 4.71 So experimentally, random assignment so that you can make a causal influence, lapse does cause relapse—not deterministically, not everyone, and so on, but significantly increasing the odds of relapse; in other words, going back to smoking with the full habit that you had before you quit. So again, it's a sign of getting begets wanting. You've quit smoking for several days successfully, you have one, feels good, and so that makes them more likely to relapse.
- 4.72 So, let's look at the free will aspect. This is something I have published on, and there's a lot of data on this, and I looked at a lot of things. If you want to read the full argument, it's in Addictive Behaviors Reports. But the answer is, they retain their free will. It's not that addiction somehow damages or removes it, even though people do have a stereotype, that somehow that's what happens. The stereotypes, of course, are sometimes wrong. A lot of the stereotypes are based on what we hear in the news and, indeed, on studies, but they're based on inpatient treatments.
- 4.73 So they'll bring people with the heroin addiction or whatever into the hospital and dry them out and get them past it and so on. But then they go back and they relapse. And so the medical people say, "Well, it's just impossible to treat addiction successfully." That's the impression there, but a lot of these people have multiple mental health issues. They're not normal, they're not stable. They're also maybe living in messed-up lives in difficult circumstances, where the heroin is the only pleasure they get.
- 4.74 So, they go to the clinic, get dried out, cured, or whatever, and then sent back to their messed-up lives, and pretty soon they resume. It's not a sign that it can't be cured. There's an interesting survey that was done, asking people, "Is it a disease that you can't help or is it a choice?" You retain your free will, and you're able to control what you do. Interesting—the addicts themselves want to say, "I can't help it." Of course, its

rationalization gives them an excuse. The paid treatment providers, usually medical people and so on, who are making money off treating them, they also want, they like the disease model, which, again, would fit with what they think.

- 4.75 The volunteer treatment providers, who are mostly former addicts themselves, they say, "No, it's a choice. You still have control over it." Researchers, they didn't include in that, but I found some compilations of different ones, and they are spread across it. I think it's only a minority that believes that you completely lose your control. They're mainly split between, you retain control over your behavior or you lose some control, but you retain some. So there are clearly different opinions.
- 4.76 Obvious points, again. The fact that you resume after you quit, that doesn't have any proof about free will. It just means, as I said, free will means you can change your mind. As I said, a lot of cigarette addicts have been very successful. You had some alcoholics that did fairly complex jobs and so on. The free action, the hallmarks of it include resisting external pressure, planning, flexible execution—all those we see with smoking. Smoking uses the voluntary muscles. It's not like a reflex or something that you can't control, an involuntary response.
- 4.77 I contrast it with, if you've ever had food poisoning and you just suddenly can't stop vomiting, it's going to happen right now, regardless of where you are. I think one of the American presidents actually vomited in somebody else's lap at a state dinner. So, yeah, I mean, obviously, that's not protocol. He would have stopped if he could, but you just can't. But smoking is nothing like that. You use all the voluntary muscles and so on. They certainly do a lot of planning and premeditation, especially these days, because it's so difficult to find a place where you're allowed to smoke.
- 4.78 There's a funny paper that looked at when the Dutch suddenly banned smoking in offices, the people with poor self-control actually quit more successfully because they couldn't figure out how to do it. The people with high self-control, figured, well, the smokers with high self-control, "I can continue. I just have to go outside, cross the street, sit on the park bench and have my cigarette there, and I have to make an opening in my schedule to do it." After a while, the low self-control people figured that, too, so the numbers converged.
- 4.79 But again, a lot of planning, premeditation. You have to buy the cigarettes and make sure you have a lighter and so on. Adjustment adaptation. It's not a stereotyped thing. If you try to light your cigarette and the lighter doesn't work, you're not stopped. You say, "OK, can we get a different lighter? Or get a match, or something like that?" Again, a sign of a voluntary free action. And if resisting external pressure—there's a lot of pressure on people not to smoke now, so you can say, "Well, that's a sign they have a lot of free will. They continue to smoke, despite pushing them to do it."
- 4.80 So, "can people quit?" is one of the great questions in addiction. Most people who have ever smoked have quit. So, many of them quit multiple times. I can say, well, they

didn't really quit, but they really did quit for a while. Usually, the pattern is, people eventually quit smoking, but they'll quit temporarily several times along the way. And each time they sort of stay off it for longer, and finally, they can keep off indefinitely. Most of them quit without professional help.

- 4.81** Indeed, to get off smoking, during the Vietnam War, lots of our soldiers were heroin addicts. Partly it was a very dangerous lifestyle, alternated with being very dangerous or extremely boring, like standing guard for nine hours. So, you're just standing in one place. It's nice to have heroin to make the time go by. And they start to realize, "Oh, my God, when the war ends, we're going to be bringing back all these young men addicted to heroin and turning them loose in our streets. What's going to happen?" But most of them just quit like that.
- 4.82** Lee Robins did these studies, and it was not a problem. Some of them even used heroin recreationally without becoming re-addicted. So, yeah, addiction was embedded in the lifestyle in Vietnam. It was something they did there, and when they came back, they stopped. One of the researchers, Stanton Peele, has a chapter in his book saying, "People who have better things to do don't get addicted." When you're a grunt soldier in Vietnam, there isn't really anything better. But you come home and then you get a job and a girlfriend and so on, then you have better things to do.
- 4.83** It turns out, the average course, indeed, of a heroin addiction is, they get addicted around age 24 and quit around age 28, when they start to get a job or a family or other things, and it's just not compatible anymore. And that happens without any professional help or anything like that. That is the norm. It's called maturing out. But it is the normal pattern. They're addicted for a while and they quit. And again, temporary quitting is common, and they can do it quite well, like in the studies of the Orthodox Jews or the flight attendants, when they can't have a cigarette, then it's OK, and the desire really goes down.
- 4.84** So, question, can people quit? Yes. Russell and Davies had the review article. "The only evidence we have that people can't quit is that they say they can't quit." So you can ask, why don't people quit then? Well, maybe you need a reason to quit. Interesting question. So who do you think has the best rate of quitting cigarettes? Older or younger smokers, and heavy or light smokers? And the older smokers will be people who have been smoking continuously for decades.
- 4.85** The most successful quit rates are the heavy, older heavy smokers, who, people have smoked for a long time. And that's, the reason is because the health effects are starting to show up. They're starting to realize it's bad for you to continue. So that gives them a reason to quit.
- 4.86** Or there's a wonderful study done. Some studies have tried paying people \$5 if they can go a day without smoking and stuff like that. That never worked very well. But Volpe, they got to work with a big company. They found everybody who smoked, and

by random assignments, and half of them just some brochures about why they should quit. But others, they sent the brochures, plus offered them money, and it was a sizable amount of money.

- 4.87** This was 2009, and probably did it around the turn of the century. I think you could get \$1,500, and today, that would be probably well over \$2,000, if you could quit and have no cigarettes for a year and a half. They had biochemical tests, so they could tell if you'd had a cigarette. Well, that was a sizable amount of money, and so a lot of smokers—and these people were not looking to quit. They had just been approached by the company with this offer, and a lot of them did it. It worked much better than just the brochures.
- 4.88** So, yes, when there's serious money involved, people can do it. In fact, I looked through the data and there's a real interesting thing, is that something like 20% of them quit smoking, stayed off cigarettes for a year and a half, earned the maximum amount of money, and then went back to smoking. They didn't have a reason to quit anymore, but they could do it when there was a good reason to do it.
- 4.89** In terms of why people smoke and why they relapse, I have to say here, my parents were in the era when everybody smoked, but they quit smoking early, and they were very passionate about it, very adamant. So I was brought up thinking, "Smokers and smoking are just so bad and contemptible. That's such an awful thing." So I was very reluctant to, in terms of my training, to acknowledge that there could be anything good about smoking. But as I read the research, I did find a few things.
- 4.90** For one thing, smokers report that they get this positive feeling, this feeling of joy. Like I said, I've tried maybe two dozen cigarettes over my lifetime. Never worked for me. I never got anything out of it. But some people definitely do report that there's this really good feeling. It's an improved cognitive function. There's a study that gave nonsmokers a cigarette, or not, random assignment, before they took an IQ test, and they did better on the IQ test. So it does improve your mental functioning.
- 4.91** It's one reason when, centuries ago, when people started more working in offices and things, farmers, they could just be mildly drunk all day. You don't need it. The water is healthier if it has alcohol in it. So they would just get up and be drinking all day. When they had to work in offices, having a break for a cigarette was better, because you've got to be able to add and subtract and other things. So that improved there.
- 4.92** Smoking suppresses your appetite. This is something people complain about, gaining weight when they quit smoking, hoping that this is just some temporary thing. But no, your appetite comes back when you're not smoking. But while you're smoking, it does indeed suppress this. This is particularly appealing to women smokers who want to keep the weight down and be fashionably slim. Nicotine is a stimulant, so it gives you arousal. So, in other words, you're happier, you're smarter, you're slimmer, and you have more energy.

- 4.93 OK, putting those together, I kind of understand why people smoke, and likewise, why they relapse sometimes. I've spoken about limited willpower and ego depletion. This is going to be a factor in relapse, probably as well. Your ability to control yourself, your willpower is limited, and it's a very well-replicated finding. And so, saying no to yourself, when you get addicted, you acquire a new pattern of wanting. You can't stop wanting. And so, again, you have to say no to yourself.
- 4.94 The urges, as we've seen, are not very strong, but there are a lot of them, especially with something like cigarettes. When you would smoke 40, 50 cigarettes every day, then you have that impulse. "Oh, it'd be nice to have a cigarette. Oh, it'd be nice to have a cigarette." No, no, no. And that wears you down. And sure enough, relapse happens, particularly when people are under stress or something like that, difficult times at the work or relationships or whatever.
- 4.95 So your willpower is going into coping with that, and that makes it harder to continue saying no to your desires for a drink or a cigarette or whatever it is. So, especially, as I said, when you quit, your cravings fluctuate. They don't go away completely. But maybe on a day when your craving is a little high and your willpower is a little low, because of other demands on it, then you will have a lapse. And as we saw, a lapse really increases the tendency of relapse.
- 4.96 Oh, another thing I mentioned on the ego depletion, when your willpower is down, you also feel desires more strongly. That's kind of a double whammy, and so increases the chances of a lapse. Again, self-defeating behaviors, strong negative feelings, not considering the consequences and so on, wanting to feel good, ignoring the downside of the risk, all those things can come together and cause relapse.
- 4.97 Resistance will be extra low, and other demands deplete the self. Resisting itself depletes the willpower, desires will feel stronger, habits will take over. So in a sense, free will does fail, as it doesn't cope with the increased demands. So in a sense, it's not because the addict cannot resist, it's because he or she can resist. But resisting over and over again gradually runs down your willpower, and then you're more likely to give in when your resources are down, over and over and over.
- 4.98 So, pulling it all together, your free will, in the sense of your mind's ability to control actions, that remains pretty much intact during addiction, but it has to serve a new master, as you have these new desires, these new wishes. You found something new that you like and that you want. So you use your free will to figure out, where can I smoke and how do I get the cigarettes and things like that? How do I get the heroin?
- 4.99 The changes are not to your conscious control, but more into the automatic system, the habits, cravings activated by cues and so on. It's a change in the unconscious, which is harder to spot and manage. Your free will is intact, but it's compromised by first discounting the future and just focusing on the present, and second, wasting a lot of your willpower saying no to yourself. So, you discover a new pleasure, you figure out

how to get it regularly or a relief from unhappiness. That becomes appealing. Desire increases with the frequent satisfactions.

- 4.100** When you quit, desire drops; it doesn't go away. And some desires, like for cigarettes, will have many cues to activate them. And you're basically in a situation of having to say no to yourself, over and over and over again. Again, it's not free will being damaged, but the formation of a habit. And so, breaking it requires this effortful override, which depletes your willpower. And a lot of people get cranky and other things when they're quitting, because they can't regulate their feelings, since they're putting all their struggle into resisting the addiction. And so the irritability and other things may be increased.
- 4.101** And so addictions are sustained in multiple ways. Multiple things contribute to it, and then when there's a lapse, that increases the likelihood of a relapse. It's not overwhelming urges; it's a lot of weak desires and habitual patterns. And again, each addiction may be a different combination of the six different ingredients of the process and outcome, crossed with liking, wanting, and doing. All right. Thank you.
- 4.102** I believe you said that the main reason why addicts don't quit is to do with the fact that they rationalize it, tell themselves, "I can't quit," or believe they can't quit. That's one of the reasons. OK. How does belief then, I guess, relate to quitting, being able to quit? If you believe you can't, then you won't, or believe you can, then you might take the steps to quit. How does belief work with that? And how does it play a role?
- 4.103** Yeah. OK, it certainly does. For one thing, one of my friends is a very distinguished philosopher, and he smoked. I think he's finally quit. But he said, "Well, I can resist any particular cigarette. I just don't think I could resist all of them for the rest of my life." And a lot of addicts have the belief that, if I stopped, the desire would get stronger and stronger and stronger, so eventually I would have to give in. So you might as well give in now.
- 4.104** It's like the same thing with people being tortured. Sooner or later, you're going to break and confess whatever it is or give away the information, so you might as well spare yourself all the suffering and confess earlier. Again, that's not what actually seems to happen, but that belief, that basically mistaken belief that desires will escalate, that certainly contributes to it. And yeah, as I said, people, multiple addicts, thought cigarettes would be harder to quit than heroin, at least a lot of them did, which surprises me, and I'm not sure whether that's empirically true.
- 4.105** But again, beliefs certainly factor into people's behavior in lots of different ways. But remember, too, there's a lot of rationalizing going on. They don't want to believe, "I'm destroying my health and wasting my family's money or disrupting everything, messing up at work," because I'm just seeking, indulging my own pleasures. So they prefer, "Oh, I can't help it. I can't." It's very hard to disentangle which beliefs are rationalizations and which have some validity to them.

- 4.106** Can obsessive-compulsive disorder be thought of as an addiction? Because you mentioned addiction is sort of a lack of flexibility in behavior. And my brother has OCD, and so he'll do compulsions for, longest has been 12 hours straight. As you can imagine, it's pretty miserable.
- 4.107** Doing what for 12 hours straight?
- 4.108** Rituals, compulsions. So it's like counting, basically in place. Sometimes he'll move. But I got reminded by that because you were looking for joyless addiction. And that OCD came to mind, because it's not that it gives him joy.
- 4.109** It's not very satisfying, no.
- 4.110** But he has anxiety, and I think it lessens the anxiety.
- 4.111** It lessens the anxiety, so it would be the negative to zero. I don't know a lot about mental health. It turns out, the next lecture, I will say a few things about obsessive compulsive, but this is mostly outside my expertise. The idea, as considered an addiction, is really interesting. From what I've read and what I understand about it, they mostly have this worry. They fear that they're a bad person and that they need to do all these things to keep bad things at bay. And so, there's anxiety if they don't do it.
- 4.112** Seligman made the profound point that a lot of mental illnesses are really exaggerations of healthy things. So, with OCD, the most common compulsions are cleaning and checking, and so those are basically good adaptive things to do, cleaning and checking. Make sure you turned off the stove, make sure you locked the door. But it gets out of hand, that you have to go and check the stove 50 times to make sure it's turned off.
- 4.113** And you go to the bus stop and then you think, "Oh, I didn't lock the door." And then you go back and check it again. And then you go again and, "Oh, did I unlock it when I was checking it?" And so, then you get fired from your job because you can't get on the bus there. How it gets established, we don't know enough about that. But yes, you do get caught up in those little activities. But the idea of thinking of it as a kind of addiction, that's really intriguing. I don't know if there's research on that. And if not, it'd be something to explore.

Lecture 5: The Self

At a glance — The self.

- 5.1 All right. Another one of the big questions in psychology is, what is the self? There's lots of research on various self-processes, self-control, self-esteem, self-awareness. But what actually is it? I mean, if the answer were easy, we'd all know it. I mean, it's a word people use all day or every day, myself and itself and herself. But when you press them to say what exactly does that mean, it's quite elusive.
- 5.2 For half a century, psychology concentrated on studying white rats, which apparently have a lot of resemblances to the human mind. Yes, there's similar patterns. Behaviors that are rewarded are more likely to happen again; behaviors that are punished are less likely, and associations form and all that. But there isn't much of a self there. Rats don't have any sign of self-awareness. We don't know if they have self-esteem or not. The self-control is probably rather limited and so forth.
- 5.3 It's not completely absent, but most of the stuff that goes with human self, you won't really find in rats. The human self is really quite remarkable. It's something—has no analog in most other species that are found in nature. For example, being an agent with a moral responsibility concerned about its reputation, we don't see animals showing much sign of that. Being able to shift among multiple different roles, the human self does that very well. It's highly self-aware, self-critical. It takes care of itself, self-serving in thinking and feeling and action.
- 5.4 Yet it's also capable of self-sacrifice, when circumstances or personal values require it, it uses logical reasoning and rational thought to guide its own behavior. Extensively self-regulating, people regulate themselves a lot, which means organizing behavior in complicated sequences across time, based on long-term goals, things like that. It changes behavior based on ideas, manages its reputation, very sensitive to how it's viewed by others, and so on, functions amid complex sets of rules.
- 5.5 But still, what is it? Is it part of the brain? You'll hear people promoting that view. Is it a soul that's not even a material thing at all? Is that the essence of it? Is it a social construction that somehow society makes it up and puts it on you? Is it a process, perhaps, something psychologically going on? Some people even say there's no such thing, that it's an illusion. It's particularly popular, I think, among neuroscientists, who study the brain, and they can't find any particular spot in the brain to say, "Well, that's your self." So they say, "OK, then it must not exist."
- 5.6 And I think they're coming around to a more sophisticated view now, but it's certainly been a popular position among them for a while. This guy, Bruce Hood, himself, I believe, a brain researcher, wrote a book called *The Self Illusion*. He said there's no such thing. Why did we evolve it? Well, "like every illusion, it serves a useful purpose."

Well, that's kind of weird, generating a useful illusion, rather than say that it's actually something that actually functions.

- 5.7** I read his book, and it was charming and covered lots of work that I knew, but he didn't really convince me that the self doesn't exist. I'm not sure he even tried. He retreated into, "Well, it's not exactly what it seems." Well, that we know. People have mistaken views about themselves. They tend to flatter themselves and so forth. That wasn't a very strong assertion. I notice he puts his name on the book, which he doesn't believe in the self, and he talked about himself a lot in the book, which if he didn't think his self existed, you would think he wouldn't talk about it so much.
- 5.8** Other people have made similar arguments. "There's no singular self. There are just modular systems discharging their functions." Well, no. The functions do coordinate and work together with some coordinated sense of self. Bruce Hood, "Most of us have an experience of a self, an autonomous individual with coherent identity and sense of free will, but that's an illusion." OK, again, there's no such thing.
- 5.9** Here's a slightly more sophisticated one. "Both the I and the me are fictions. In many instances, they're high in pragmatic accuracy." So they're similar to something that does exist, even though they don't exist. Well, that's kind of weird. Why don't you say the thing that does exist? Maybe that's the essence of the self. These views are philosophical views, Eastern philosophy and so on. Let's say, as you engage in spiritual practices, your self disappears.
- 5.10** A study some researchers did, taking Zen monks and other people and asking them questions like, "If you had a disease and there's only one dose of medicine in the world, would you give it away to someone else or would you take it for yourself?", the Zen monks were actually more selfish than other people, which, when the researchers went back and presented them, then the monks were mad, because it goes against it.
- 5.11** There are other findings that people who sign up for a yoga class or a yoga club or whatever, their self doesn't go away. After a while, they start to think they're better at yoga than the other people in the class, so this egotistic inflation. So this idea that it's an illusion that doesn't exist, I find rather dubious. There's something fundamentally real there, and we know it. We say, OK, next time you take an airplane flight, don't take any identification. Just explain to them that selves are illusions, so what does it matter? And see if you get past security.
- 5.12** In terms of what the self is, at one point, I was tasked with writing an overview of what psychology knows about it, a real long one for the Handbook of Social Psychology. Well, if I could just find out, what is the original basic experience that gives rise to the self? Well, I couldn't find it. I actually came down to three, at least—three rather fundamental experiences that can organize all the research on the self.
- 5.13** The first is that consciousness turns around toward its source, that we can become aware of ourselves. It is complicated. You're not aware of yourself the same way you're

aware of the table. You sort of catch yourself in relation to the world. I think some of those philosophical arguments that the self is an illusion are really making this point. It's not that there's no such thing, but it only exists in relation to other things. It doesn't have independent existence. Anyway, how the self knows itself, there's lots of research on that. What happens when people are made more self-aware, research on self-concept or beliefs about the self, self-esteem, things like that. Yeah, plenty of that.

- 5.14** The second, though, there's the interpersonal being. The self doesn't grow up in isolation, sitting under a tree, contemplating your navel or anything. Rather, you learn who you are from interacting with others. The baby learns pretty fast that it belongs to this particular family. It's a boy or a girl. There are basic aspects of that. And much of life is spent being a member of a group, belonging to relationships, forming them, sustaining them, and so forth. So there's a very important interpersonal dimension to the self as well.
- 5.15** And third is the executive function, that you exert control. I mean, with the other two, the self could know what it is and carry on relationships and so on, but it couldn't actually do anything. So the self exerts control over the world, as in making decisions and choices, gaining control—also control over itself, self-regulation, a very important aspect of the self.
- 5.16** In terms of the basic defining properties, I think those three, again, they organize a lot of the research, what people study about the self, and it may be not complete. Again, psychology doesn't have a way to say a list is complete. But those are the big things, and those are, I think, the main things that we need to do. Some people argue, well, there might be multiple selves, but I think the essence of the self is unity. It's indeed the creation or construction of unity, imposing unity, is an important way to understand it.
- 5.17** You are an individual, separate from others, continuous across time. There's some integrated organizing system that makes you the same person that you were last week and last year. We learned to put our shoes on our own feet. That's a real difference. Even fundamental things like that, the self operates pretty well in practice, but as a distinct individual, separate from others.
- 5.18** The self is not a lump of stuff. It's not a physical thing. Again, this, I think, dismayed the brain researchers. They wanted to say, "Oh, this little spot here, that's it," as if eventually you'd be able to cut into someone's brain and cut that part out and plant it in another person, and then you'd be that person, or maybe plant it in a giraffe or a tree, and that would be your new body. But that doesn't seem to happen. They couldn't narrow it down.
- 5.19** And so, in particular, the self is not a piece of the brain. It's not a thing in that sense. Now, don't get me wrong, nothing in psychology happens without the brain, so the brain is certainly involved. But rather, I would think, the self is more something the

brain does than that's something a part of the brain is. So the brain performs roles. There's a process, maybe an organizing system. Those are all real, but they are different from being a physical thing.

- 5.20** One reviewer, commenting on one of my manuscripts, made a profound part, and I wish I knew who it was. But he said, "You can look at different fields, compare them, different subfields, say, in psychology, as to how much they believe in the self." So, as I said, the brain researchers don't really see it and don't really need it. Likewise, cognitive psychology, which looks at particular thought processes, little sub-routines, if you will, inside the brain, they don't need the whole self.
- 5.21** But as you move into the more macro ones, the self is indispensable. I mean, social psychology, you can't really talk about relationships without the two selves who are involved and how they are treating each other. Developmental psychology, how the child learns what it is and so forth. The self is important there. At the larger social systems, I mean, economics, you can't do economics without selves. How could you own anything? Why would there be a market? Why would there be trading, without separate selves?
- 5.22** If you're not going to continue to exist and own the thing that you bought, there's no point in buying it. I suppose you could eat it right away, but in general, you know what I mean. Let's look at it more as a process or a performance. It's not a material thing. People say the self is an illusion. Well, you could have a story without selves. I guess they say there's no such thing as a true story. I mean, you cannot have a story without selves. Stories are about particular individuals doing things. I mean, it really doesn't add up to do it.
- 5.23** But looking at it as a process, this guy, Dan McAdams, has spent much of his career studying the stories people make about their lives. People do understand themselves. Sometimes it's called the narrative self. So you're a character in this, and you have these relationships and these challenges and these adventures and successes and failures and so forth. Then he talked about several steps as children grow up, to develop this narrative understanding of who they are.
- 5.24** The first one is just the social actor. The small child learns what it's supposed to do. You can see some of this structure in the games children play. So small children play games like playing house, where they just act out the role, and you get to be the mommy and you the kid or whatever. And you're going to be the mommy because then you get to boss people around. That's what they think the role of the mommy is. But basically, you're following a script. You learn how you're supposed to act.
- 5.25** Then, as you get into later childhood, you'll become more of a motivated agent. So you're not just following a script, but you're starting to make choices. You're still acting within a story, but you get to decide to do this or that and steer things this way or that

way. And so, your own hopes and feelings and fears and whatnot come into play more, and you realize these more.

- 5.26** And then the third and final stage, it starts really more in adolescence, as you even start to become the author and you're deciding what script your life is going to be about. And obviously, in late adolescence, you start on a career path somewhere, and there are many possible ones, especially in the modern world. In the ancient world, most people were farmers, and there wasn't any other option. But now, there are quite a few, and you decide what romantic life you're going to pursue and so on.
- 5.27** So, essentially, you begin—I mean, you're still acting in it and making choices and so on, but you've added that you're exerting control at the more macro level over what the story is that you are acting in. I think that really illuminates the self as a process much better. And if the self is not a lump of stuff, maybe it's the organizing system. We know matter does lend itself to organization. Some people go down to the molecular level and say, "We don't find the self in the molecules."
- 5.28** Well, it's not there. Most psychology things disappear when you get down to the level of molecules, but they become part of organizing systems. There's a big philosophical issue here. Is really everything about the laws of physics? Physicists, as I understand, used to believe that, eventually, physics would be able to explain everything, but they don't believe that anymore.
- 5.29** And part of the change was this classic paper by Philip Anderson. And if you're interested in the physics and metaphysics aspect, I suggest looking up that paper. It's a classic. It's only three pages long, and really, you get all you need from the first page. But he explained, more is different. The way he described it, he said, well, the sciences exist in a kind of hierarchy. They're not just all separate, all over the place. The most basic ones, physics is the most basic, the most fundamental, and the others build on that.
- 5.30** The point he made is that at no level can you violate the more fundamental ones, but you can—new causes come into play at the higher levels. So chemistry is not reducible to physics. It can't do anything that violates the laws of physics, but it adds new kinds of causes that physics itself can't really explain. The most obvious, I think, is to go from chemistry to biology, because life is a big change. But life is not just a bunch of chemicals.
- 5.31** And again, life has to obey the laws of chemistry. You eat the wrong chemicals and you'll be dead. But there is clearly something new, that's something organized in a higher system, and a whole slew of new causes come into play at higher levels. So to understand the self, I mean, this is called emergence in science; and so, self is an emergent property at one of the higher levels of psychological ones. Again, it's constrained by the lower ones, and you can't do anything to violate the laws of physics and chemistry, but new causes come into play that are not reducible to the others.

- 5.32 Self exists at a high level of organization, and self-processes may be integrated into this emerging unity of the self. The higher levels have these new organizations. It's also notable they become more statistical and probabilistic than deterministic. I mean, Newtonian physics is the classic, the thing that determinists always want to go back, "Well, the cause guarantees the effect. Nothing else can happen." But when you're dealing with human behavior or economic systems or whatever, there's very little determinism, it's changing the odds of things and highly complicated probabilistic causes.
- 5.33 It's worth reflecting, for example, on mountains. I mean, I live up in the mountains these days, myself. But a mountain doesn't really have a self. There's no boundary. What happens on one side of the mountain doesn't affect necessarily what happens on the other side. If society divides it up into separate things, the mountain doesn't care. Really doesn't care about anything; it doesn't have a capacity for caring. There are lots of molecules. Mountains are great big things. But there's no coordinated system of producing behavior in some fashion.
- 5.34 Another way to talk about this problem, most treatments of the self at some point talk about the classic parable of the Ship of Theseus, it's called. So ancient Greece or whatever, the ship went out on a voyage and, coming to the first port, it bumped into some pieces of some rocks and things like that and damaged some timbers. So when it pulled in the harbor, pulled out those timbers and replaced them with new ones. Then it went on its way to the next one and had to do more repairs and replace more.
- 5.35 After a while, all the boards of the original ship were replaced. It was an entirely new set of wood. Not one piece of the wood that had started the voyage was still there. So was it the same ship? Well, physically, it was all different molecules, but the organization was the same. It looked like, it acted like the original one. It has the same name as the original one. It's following the same voyage plan. It has the same owner, even though that was there. So what's consistent was the organization. I mean, it was built the same. It looks like the same ship. It's doing the same job. It just had replaced.
- 5.36 And then the same thing happens with people. I mean, we're born as a tiny little baby. By the time you're a 60- or 70-year-old man or woman, it's not clear that any of those original molecules are still in your body. Certainly, there are a lot of ones that weren't there originally, and a lot of the ones, or most, if not all of the ones that were there, are gone. And yet, society very much says you're the same person.
- 5.37 When they ask you where you were born, you mention where that little baby came into the world, and you have the same name and tax identification number and so on. So the sameness comes from how the culture understands you and from being a continuous organizing system, not from being the same molecules. So again, the self is not the physical thing; it's the organizing system that is imposed on them, guiding how the molecules work together.

- 5.38 And the molecules may come and go, but the system endures. The self, again, is not the body. The death puts an end to the self, even though the body continues to exist. It's not working as well as when you were alive. But the thing is, the body is still there. Again, continuity comes from social environment, from culture, from some psychological processes, not from the continuity of the physical thing.
- 5.39 More essentially, one key to understanding the self is that the brain learns to operate a role in the social system or several roles. And that's the key to understanding it, as I've said before. The basis of my thinking is a human mind created by nature for culture. To take part in a cultural system, to be a member of it, you have to perform some role in it. As a small child, the role is pretty simple. You're just somebody's kid. As you get older, you take on more responsibility and more obligations, and so on.
- 5.40 That is how humans survive. We work together in these complicated social systems. The brain has to learn to do that, and that really is what imposes continuity of self much more than any inner requirement, anything like that. In particular, society thrives by continuous self—ownership has to imply continuity over time and interpersonal roles, things like that, moral reputation.
- 5.41 The self has to regulate itself as well, and as I talked about, to adjust to the rules and expectations of society. So the brain learns how to behave in ways that cooperate with others in the system, to make the system work and enable people to live and do well. I mentioned in my free will lecture, responsibility and autonomy are the two things that society wants the individual to have in terms of free will. That is, being able to govern yourself and decide what to do and take care of yourself and accept the consequences of your actions.
- 5.42 So, I think unity is not automatically given. It's something that's gradually created. And this may go back earlier in evolution, indeed, at least the starting point. Every living thing delineates a boundary between itself and its environment. You can dig up a tree, wash off all the dirt, and move exactly the tree and nothing else. And so, that's one essential piece of life, is that boundary.
- 5.43 Of course, it's a permeable boundary. You eat and drink and whatnot and eliminate waste. So stuff crosses it, but it is a definite boundary, and stuff is very clearly inside of you or not inside of you. Even very simple animals act as a unity. Plants have something of a self, too, but they have no central nervous system that coordinates the actions. It's done more locally. But animals generally with a brain, the brain is the center, of course, of the central nervous system, and that controls the arms and legs. Actually, digestion brain plays a big part in that, too.
- 5.44 So you need an organizing system to get all the molecules to work together as a unity. You have more self with animals than plants, and the brain understands this organizing system to some degree. So, the first thing is just in the immediate present, though; I call it the right-now self. Other people have other terms for it, but that's a simple way

of understanding it. This, a lot of animals get there to have a kind of right-now coordination and unity.

- 5.45** So the brain assesses the situation, gets an impulse or motivation to do something, this or that, and then tells the arms and legs or paws or whatever to execute that action. Learning to crawl is an interesting step in the emergence of self. We're watching a baby learn to crawl, and it's actually complicated because you do right hand and left knee, plant those, shift your weight onto those, and then left hand and right knee, and shift your weight onto those.
- 5.46** Watching the baby doing it, she had this look like, "Wow, this is going to change everything. Now I can go where I want." If you're parents, you discover you lose control over the situation as soon as the baby masters crawling. But anyway, it's a rather complicated system. You have to understand front and back or hands and knees, left and right, and coordinate them diagonally and alternate doing that.
- 5.47** It's not that complicated a system. A lot of fairly stupid animals obviously can walk as well. But that's a big step, I think, in the emergence of self. It's the brain imposing a system to create unity, so that the organism can move itself around, which is the essence of being an animal as opposed to a plant. Unifies the body to enable movement. The brain takes information from the senses and then dictates which direction to go. But the organism is responding as a whole unit. Again, walking, moving, you're moving all together.
- 5.48** Then we go from the right-now self to the extended time version of it. This is also what people call the narrative self; across time, whatever you're doing is in a story aspect. This appears to be mostly unique to humans, so it integrates past, present, and future. Again, time is a much bigger aspect of human life than for other animals. Present behavior, what you do now, is guided by the past, promises you made or obligations, constraints, and future goals and things you're striving for.
- 5.49** Self-regulation often means not indulging in some present pleasure, in order that you'll be better off in the future. Saving money, for example, or quitting smoking, all those things might not be appealing in the moment, but it's highly adaptive to integrate that into there. Morality is very good since humans depend on cooperation. You need other people to cooperate with you. That means you need to be trustworthy. You need to have a good moral reputation. As we emerge in modern life, things are more complicated; there are more stranger interactions.
- 5.50** That's one argument of laws and morals tend to enforce the same kinds of behaviors. And there's, over time, a gradual shift to more emphasis on laws, because moral reputation depends on other people knowing you. And so in a small village, everybody knew everybody; and if you lied or cheated or something, people would remember that 20 years later, and it would damage your chances. But in a big city, you could cheat somebody and not get away with it. So that's why they need laws to regulate things.

- 5.51** Stable roles in the system, maybe you're having a job and so on. Again, the extended time self cultivates the credentials for this job, and then takes it and takes a commitment. People work at the same job for many years, for a while. My father's generation, they spent their whole career at the same company. That's much less common now. But still, it's over a substantial period of time that you do it. And so, understanding of self as a character in these ongoing stories, that helps extend the time as well.
- 5.52** So, in terms of how did this happen in evolution, I think a progressive increase in the organizing system, getting more and more complicated as we move through evolution, we can see that's what led, ultimately, to produce the marvelous human self. So, the first step, as I said, was just that each living thing marks a boundary between itself and the world. And you can say already there's a self. It doesn't know it, necessarily. It doesn't have any meaningful understanding of it, but there's definitely a boundary.
- 5.53** And even a plant, you water the roots, and the leaves grow up top. Differentiating self, so to speak, from the world. Central nervous system is the next step. It's a brain to impose systems to enable the animal to operate together. Agency, making choices, making decisions, exerting self-control, exerting control over the environment—that becomes more important, too.
- 5.54** Then as we get social, things gain a great deal of complexity. In social life, there's ownership, which is actually a social thing. It just means, it's mine and not yours. If there were nobody else, you wouldn't need to have any sense of ownership. Belonging to relationships, you have this role, that role, whether it's in a marriage, a husband and wife, or whatever.
- 5.55** Agency gets integrated into longer time sequences. You get a job and you sign up a retirement account that takes some of your salary and puts it away for you to have many years into the future and so forth. You're aware of yourself, aware of what you're doing, but you also become aware of how you're perceived by others. This is very important because, again, you need to get along with others.
- 5.56** So, humans are much more advanced than any other species we know, and understanding how other people perceive them and doing a lot of things to create or maintain a particular impression on other people. So, self builds on the earlier structures and so on. As I said, the unity of self exists long before anything that a human, the most basic living things have it. Also, differentiating self from the environment.
- 5.57** Agency, we see in animals. They do make choices and make decisions and so well. The need to get some approval from others; well, some animals have that, and the need to be accepted is much more central to human life than in others. And so human self uses its agency to get social approval and basically integrate your animal body into the social system, into society.

- 5.58** You want people to like you. You want people to respect you. Respect is basically two dimensions, that you're competent at doing whatever chores and tasks you're supposed to do and that you're moral. So a well-respected plumber, someone who will actually fix your plumbing when it's not working, and will be ethical and will charge you a fair price and not cheat you or whatever.
- 5.59** In terms of looking for evidence about what happens when the self goes wrong, I found this source. It was a lovely edited book with many different experts talking about many different psychological disorders. And some of the problems, you think, well, maybe there's a piece of the self that's wrong. And well, there are some of those, but that's not that common. The most common would be some kind of low self-esteem, that you think you're bad or you have some negative view of the self.
- 5.60** But the majority of them were about organization. Again, there's something wrong with the organized system, so the different parts of the self aren't working together as well. And some are also about deficits and self-regulation, but some of those have organizational aspects as well. So I think it's some of these, just as examples. The bipolar, the self split into two separate sections, rather than being fully integrated. The one is super confident and thinks it can do anything. The other is depressed and thinks everything is hopeless.
- 5.61** The obsessive compulsive ones that we were talking about earlier, they have the view that there's something wrong underneath, and they have a bad self that they're worried about, and they need to do these things to maintain control. And so, these activities and doing the activities, checking whether you turned off the stove 50 times, doesn't really solve the problem, and they kind of know it, but they just feel bad. If they don't, it's like something else is happening.
- 5.62** Post-traumatic stress disorder. There's usually some organizational problem that predisposes to that. I think I've mentioned that post-traumatic growth is more common than the stress disorder, but the stress disorder is real, too. A lot of it is wanting to go back to the self before the trauma. There's a sense of the two versions of the self, and there's a damaged one now, and you want to get back to the other.
- 5.63** Schizophrenia, people think that's multiple personalities, but it's not, but it is a deficit in the organization. The sense of me-ness isn't there the way it normally is. And so, we feel we own our thoughts and own our feelings. "They belong to me." And the schizophrenic is less likely to do that, which makes them more susceptible to think other people are controlling their thoughts or planning thoughts. Anyway, more of the same show that aspect.
- 5.64** So, let's turn now to why. Why does the self emerge? Why does the organism become organized with a self? Why does it find it useful? Well, it is worth reflecting a little bit on what the self does. What are the benefits we get? What are its main jobs? I can list

some key ones. First and most fundamentally, everything at the biological level, the evolved purpose is to survive and reproduce better.

- 5.65 As I said, we humans do that in an unusual way. We do it by belonging to a cultural group, sharing information, performing roles in the system, cooperating and communicating. So, survival and reproduce, or reproducing, are the most basic things to continue life. For the human self to do it, you have to be part of the group. Very few people can survive very well, totally on their own.
- 5.66 Even these days, lots of people live alone, but of course, they're well integrated into the system, and they get their food from the supermarket, and get their money from the job, and so forth. So you need to be accepted. And one big thing for the self is to secure acceptance, to get yourself hired in a job, to get yourself accepted by a group, integrated into a family, whatever.
- 5.67 And for that, you have to perform your roles within the system, do what's expected to you. So you have a job, you have to perform it well and be reasonably competent, do what is needed to keep the company system, the whatever, the employer, working and functioning as it does. You do your part, every other does their part, and that works. You've got to take care of yourself, feed yourself, and so on, within a complicated system with lots of rules.
- 5.68 And even, say, the hunter-gatherers is a much simpler system. It had strong rules about how the food was going to be divided. Often, the person who killed the giraffe doesn't get to apportion it. Somebody else gets designated to chop it up and give different parts to different people. And anyone who violates the roles is quickly rejected and ostracized and so on. So, yeah, the self is a big rule-follower, needs a good reputation.
- 5.69 There's always been discussion in my whole career about self-esteem, but nature doesn't really care that much what you think of yourself. The benefits of high self-esteem are rather limited, not nothing. But other people thinking well of you, that's extremely important, and everybody needs that, and few people can get by very long if nobody likes them or respects them.
- 5.70 You make choices, not just whatever serves you well, but also what you can justify to others, and yet still manage to take care of yourself and get what you want. Society expects consistency. Over time, you have to do the same things. There's consistency, there's also flexibility. Sometimes change is required. You get promoted, even you get a new job, so you have to take on a different role, but still, there's a consistency.
- 5.71 Even when you're old, you still have your high school diploma and other things that are there, and you might still have ownership of some positions that you had. So, a lot of the functions are indeed highly social. While I was working on this, I came across this book. Actually, there are a couple of things like this. One was a guy who was having a difficult time in life, and he got a chance to live out in the country, in this very isolated

cottage in the western part of the United Kingdom. And so he did it. There were no roads there, there was no electricity, there was nobody else.

- 5.72** So he said, "Great, I'll be able to live alone for a while and really get to finally understand myself and resolve my problems." And he took notes and he lived there several years. And he came out and he started getting ready to write his book about what it was like. He said, "I didn't really learn anything about myself. I stopped talking about myself. By the second or third year, I didn't say anything about myself. My notes were about what birds I saw that day and what food I was able to harvest or track down."
- 5.73** Other people reported similar things. There was a guy who went off the grid in New Hampshire or somewhere in the Northeast there. He lived out in the woods and would get food by breaking into vacation homes and taking stuff from their refrigerators. It was a long time before he was caught. Finally, they talked to him. I remember at one point, one of the journalists asked him, "So, what did you learn? What were the deep insights you learned?" And he said, "Huh. Get enough sleep." That was his big lesson. It's good to have enough sleep.
- 5.74** So, yeah, the self, you don't really need much of a self when you're by yourself. The self, really, most of its functions are within the social system. I mean, think about it; suppose you live totally alone. You wouldn't even need a name, right? What would you need a name for, address, job title; you wouldn't have a job title. Any role identities, you wouldn't be somebody's husband or wife or anything like that.
- 5.75** Ownership, there's no point to ownership. If you had a toothbrush, it would be the toothbrush; it wouldn't be my toothbrush. It only has to be mine when you're living in a home with other people and you don't want other people to use your toothbrush. Self-knowledge, again, most of that, you don't really need to know. I mean, you need to know how to bandage a wound or something like that, and what kind of clothes make you feel good or something like that.
- 5.76** But most self-knowledge is highly social, and so most of that would be unnecessary. It wouldn't have a reputation. It wouldn't be sensitive to what other people think about you and all the stuff we do and think, "Am I making the right impression?" and all that. Moral responsibility, no point in morality if you're by yourself. Morality is for regulating people's interactions with each other, not by yourself.
- 5.77** So, it's not that you'd have no self. Again, you'd have to still put your shoes on your proper feet, and you'd still walk and do those basic animal unity things. But most of the self will become unnecessary. My point is, you need the self within the human social life. That's why human social life, being so much more complicated, is why the human self is more complicated.
- 5.78** I've made the point before, culture is our biological strategy. It's how our species solves the problems of survival and reproduction. And so, culture is a new and better way of

being social, but it places then more demand on the self, as well as bringing more benefits. Advantages of culture, talked about in the first set of lectures. You have language so you can communicate better and also think more effectively, store information and so on.

- 5.79** Progress, that knowledge is stored in the group, not the individual mind, so each generation can build on what it's already learned. Division of labor improves the efficiency of the system. People can specialize in one task and get really good at that. Trade, economic exchange, benefits everyone. Those are some of the big advantages.
- 5.80** So the question then, to what extent is the self a product of the cultural group or system? Would the system benefit by having more differentiated selves and identities, which would mean humans, in our more complicated system, would need much more of a self than others? And the short answer to this is yes, and the way I stumbled on this was a bit of a story.
- 5.81** I have an undergraduate textbook for social psychology that I wrote with a co-author some years ago. It's done quite well. It's like a sixth or seventh edition now. But anyway, we were doing the first one; that's the biggest job, is to do all the research. We planned to have a certain number of chapters and divided them up between ourselves as to who was going to write the first draft of each one.
- 5.82** I didn't really know anything about groups. I'd always—chapters have to have, textbooks have to have a chapter on groups. Most of them do, but I'd never taught that in my class, and I'd never been interested in the groups literature. So I made sure that my co-author was assigned to write that chapter, and he was happy to do it. Except when he got to do it, he said, "Wait, I can't do it. I don't know anything about groups." And so he wrote to me, "Please, would you do the groups chapter?"
- 5.83** He offered to give me a share of the royalties for that. I wasn't going to do that. He'd been a champ. So I said, "OK, all right, I'll do it." Now, you're writing a textbook and you get to the chapter you don't know anything about, what do you do? What a lot of authors, I think, do is just look at other textbooks and kind of do a new version of what they did. But I didn't want to do that.
- 5.84** I thought, OK, why don't I just take some time and actually read the literature on groups and learn, and I'll discover some patterns? So I read a whole bunch of—I didn't read thousands of individual studies, but I read a lot of the big summaries and literature reviews and learned a lot about groups. It was interesting stuff. And so, I looked for a couple of big patterns, and I found two that kind of stood out as coming up over and over in these.
- 5.85** First one is that groups are good. People like being in groups. Groups can do things that lone individuals can't. So they really expand the power of the human brain and arms and legs and all that, by working together is very efficient. And as I said, people like being accepted in groups and being part of them, so that was one. The other is that

groups are bad. Groups do terrible things. They do much more nasty things than individuals.

- 5.86** Even cooperation—one way to think of this is, suppose people of different races, like a white man and then a black man, have some disagreement over an issue. They sit down at a table, they can probably work it out. But suppose it's four white men and four black men with the same problem, sitting down at the same table, it's going to be harder for them to do it. Sure enough, the groups more square off against each other and make it more difficult to get along.
- 5.87** There are nice lab studies showing this effect. Of course, world history, we had our lecture on evil. The groups doing bad things are by far the worst. OK, so I have my two themes, but unfortunately, they contradict each other. Groups are good and groups are bad. So it got me to wondering, what's the difference? And there are lots of findings for both aspects. So I wanted to take a closer look, and not necessarily for the textbook, but for my own understanding.
- 5.88** What is it that makes a group? Some groups are more than the sum of their parts, some are less, and some are better than the individual alone, and some are worse. So, looked at some of the bad things groups do, the pathologies that psychologists have shown negative aspects of being groups. First, there's diffusion of responsibility. Nobody takes responsibility.
- 5.89** A problem comes up as one of those classic findings, that a victim in an emergency is better off to have one person see them than to see a crowd of 10 people, because the crowd of 10, everybody thinks, "Well, it's not my job. Somebody else will do something." So there's diffusion of responsibility. People get lost in the group and start going along with it. Mob violence sort of things. People get carried away there. Groupthink is that people just go along with what others in the group are saying or judging, or their opinions, and stop thinking for themselves. And groups can get carried away into really stupid or wrong things.
- 5.90** There was research on brainstorming for a while. This is something the advertising agencies had originally come up with, that you put a couple creative people in a room and they can bounce ideas off the other and supposedly inspire greater creativity. But the research was not kind to brainstorming. Several people working alone, individually, will produce more creative ideas than those same people in the same room, because they interfere with each other and distract each other and so forth.
- 5.91** Conformity. There are those early studies that people can see something is the right answer, but everybody else gives another answer. They to go along and give the wrong answer, just to conform to the group. There are the obedience studies, where somebody tells you you have to keep delivering shocks to this person, even though they're banging on the wall saying, "I quit the experiment. I don't want to do this anymore."

- 5.92** Social loafing, when people are working in a team, they often reduce effort. They don't work as hard as they do. There are even nicely controlled studies about swimmers, in a relay, they don't run as fast as they do when they're swimming in an individual event, when they'll be timed, and someone will know exactly how fast they swim. If they do keep the timing of each individual, the split time, it's called, then they do swim as fast as others, but otherwise, they don't.
- 5.93** And committee decision-making, a whole bunch of interesting findings there. What these all have in common is submerging the individual in the group. So you're not being a separate autonomous sense, and the group self produces the negative effects. And say, like with the swimmers, even if you're in a relay, if your own time is going to be posted so everybody will know, then you somehow swim faster than if—you're still swimming in the relay, you still want to win, but if they only have the group performance, not the individual, that takes that away.
- 5.94** So, identifying individuals counteracts a lot of these negative things about the group. And then I turn to, what are the things where groups actually do best, that make groups really more successful than individuals? Division of labor is one where people split the tasks into different people perform different roles.
- 5.95** The assembly line to make cars produced cars much faster and cheaper and better, even though the members of the assembly line didn't have to know nearly as much as, previously, one mechanic or two guys would make the entire car, in which case they had to know everything about a car. But the assembly line worked better. It's one of the fundamental things in economics. I think I've mentioned it before.
- 5.96** Accountability. If you're responsible, you'll have to explain what you did to others or justify with others. Then behavior comes better, thinking becomes better, people perform better, and so on. So yes, being in a group where you're accountable to the group does improve performance, and it's better than alone. There's the wisdom of crowds effect, that if people make judgments about something, the average over a lot of people is remarkably accurate.
- 5.97** A famous story, there's a book called The Wisdom of Crowds. Starts with a story by Francis Galton, one of the great scientists of the 1800s. He went to a county fair where they had a big steer, and people were supposed to guess how much would it weigh; and not even just the steer as it was, it was after it was killed, and the blood drained off and whatever, throwing away some of the parts. So how much meat would be there? So I mean, who knows? No one has any idea.
- 5.98** So he got all the guesses. People would pay a small price and put in a guess, and whoever was closest would win a prize. And he wanted to show how stupid people were. So, he got all their guesses and went through, and he was shocked that the average of all those guesses was almost exactly what the right answer was. It was within 2 pounds, when it was over 1,000 pounds. Extremely accurate. So a lot of

people, they were just on a whim, guessing, whatever. Probably a few people were butchers who had a better idea. But still, they were able to get close.

- 5.99 Other findings have found the same thing, that even in sports betting, the average of all the bets often is more accurate than what the best experts can predict. Experts have supposedly studied everything, the injury reports and who's on the bench and all that stuff. The crowd wisdom is more accurate. Now, it's required for this that everyone make an independent judgment. So the wisdom of crowds is a very delicate effect.
- 5.100 It doesn't show up, especially if people know what other people are saying or doing. But it can work, but again, it requires each self to function as an autonomous individual, independent of the others. I mean, if there's a leader saying, "I want you to say this, or this is what I think," that can bias the whole thing and throw everything off. But as long as people are operating as individual selves, that can work.
- 5.101 I mentioned the social loafing, but sometimes people do better when there are other people there, when they're competing with others, or others are even watching them; they perform better if they're identified. How do we put these all together? Well, it's when people are functioning as a separate self that the group does better.
- 5.102 Instead of being submerged in the group, which produced the bad effects of groups, being individually identified, that's what enables them to do better. Being, again, being accountable, if you have to explain your things; being responsible, other people will know what you did; performing a unique role and so on. Again, with the wisdom of crowds, you have to form your own opinion independently of all the others, and then somehow that works.
- 5.103 So, are groups good or bad? Well, it depends a lot on the differentiation of selves. Groups are sometimes good and sometimes bad. But the individually identified, autonomous, separate-acting self is a vital ingredient to producing the good, whereas being submerged in the group and kind of forgetting and going along with the mob, that's what produces the bad results.
- 5.104 It's not just anonymity because sometimes you have to be responsible. Anonymity sometimes helps. Anonymity does free you to act more on your own, which is why anonymous voting is better, and you want to keep voting secret and confidential. So there it benefits, but other times being responsible and being accountable so other people will know what you did, that contributes as well. So the key is operating as an individual.
- 5.105 Implications for self-theory, then. The cultural group benefits from differentiated selves, differentiated on a variety of dimensions. You're morally responsible, so what you did, right or wrong, you accept consequences for that, and other people will praise you or blame you according to that.

- 5.106** Contributing in group tasks, especially where you perform a separate role in there, and you do your part, and they do their part, and then the system works. Thinking independently, thinking for yourself, rather than going along with what others suggest or what other people seem to be thinking, and integrating yourself with your contribution into the group.
- 5.107** To conclude then, why does an animal body need a self? Well, first, the requirements of the kind of advanced social systems that humans live in and that we thrive in, that have produced this burgeoning of the human population. Ultimately, on the biological level, you survive and reproduce better by operating a self in one of those systems than being out on your own. And so, the self really exists at the interface between the physical animal body and the social-cultural system.
- 5.108** To review, before closing here, so it's not something that's inside the brain. It's more something that the brain does than some part of the brain is. Integrate yourself into, integrate separate processes. The self is about the creation of unity. It's required to take part in advanced social systems, which is how our species thrives. The brain, as I said, essentially learns to operate an identity, or a couple of them, within the social system, and that's what enables it to work effectively.
- 5.109** So, self is definitely real. It's not a thing; it's a process or it's a performance, or a process of performance, perhaps. Emerges only at a high level of organization. It's not a lump of stuff or a physical thing. And it's progressive integration of multiple systems. The creation of unity is another key factor of it.
- 5.110** Exists at the interface between the animal body and the social group, so it's not inside you so much as in between you. Again, it's a requirement of social systems. The brain requires the skill to perform the roles in the systems. So that's what works for us. All right. I haven't totally answered what is the self, but I think we're closer than we were before this. Thank you.
- 5.111** Do you conceptualize hobbies as part of the self? And if you do, where do hobbies or passions arise from? Like, psychological, maybe not needs, but maybe? Maybe there are psychological wants, things that people find that, if they don't do it, they wither on the vine. And do you think hobbies come from nature or nurture?
- 5.112** I don't know a lot about that, so I don't have a strong answer. I can just, I can just speculate. Earlier, you mentioned twins. It would be an interesting project to see if identical twins have more of the same hobbies. I would guess they would, not entirely. But they have the same feelings and the same talents. And so, a hobby is clearly something that's learned and acquired. It's not built-in.
- 5.113** But your ability to enjoy it, and your tendency to enjoy it and want to do it, and getting begets wanting, would depend on your innate temperament. I remember reading that liking for jazz is highly heritable. The way they look at heritability is its strong genetic component, and yet jazz can't possibly have genes. It's only existed for a century and a

half. There's no such music if you go back two centuries; that's way too short for there to be any kind of genetic encoding of jazz itself.

- 5.114** But the kinds of things that would make you react favorably to jazz could be innate. Obviously, you have to have good hearing. If you're tone-deaf, music is not going to do much for you. For jazz, you like a lot of complexity and a lot of changes and the interplay between the chord progression repeating itself over and over throughout the tune. Even though it might be played differently, it's still the same set of chords, whereas the melody is improvised during the various solos.
- 5.115** All those things, some people might just like nice little simple tunes, and they're not going to go for jazz. As I said, people who can't have good hearing wouldn't be attracted to any sort of music. And you'd probably make the same arguments about bicycling or swimming or building model airplanes. Plus, you have a twin who's there, and you might see that they do it; that could reinforce you, and it'd be fun to do something with your twin.
- 5.116** So, it's not, I mean, hobbies change over historical periods, and so they're not obviously deeply rooted. The hobby itself isn't something rooted in genes. But the predilection, the tendency to like this sort of thing and not like that sort of thing, that could have some genetic basis, and then it has to be there in the environment as an opportunity for something to do.
- 5.117** I mean, our hunter-gatherer ancestors, none of them had a coin collection because there weren't coins. So, coins and stamp collections were popular for a while, but nobody uses stamps anymore. So I imagine the stamp collectors of the world are dying out. So, it's some combination of what's available in the environment.
- 5.118** But what you're going to find fun, what's going to be appealing to you, that could have some inborn basis. As well as, sometimes you can just learn things and take things up. You see other people liking it, and you do it because you want to be part of them or be with them. Or it looks like fun, and so you give it a try.
- 5.119** So, with your understanding of what we know about the dimensions of self right now, what are you most looking forward to and where science is going towards revealing more about what exactly the self is?
- 5.120** I mean, the obvious answer many people would bring up would, if you want to know some of the basic processes, like what goes on in the brain. As I said, I've been skeptical of brain research in this because they thought there'd have to be a particular place in the brain that they could say it. But once they understand more how networks operate and networks of brain cells firing together, and really understand that the self is something the brain does, then I think they maybe become quite excited about showing the basis for that.

- 5.121 Now, whether that, how much that will add to our understanding of the brain, I don't know. So far, the yield of brain studies for other psychological processes has been disappointing, frankly. It's not nothing, but we'd hope to learn a lot more. But it doesn't change how we understand attitudes or relationships or anything like that. So, of course, it's an important part of the story, but maybe not the most interesting part.
- 5.122 Thirty years ago, when I was getting going on this, the big deficit was the agent. How does the brain make choices? How does it regulate itself? And it was common for people to say, "We just don't understand that part of it." But there have been big advances in that since then, so that's come around nicely. I think change in self is an interesting project that we need to understand. Of course, the first thing has to be the continuity, that you are the same person with the same name and all that, in the same role.
- 5.123 But when you do make changes, how does that happen and what are the consequences? Some people have lives that are continuous and all seem like a coherent story. Others just have a phase of this and then do something completely different. We don't know enough about what the effects of those things are. My father's generation, he worked for the same company his entire career till he retired. That's much rarer these days.
- 5.124 So that's something we're going to need to understand more about the self, as the business world continues to be more unstable, and companies come and go, and people leave jobs and go to new ones. What are these adjustments to adapting to a new set of circumstances and roles? Is the self, is this placing new demands on the self that weren't necessarily there? My father had promotions and things, and a lot of people do that.
- 5.125 In some countries, I understand in Japan, it's still common to stay with the same company for your full career, even though you might change jobs a couple of times. But with much more requirement to adapt to new circumstances, what will that do to the self? I think that's going to be a really interesting thing for research to zero in on.

Lecture 6: Men and Women

At a glance — Men and women.

- 6.1 All right. Well, this topic will be trying to come up with a more positive account in theory of male-female relations. We see all signs that young men and women aren't getting along as well as in the past. There are fewer dating, less sex, fewer marriages or delayed marriages, having children, things like that. Indeed, the public discourse about gender is filled with negativity. I mean, look at some of these terms. If you read reports on men and women, you see these kinds of things repeated.
- 6.2 They're all very negative. You could easily get the impression that men and women are fundamentally enemies. But are they, really? I mean, there's so much good that goes on. First of all, lots of happy marriages. The United States government's General Social Survey asks people, "Are you very happy? Kind of happy? Not too happy?" Well, you can be very happy if you're single, but the odds are double of being very happy if you are married. This is true for both men and women. It's a big boost there. So, something good is happening right there.
- 6.3 Men and women work together really well. Great coworkers. I can't imagine what my career would have been like without all the fine women scientists that I had the good fortune to collaborate with on projects and learned plenty from. If we stick to merit, and that's a big if, if we stick to merit, the talent pool is improved if we have both men and women available to do all sorts of jobs. And on a fundamental level, men and women generally like each other. Lots of positivity and often ignored, but it's good, too.
- 6.4 At one point, my wife and I were tasked with organizing a series of parties. It was not in our natural skill set. Of course, we had to study what makes a good party. Well, having about equal numbers of men and women is one thing that people do appreciate; and you get farther off from that, things don't always go as well. And the last one is the most important. Ultimately, our species is doing very well. Remember, biology measures success by, is the population increasing or decreasing? Going extinct is failure.
- 6.5 But ours has this tremendous success. In 1800, there were a billion humans. Now, two and a quarter centuries, there are 8 billion. No other mammals can match that. Last I checked, babies are made by males and females, doing some things together and raising them. Any account of male-female relations should start with the success of our species, with the biological flourishing. Human males cooperate, females, better than the other mammals in terms of keeping the population going, getting large, and staying large.
- 6.6 We're certainly in no danger of going extinct, unless we blow ourselves up with nuclear war, but that's an unusual problem. Of course, yeah, there are conflicts, and social life, there's always going to be some negativity, but we should start with the focus on what

we are doing right, what goes well. I'm going to propose an alternative to the standard theories. Alternative theories are scientifically good. We should have them. Even if the alternative is completely wrong, you can test them, and that will provide more evidence for what the truth is.

- 6.7 If you don't have alternative theories, any mistakes in the one theory you have will never be noticed and challenged and corrected. The issue here, I'm not going to try to persuade people that this is true. All I'm going to ask is, let's create another theory that perhaps should be put in play and be allowed to be considered. And there are people who say, "No, we shouldn't be allowed even to think any other thoughts." But suppression of dissenting views is basically anti-scientific.
- 6.8 It's usually, in history, it's a result of some powerful elite that's protecting a bunch of, probably lies, a false narrative that it benefits itself. So whenever you start to hear, "Oh, you're not allowed to say that," that should be a red flag. So the dominant view that we hear a lot is that men are basically oppressors and women are victims. This doesn't seem very promising for explaining what we're doing right, how our species is flourishing, anything like that.
- 6.9 And moreover, as a number of people pointed out to me, this story is unhealthy for both men and women. Neither oppressor or victim is a very positive role. So this may be useful for activists to use the victim-oppressor thing to advance their own careers and extract goodies. But it may be doing a considerable harm to both our young men and women in contributing to the problems that we have.
- 6.10 The Pew Foundation had this survey a couple of years ago that they published and asked, "Have you ever been diagnosed with a mental illness? Has a professional ever told you you have something?" Now, the chance that something has ever happened can only go up over time. It cannot possibly go down. And yet, the cross-sectional data show the opposite. It goes down. Old people say, "No, that's never happened to me." But a lot of young people are doing it, so there is indeed plenty of evidence of a mental health problem, some people call it a crisis, among our young people.
- 6.11 And where is it worst? The highest number is the bar on the far left. The blue are the women. So these are young women, adult women under the age of 30, with strong left political views. Now, why is that? What is doing it? Jonathan Haidt had this book, *The Anxious Generation*, and he says, "Well, it's because of phones. People are spending too much time on their smartphones and stuff." And he makes a good case, and things like this are, no doubt, multiply determined.
- 6.12 But the conservative young women have phones, too, and their mental health is just about average for the population, so I don't think it explains the extreme. Over half of young women under the age of 30 in this national survey are reporting that they have been told they have some mental illness, usually anxiety or depression. I haven't seen anyone put forward a theory as to why them in particular, so let me suggest one.

- 6.13** The far left is where the "woman is victim" ideology is most strongly asserted. The thing is, there's no such thing as a happy victim. Those are not compatible. Victim is an inherently unhappy role. So the more you embrace the idea that "I'm a victim and all my friends are victims," the more you preclude the chances for being happy.
- 6.14** I mean, it's hard to imagine a young woman saying, "Oh, society oppresses me and all my friends and blocks our advancement and holds us down and takes advantage of us and all these things. But my life is great, and I'm really happy, and it's wonderful being me." Those don't really seem to go together. At least clinical observations for this, I have a friend who's a clinical therapist in France, and she's, by all counts, very good at it, treats a lot of people with different problems and very successful. Her patients love her and so on.
- 6.15** But she told me recently that she had failed with several people. She had several cases where she just had to terminate therapy and send them away and say, "I can't help you." She said they were all young women with strong feminist views. They were really unhappy, but really close-minded. She does therapy by getting people to look at their lives in a different way and think about how you could take control and manage things and maybe make things better. She said no, she said they would not let go of the victim role. She finally had to say, "Sorry, just stop therapy, and I can't help you."
- 6.16** That's the women. I don't think it's good for her. The young men, too. There was an article in the Free Press recently that put it this way. "Think what it must be like to be a white boy growing up in a culture that tells you you are what's wrong with the world. You are not only demonized by cultural elites and institutions, not because of anything you believe or have done, but because of who you are. Not only that, you'll be denied opportunities for educational and professional advancement. The power-holders who treat you that way will consider themselves virtuous for doing this to you. You see women and non-whites openly embracing their identity groups, while you're told this is off-limits to you."
- 6.17** I've heard other people reflect on things in this way, and this has become extremely common in the schools, as far as I can tell. This is the victim-perpetrator account of male-female relations that's being widely taught, so it's probably not having a good effect on young men, too. A young man wants to grow up to be a hero, and it tells him, "Well, you're just going to grow up to be an oppressor, and we'll be watching you. If you say anything wrong, we'll pounce and shame you. You really probably should just be apologizing for being a man your whole life."
- 6.18** Well, that's a loser deal. No wonder a lot of the young men are withdrawing into alternative worlds of video games and things like that, where they can be a sort of a hero. So, can we make a positive theory? Look for an alternative way of thinking? Again, I'm not trying to have a political agenda here or change things as well. I'm just trying to figure things out, but let's put in an alternate theory that we should at least be allowed to consider it as a possibility.

- 6.19 So this, for me, is an intellectual exercise. As I said, my project is to try to figure out what is the essence of being human, what sets us apart, what are the essential human traits. I try to figure out, I want to end up knowing the truth. I keep revising my opinions and have to accept things that I'm not necessarily fond of. Wading into the gender literature is a difficult one, but I can't really do without it. Gender is part of everybody's life every day, starting with, you get up in the morning and put on your clothes. So, what clothes do you put on?
- 6.20 It's difficult because at least there's a suspicion that there's bias that might distort the research field. Even I can't totally believe all the research findings. That raises the question, is there indeed widespread bias? In the United States, there are 340 million people, something like that. Of course, you can find somebody with any sort of prejudice or bias. I know a guy who doesn't like Canadians. Sure, lots of different prejudices exist, but is there a widespread one that's widely shared through society that really could potentially warp how our understanding of the literature is?
- 6.21 Well, one of the definitive studies was done, published two years ago. It was really impressive, measures of implicit prejudice that people had, comparing four of the big ones. They had two national samples and two student samples, but still four different studies looking to compare black-white racial prejudice, male-female, sexist prejudice, old-young, and rich and poor. Only one of them really turned out to be common.
- 6.22 Again, there are traces or hints, and you can probably find somebody with all these prejudices. If you haven't read the study, try to make your guess in your own mind which one was the real prejudice that's shared in society. If it makes you feel any better, one of my friends saw the preprint for this and called me up and was telling me about it and asked me to guess, and I guessed wrong. I thought it'd be ageism because, come on, everybody knows it's better to be young than to be old, right? So I thought that would be the most robust.
- 6.23 But no, it turns out, lots of old people think lots of young people are messed up, and lots of young people just never think about old people at all. They're irrelevant. So they don't have much of a stereotype there. Anyway, the answer was male-female. So if you picked sexism as the one real prejudice, give yourself credit for it. Specifically, anti-male, pro-female. The prevailing bias in our society is against men and in favor of women.
- 6.24 If that surprises you, you should probably look around more. I didn't pick that one because I didn't think they would be allowed to publish it, not that I didn't think it exists. But that was very consistently found. Sure enough, even in our own data, for example, both men and women think women are more intelligent than men on average. We have mountains of data showing that's not true, but it's widely believed, as I said, by both men and women. It is what you see on the media, in the movies, and people get a disturbing amount of their information about the world from Hollywood.

- 6.25 Anyway, the reason this is relevant is that, if someone were to come up with a perfect, totally accurate theory—and I'm sorry, please have no illusions you're going to hear that theory today—but if someone did come up with a perfect theory that had no bias whatsoever, we would think it was biased, because perfect theory is going to notice the strengths and weaknesses of both men and women. We're used to hearing criticism of men, but criticism of women is not allowed and is suppressed.
- 6.26 And so, any true theory that actually recognized women are ordinary people too, with their faults and negative sides, that will set off people's alarms, and often people will try to suppress it. So, I would suggest men and women, are not fundamentally enemies or opponents. More importantly, they're partners. They work together to produce this flowering of the species. This idea of constantly struggling against each other, I mean, yeah, as I said, there's some conflict in any long-term relationship.
- 6.27 But most often, and more importantly, men and women cooperate. That's what enabled our species to survive. The great Helen Fisher, a pioneering anthropologist studying gender, I can't improve on her analogy. She said, "Men and women are like left foot, right foot." I like that because left foot and right foot are similar, clearly. They're also different, and the differences complement each other. So you need both to get anywhere, to get ahead; that one is half of two, but one foot is not half as good as two feet. Really, having both really enables you to walk much better.
- 6.28 It's just one example among many. In the traditional farm, there was men's work and women's work. There were no singleton farmers. You had to have both partners in order to get all the jobs done. There's always a lot of work on the farm. Indeed, I saw the Economist had an article on lonely hearts advertisements from the 1800s. It says, "Oh, we'd like to place an advertisement, saying he wanted a new wife because his last one was dead, and he wanted someone to look after the pigs while I'm out at work."
- 6.29 Now, I haven't done these modern online dating things, but I'm guessing this is not the sort of profile that you would see to attract people. But he apparently thought it would work, and it might just have done it, because people died at all points in life, and there might be a woman with a farm, and her husband died, and so she really needed a man. And she'd say, "OK, well, I can look after the pigs while he goes out in the fields. That could work."
- 6.30 Also crucial, and this is another thing Helen Fisher told me—she died, unfortunately, about a year ago—but this is another point she made. I was able to talk to her there, and she said, "We need to place more emphasis on female agency. This idea that women are always passive, that things are done to them, they don't have any control, this is just absurd." Women are taking responsibility, or shaping their roles as well, and actively taking part in making society the way they want it to be. So they want support for themselves and for their children and protection and so on.

- 6.31** One of the most important papers on psychology of women in the last couple of years was by Joyce Benenson and her colleagues at Harvard in the behavioral and brain sciences, making strong case that the desire for safety is a really central aspect of female motivation, different from males. It's one reason women live longer. It's very adaptive. And particularly for the children, losing their mother is probably worse than losing their father, so biologically, selecting in favor of women who are seeking safety.
- 6.32** If you look at people who take chances and get themselves killed, there are a lot more men than women that do stuff like that. I mean, it's not absolute, but in general. And so, I think understanding the importance of safety, both physically and socially, I think sheds a lot of light on a number of the ways that men and women relate to each other. And they have things they can offer the man in exchange for getting provided for and protected from dangers. Again, we're all apes, carving out niches for ourselves in this new and complex social environment that humans have created.
- 6.33** All right, I said this is going to be an alternative theory. So, let me say what I'm not going to talk about, some of the things that you hear about a lot in discussing gender relations. Honestly, is there any scientific evidence that any of these things really exist? I think if we held them up to the usual standards of scientific rigor, evidence is extremely thin, vacuous to negligible.
- 6.34** Now, there's this new term, motte-and-bailey, which I don't fully understand, but I know the phenomenon; I've seen it for a long time. Activists will have two different definitions for something, one that's very inflammatory and calls for change that they can campaign on, and the other is sort of trivially obvious. So when they're challenged, they can replete, they can retreat to that. But they're not seeking the truth. It's scientifically that's bad, but they're trying to campaign. So it's a useful way to do it.
- 6.35** So, patriarchy—the idea that men band together and work together to oppress women. This is basically a conspiracy theory, and conspiracy theories are generally wrong. So, this idea that men are working to subjugate women and dominate them and oppress them and exploit them. Evidence for this happening, as I said, is extremely rare. As psychologists, we hear this "lived experience" thing, and we're rightly suspicious of that. But still, sometimes a reality check is useful.
- 6.36** I have to say, I'm an old man. I've lived a long time. I can't recall a single conversation with other men about how can we oppress women or hold them down or block their advancement. Lots of the other kind, of how can we help women, junior colleagues, students, or even women in general, advance their cause. But if the patriarchy exists, they never reached out to me. There should be lots of evidence of men doing this. If any of you are members of the patriarchy, please do let me know and tell me what you guys are up to.
- 6.37** Misogyny is another one. Again, with patriarchy, we have the two definitions that men are working together to oppress women and block them. If they're challenged, they can

say, "Well, look, men tend to be in charge of everything." Well, there's another explanation for why men tend to be in charge of things, which I'll come around to, which is men created the social systems. But the idea that there's this anti-female conspiracy, again, if you hold it up to the usual standard of scientific rigor, I'm not sure how much you'd find.

- 6.38** Misogyny is, literally, hatred of women. Again, I've never known a man who hated women in general. I've known men who hated specific women, often for understandable reasons; but hating women in general, I'm not sure. Actually, there are a couple articles recently that looked hard for evidence of misogyny, even a more watered-down version of it, of thinking women are generally inferior and so on. They just couldn't find anything. As we saw, the prevailing prejudice, if there's any gender prejudice, it's against men, not against women.
- 6.39** Exploitation is part of, supposedly, what the patriarchy is doing, exploiting women. It's a loaded term that can have multiple meanings. But we can use the scientific definition of exploitation, which is to extract resources. You use that in biology and in economics. It's the same thing. But are men really extracting resources from women? I've seen a couple big analyses by economists for a couple different countries, looking at relations between the individual and the government, and money flows back and forth.
- 6.40** Like, I'm old. I get a social security check every month. I've also paid a lot of taxes. I still pay income taxes, send them back some of the same social security. So it flows both ways over the course of the life. Anyway, they added up across the entire country, and the bottom line was, the average man pays a couple hundred thousand more to the government than he gets back, whereas the average woman gets back a couple hundred thousand more than she puts in.
- 6.41** Some of that is, no doubt, the single mothers who don't, unemployed, who don't pay any income tax and subsist on government support. So professional women may certainly pay a fair amount. Also, women live longer at the end, so the old age pensions, they consume more. But anyway, it's still hard to argue that men are exploiting women when the average transfer of money through the government, at the tune of a couple hundred thousand dollars or euros or whatever, goes the other way.
- 6.42** And that doesn't even count men paying for dates, buying gifts, supporting the family, earning the higher salary and all that. So money, as the biggest resource, seems to flow pretty consistently from males to females. I'm going to argue, that's probably the way it has to be, since it seems to be true everywhere in the world. It's hard to find any contrary ones.
- 6.43** So, coming to the, one of the central issues here, are we looking, what is the version of man? Is the man basically an oppressor or a provider? In other words, the human male, what does the man go up? Does he acquire his sense of who he is as a man, is it

basically to oppress women and hold them down and block their advancement and exploit them and so on, working with other men, perhaps to do that?

- 6.44** Or is he basically a provider, whose deepest impulses are to take care of his female partner and to protect her and bring resources to her and to the children that they produce together? And providing resources and also protecting from danger. As I said, since I've been working on this, I've become sensitized to Benenson's work on the importance of safety. So here's the question, what is the essence of manhood? So again, oppressor, I'm going to try to build a theory around more the provider, protector, achiever.
- 6.45** Indeed, if there's patriarchy in the sense of men working together vis-a-vis women, it's probably for a more positive thing. I would guess, if you really looked at what the men are doing and saying, not the way the feminists are interpreting what the women are feeling, it's that the men are doing things to protect the women and take care of them, what it was to create the resources, create more resources to share with the women, and so on.
- 6.46** What happens in the hunter-gatherers when a tiger comes to the camp, how many cultures do you think the men herd the children off to safety and the women grab their spears to tackle the tiger? I'm guessing there are not very many of those. And again, female agency, women would have accepted and encouraged that if safety is their goal; they would have looked for a system and been accepting of a system in which the men worked together to protect them and provide for them.
- 6.47** As a generalist, I come through many different areas and I encounter a lot of ideas I haven't heard before. And rather than deciding right away whether I agree or disagree, what I do is try them out for a while. So I'll try out an idea for six months, trying to understand the world that way. So I'm inviting you to try this out today, at least for the next half hour or so—try to look at the world with this view and see if it can make sense down the road, at some point, to start and decide if you agree or not.
- 6.48** So, ready to get started with the theory. Again, the basis of my thinking is, we are cultural animals; the distinctively human traits are the result of biological adaptations to enable us to participate in culture, language being the first opposite one. As I read the evolutionary literature, communication and cooperation are the two main spheres where we go way beyond the other great apes. Both males and females have this, but they may use it in somewhat different ways.
- 6.49** Now, we name our species *Homo sapiens*, Latin for clever dudes, and so our intelligence is supposedly there. And let's reflect on the brain a minute. I'm rather convinced that the brain evolved big because of communication, because there was more information suddenly in the environment, so there's more for the brain to do. So, plus size of the big brain, it obviously enables culture better. You can handle more

information, understand more complex systems as well, communicate and cooperate better.

- 6.50** It takes more energy, but you can learn more and process information more thoroughly. And the other, the brains can pay for themselves. There's a simple-minded kind of evolutionary thinking which says, well, if two animals are competing for the same reward, the smarter one is more likely to get it. And so in the long run, in the aggregate, it's more likely to survive and reproduce, which means that all species will evolve gradually to become more and more intelligent.
- 6.51** But that's obviously not true. So why not? Well, the human brain is a very expensive organ. Actually, all brains are rather expensive. And so, if some mutation produced a squirrel, say, with a great big Einstein brain, and even if it had the muscles to lug it around, it would starve to death because it wouldn't be able to produce enough calories, it wouldn't be able to find enough more nuts to feed the demands of this giant thing.
- 6.52** So, human brain, the usual statistic, it's 2% of your body mass; consumes 20 to 25% of the calories you take in. So again, it needs a lot of calories to keep it, but solitary organisms probably can't make use of that. There's no payoff for being that smart, but in complex social systems, more resources can be produced.
- 6.53** And Dunbar compared various different species on brain size as a proportion of body size, that the bigger brains are mainly for the social environment. It's not for dealing with the physical environment, that bigger brains go with more complex and larger social networks. He calls it the social brain theory. All right, so that's what the brain is all about.
- 6.54** Now, turn to the minuses. OK, first of all, it requires more energy. I've already mentioned that. Giving birth to a large head also is going to be hard on the mother. It is already fairly difficult. We talk about premature babies, but in ape terms, all human babies are extremely premature. Because otherwise, what is the big brain going to grow inside the womb? The woman is, it's enough of a hassle to be pregnant for nine months. She's going to be pregnant for nine years, and then squeeze a nine-year-old boy out of her private parts? That would kill her.
- 6.55** Plus, she could only have one baby per decade, even if she did somehow survive. The only solution is for the brain to grow after the baby is born. It has to be born with the potential, but then grow it over a period of many years before it's fully functional. Again, all human babies are born way premature. Gorilla baby starts to feed itself around five months old and is pretty much self-sufficient by three years.
- 6.56** But of course, there's no country in the world where a three-year-old human can function as an adult. You're just helpless for a much longer time, requiring protection, requiring provision. The longer dependency needs to be fed and protected. Mothers'

milk is probably not going to work either for 10 years. You can feed a small child, but there aren't going to be enough calories in there. It's going to need protein as well.

- 6.57** It's going to need somebody to do hunting for it, as well as to protect it. The only essential solution that our species came up with is that the human man took on the provider role. I can't emphasize this enough. This is a radical departure from the other great apes. Basically, none of the other great apes take fatherhood seriously. I've talked to a bunch of primate researchers who really know this stuff, and they all have a story about, oh, here or there, there was a father, especially if the mother died, and the father did take care of the child a little bit.
- 6.58** And it's not even entirely clear they care whether it's their own child or just a member of the group. But certainly nothing like providing food and shelter and protection on a daily basis for a decade or more that human men all over the world do. So, this is a huge evolutionary change, converting the male to accept the provider role—one of the keys to human evolution, one of the defining traits.
- 6.59** I rather doubt men came up with this on their own. I suspect it's much more women selectively breeding with men who would accept that role. No doubt, as the early stages of human evolution, there were males who said, "What? Just because we had sex a couple of times, I'm supposed to feed you every day? That's ridiculous." And acted like any self-respecting gorilla would and did this. But those guys took themselves out of the gene pool. We are descended from the ones who said, "Sure, yeah, I'll do it. I love you. I'll take care of you. I'll provide for you and our children."
- 6.60** And it's true, still in modern-day appeal, you can still see the same differences in there. Now, don't overstate the differences. Men and women want a lot of the same things in a partner. They like someone who's kind, who has a good sense of humor, who's intelligent, and so on. That is similar to both genders. But in terms of the differences, the woman's physical attractiveness or sex appeal matters much more. The man's financial status, how much money he has, carries a lot more weight.
- 6.61** Women can find a mate if they don't make very much money at all. But men who don't have a much harder time, and likewise, a medium, it's hard if you're really ugly, but a medium attractive man, if he has a lot of money, he will get plenty of female interest, whereas it will take away somewhat from the female's sex appeal. So the bond is achieved by matching the man's provider capacity with the woman's sexual and reproductive capacity. And sex appeal is highly correlated with what enables you to bear a lot of children, being young, being healthy, and so on.
- 6.62** And remember, the bond has been very successful. Our species' population is increasing and doing quite well. So, for these cultural animals to evolve, what had to happen? It's worth reflecting on what nature did to bring about these radical changes within the great ape family to set us on a different path that's ultimately made us so vastly more successful than the other great apes in biological terms.

- 6.63** So they had to arrange for the male and female to form a lasting bond, in which the man provides protein-rich foods to both the children and the mother of the children, and also provides protection. He has to achieve, in a males' group usually, to rise in status, to do a good job of that. So, multiple changes to the male and the female psyche occurred, compared with the other great apes, to promote this bonding. It's worth talking about a few of them.
- 6.64** Passionate love is one. Other apes don't seem to have anything like this, but it's a universal experience. Humans all over the world fall in love, and there are different kinds of love. But I'm talking about that real feeling that, when the person walks into the room, you automatically orient toward them, and you think about them all the time, and you love being together, and you want to be pressed against each other physically, all that.
- 6.65** In the movies, that lasts forever. In the data, it lasts for six months or a year, two years, if you're lucky. But that's this real altered state of consciousness—obviously, a good state for bonding the male and female together, perhaps almost like a kind of addiction.
- 6.66** Sex. Most of the apes do it rear-entry, called doggy-style, but might as well just call it gorilla-style. From what I'm told, I'm not an expert, but I'm only told this is more effective for actually planting the seed and making pregnancy. So it makes sense why nature would arrange for most animals to do it that way. But then there's a question, why do the humans turn the female around? But obviously, that's really good for bonding the two, for establishing an emotional connection to them. You're having this ecstasy, you're looking into each other's eyes.
- 6.67** Other things go with it. The kissing, the eye contact during sex, female orgasms. Human females have a lot more orgasms, so that she's having bliss, too, bonding with the male. Even the male concerned that the female like it. I mean, for a while, the feminists were complaining that, oh, the men put pressure on women to have an orgasm, and that's some kind of oppression or exploitation.
- 6.68** But come on, it has to be nicer to have sex with someone who cares whether you enjoy it or not. I mean, is your ideal partner a gorilla? A male gorilla doesn't care whether his partner enjoys it or not. So a female gorilla doesn't get that many orgasms. So again, that has to be nicer for forming a lasting bond between these two people.
- 6.69** And fathers loving their children is another human innovation that we don't see, really, signs of among the other great apes. My child was born, I was already in my 40s. My philosophy of life was already fairly worked out. So it was against my principles to love my child more than my wife, because obviously my wife and I had chosen each other out of all the world for meaningful reasons.
- 6.70** The baby just showed up by accident of birth, and that held up for about the first year. But by the middle of the second year, it was clear, I loved this little baby more than

anyone else I ever had. Lots of fathers will tell you the same thing. Now, my point is, again, this is a huge innovation in humankind. We don't see signs of it in our biological relatives. But again, crucial for the man to accept the provider role and take care of the child through this extended dependent period, until it can become a functioning adult.

- 6.71** Now, David Buss, one of the great researchers in evolutionary psychology, he commented, men and women complain about each other a lot, but in a sense they shouldn't, because they have shaped each other biologically through selective breeding. For a man to say, "Oh, women, they have this problem or that problem, I don't like them," in a sense he's complaining about his great-great-great grandfathers and their friends, that if they had mated differently, the women today would have different traits.
- 6.72** So men and women both bred each other selectively, but not necessarily to equal amounts. And indeed, I think the evidence is pretty clear that women selectively bred men much more than men bred women. I mean, one sign of it, and you can always get a lively cocktail party conversation going with this, that if you look at the human population today, what percent of our ancestors were women? What percent were women? Well, the DNA researchers figured that out about 20 years ago. It's 67%, two-thirds of our ancestors are women.
- 6.73** Now, people are surprised by that because they think, well, equal numbers of boys and girls are born, which is true. But to be an ancestor, you had to do more than be born. You had to reproduce in a continuous this line down to the present. And so, far more men have been dead-ends than women. And one reason for this was the polygyny, or one man with multiple wives, the rich men had several and had lots of children, and the poor men often had none.
- 6.74** Or if that's too complicated, an argument is just also very clear that women are much more selective about who they go to bed with than men are. Men tend not to be picky on that. Women have basically bred men more than the reverse. This is a major contribution to the flourishing of our species. Again, giving credit where it's due, this is something that ancestral women really achieved.
- 6.75** It's interesting to reflect on, what would be the traits that they would breed in favor of? What would they choose a man for? In a sense, you can ask, if a woman could design her ideal husband, what would be the traits she would choose for him to have? It'd have to be traits that men actually have. It's like, "I want a husband who can fly." Well, good luck. But among the traits that actually exist, which ones would have been encouraged and which ones would have been preferentially bred, insofar as women had choice?
- 6.76** I think the top things would be the ability to provide and protect, and the desire to provide for them. The generous sense of chivalry, sense of obligation, and also loyalty. She wants, the woman, the woman wants the man to stay with her, rather than

running off with someone else. Same for protection, that he would have to have the ability and the wish to protect her. So, some kind of innate broad desire to take care of women, which I think lots of men do indeed have.

- 6.77** My guess is that men were selectively bred for higher sex drive. I'm not sure. I can't prove it. I don't even know how you would compare sex drive across species. But sex is one of the main things that attracts men to women. And so, the men who didn't care as much about sex might not be willing to put up with all the obligations and duties that come with that. And some women are wonderful. I think my wife is close to being a saint, but I've been around enough.
- 6.78** Not all women—some are very difficult, but nature wants them to reproduce, too, right? And so, what will keep a man going for it? Well, maybe she's difficult and high-maintenance, but if the sex is good, and you hear men say this, "A lot of trouble, but it's worth it." Anyway, I'm just guessing; this part is speculation. Loving his children would be another positive thing that they would like.
- 6.79** She'd also want to be able to control him and influence him to some degree. He has to be somewhat pliable. Things like, men who don't want to argue with women, who will take what the woman says and go along with it. Men are better than women at reconciling after conflict. There are good data on that as well, because that will help sustain the relationship. There may be a bunch of things like that. Anyway, it's not a complete list, but I think some of that probably was what happened.
- 6.80** Now, one thing that's pretty clear is that, if the man is going to be a provider, every man's workload increases. This is a crucial part of the theory. If the adult male gorilla spends three hours a day foraging to feed himself, now he's got to forage for a wife and a couple of bambinos. Bambini? Bambini. Well, then three hours, maybe it's going to be six hours. That's a lot more work, and that's a hard sell to get the men to put up with.
- 6.81** So, notice, though, most men in the group will have the same problem. They have a wife and children. So there's a big pressure in the men's sphere to reduce the workload, to figure out ways to cooperate with other men, to produce more resources without totally doubling the amount of work that they have. And so, male groups—and actually, male groups show some cooperation even in the other great apes, much more than females.
- 6.82** Adult female apes don't cooperate much at all. But the human males do it quite a bit, and there'd be an incentive for them to figure out, how can we work together to produce more resources, especially using communication and cooperation, so to lighten the extra burden that we've all taken on? So, the incentive pressure to innovate collectively, specific to the male sphere, less need for the women to do that.
- 6.83** For example, there's a style of hunting, where the hunters form two lines, one upwind and one downwind. And the upwind ones, they just make a lot of noise and slowly advance forward; and so, all the little animals hiding in between, they go running away

from the noise, right into the arms of the other ones who are waiting with their nets and spears to catch them. Chimpanzees would never do this because the noisemakers wouldn't get anything to eat, because they haven't figured out to share the booty.

- 6.84** But the human men came up with this early. "We'll work together, work, instead of having to spend all afternoon, we can make one sweep and catch a whole bunch of stuff, and then we have enough for dinner tonight." So the men started doing this. There's less incentive pressure on the women's sphere for them to work together, to adapt, to develop a social system. So men began to develop these cooperative social systems, ways of working together productively, while women mostly did not.
- 6.85** It's important to keep in mind, too, that it's something we intuitively don't get today, where men and women are all in the same places everywhere. Through most of history, men and women lived separate social spheres. And yeah, they did come together at certain points and maybe sleep together and make babies and so on. But mostly, the men spent the day with the men, and the women spent the day with the women and the children. And at some point, the boys moved over as they got bigger, moved over into the men's sphere.
- 6.86** Indeed, this is probably true well into the 1800s in most places in the world. We don't see it now. So again, intuitively, it's a surprise, but we need to keep that in mind. And so, what was going on in the men's sphere and the women's sphere was largely independent of them, and so they were separate. Now, the hunter-gatherers were extremely egalitarian. Everybody says that they had no power hierarchy. In fact, that's one of the interesting transitions in social life, because the apes had hierarchies with the alpha male dominating the group or whatever.
- 6.87** But the hunter-gatherers, which is most of human history, prehistory, were extremely egalitarian. So it wasn't like the men were lording it over the women or the men were all equal. Anthropologist was talking to another couple of anthropologists about the time they'd spent studying modern hunter-gatherers, and they agreed they'd never seen any adult do anything that he was commanded to do by another adult, he or she. They just don't have that.
- 6.88** The closest is, maybe an old man will say to a younger man, "You know, it would be good if somebody were to do this," and then the younger man decides to do it. But no commands, no hierarchy. Males and females were pretty much equal, and this gets to one of the fundamental issues in theory about gender. Gender inequality is an empirical fact. I think, in every country in the world, the men have higher social status, more power, more resources, and so on.
- 6.89** What was the source of that, coming out of the egalitarian prehistory that was—if human civilization had existed 150,000 years, 140,000 was hunters and gatherers. Well, the feminist account is that the men banded together to push women down and oppress them and so on. It created this patriarchy. Again, it's a conspiracy theory.

There's no evidence. It would have had to happen in just about every culture in the world, but we don't find any signs of it occurring. There certainly aren't records. Conspiracy theories are generally wrong.

- 6.90** We propose part of the alternative theory. A more plausible one, I think, is that the cooperation among the men produced cultural progress, whereas the women weren't cooperating in this way. There was no incentive for the women to figure out how to work together in large groups. So, ultimately, wealth, knowledge, and power were created in the men's sphere, and that's what raised the men above the women. And they didn't push the women down; they actually raised the women up, too. It's just, they raised themselves up more because, again, wealth, knowledge, and power, they were coming from the men's sphere.
- 6.91** You can think of it sort of like a race. At a prehistorical point, the men and women started off separate but equal. After an even start, after many centuries, the men ended up far ahead of the women. So, did the men compete better or did they cheat or some combination of both? The feminist story wants to be, "Oh, the men cheated and oppressed the women and prevented them from inventing chemistry," and things like that. But really, I think it's more plausible that just progress emerged from the large-group hierarchy mentality of the men.
- 6.92** Progress in culture improved survival and reproduction; the population increased. It was not necessary for both sexes to produce technological advances or innovations like that. Once the men were doing it, that was enough, as long as they shared it to some degree with the women. So the system challenges to have the men create resources and transfer them to the women and children. This idea, men and women are basically enemies, and men at some point banded together to oppress women, I find, again, the evidence very thin and dubious for that.
- 6.93** But indeed, the framing of this is somehow men pushed women out of the public sphere. Well, the argument, when you think about it, it's like there's men, women, and then there's the public sphere that they both inhabit, that the men gradually took over. But really, what more likely happened is that the men created the public social systems, and the women didn't. That's why the women weren't in there, and that's why the men tend to run things, is that they created these systems.
- 6.94** So, positive view of men and women are basically partners perpetuating the species, taking care of each other. The men banded together to work together in large groups to protect the women, provide for them more easily, lifted themselves and women up. And the reason the men predominate is that they created these structures. Now, women do plenty of wonderful things. They're essential partners in this, too. I'm not saying there's anything bad. It's more, again, the right foot, left foot—that men and women are making separate contributions.

- 6.95** So, human evolutionary advances or communication cooperation, men learn to work together in these, in large groups. Like, again, the coordinated group hunt, communicating plans, doing things like that, developing cooperative hierarchies where some people can say, "Well, let's go this way and that way, or you guys go over here and we'll go over here." And ultimately, that's what led to the progress in culture and civilization, from the Stone Age to modern Western civilization. Most of the innovations were from men.
- 6.96** You start to think what came out of the men's sphere, of men interacting with men—religion, philosophy, technology, construction, in terms of building shelters and things like that, science, and then medicine. When women did do a lot of the early medical stuff, what plants would do this, I mean, the women's contribution is not zero, but it's not close to this; and the science-based medicine is clearly superior.
- 6.97** Military organization, governments, so public policy, legal systems, economic marketplaces, schools, universities—men basically put those together. Again, women didn't do nothing, but it's not commensurate. That that's not women's role in producing progress in culture. Probably the biggest contribution of women was clothing, which enabled people to live in much more difficult climates and things like that, and that's a substantial one.
- 6.98** It's an interesting question, why didn't women cooperate with each other to produce progress? Because most of history is groups of men competing against other groups of men, whether in business or in the battlefield or exploration or whatever. I mean, 50 women could have built a ship and sailed off to explore unknown places, too, but they basically didn't.
- 6.99** As I said, once the men started doing these things, there's no need for women to do it. Plus, on the ship thing, women are interested in safety, and it's risky to sail off into the unknown, so that's not the kind of thing women will want to do. It's one of the puzzlers I've wondered about for a long time—why didn't women cooperate with each other to produce progress? Again, the history of progress of civilization is almost entirely driven by groups of men.
- 6.100** If you're not convinced by the history of civilization, there's a meta-analysis of cooperation in one of psychology's top journals. A meta-analysis combines results of, I think it was close to 200 studies, using the modern economic games where you choose whether to cooperate or look out for yourself. With large numbers, you always get a significant difference, but they didn't overall. The omnibus analysis, the total cooperation, men and women were equally cooperative, which fits my view that this is one of the basic human traits, because that would be resting on lots of genes, and men and women would share those genes.
- 6.101** But the patterns were different. In particular, men cooperate with men, men cooperate with women, women cooperate with men really well, but women don't cooperate with

other women nearly as much. Also, over time, if anything ever goes wrong, the men are more forgiving and they'll give the person another chance. The woman, once something goes wrong, they cut it out. "It's over, we'll never speak to her, the other person, again."

- 6.102** I've thought about these results. It's based on a lot of data and it's certainly true, but it is cooperation among strangers. Before we conclude, women are less cooperative with other women. Well, they may cooperate a lot with other members of their family, with other females in their friendship network. That would not show up in these data. Of course, men do that, too. I don't know if there's a gender difference in that.
- 6.103** Again, overall, the total cooperation was the same. Anyway, there is woman-to-woman cooperation, but probably in a much smaller social circle. And so, again, the woman-to-woman cooperation, but that's what's required for building the social system. And again, relationships among women, they're more unstable than, say, relationships among men.
- 6.104** One of the early studies on relationships came to a shocking conclusion. Everybody thought the female-to-female, the lesbian relationships, were the most intimate and the strongest and the most durable. When they followed them up, a year or two after the initial interviews, the lesbians had the highest breakup rate, and other subsequent findings have replicated this.
- 6.105** So, obviously in heterosexual marriages, too, wives initiate divorce more than husbands these days, though there might be multiple reasons for that. There are even several papers on college roommates, when you're randomly assigned when you first come to college, just to share a room with somebody of your same gender, and they look at, well, do they stay together for the second year? And the pairs of men often do much more than the pairs of women.
- 6.106** So again, there's some instability in that, and that might weaken the chance for long-term progress as well. Males can produce cultural progress more easily as the longer-enduring relationships. They also have larger networks. There's some papers on what are the social life things that—they're initially commenting on an article that said women are more social than men because they are more interested in relationships. So, men are more willing to fight because they don't care about relationships.
- 6.107** And I voted to publish the article. I was a reviewer on it because they made a good case. But I said, you could look at all the data in a different way. That's just the one-to-one relationships. If you look at things that people do in groups, then men are more social, and there are plenty of studies on this. Even, they have two children playing together, and then the researchers bring in a third one. If it's two girls, they don't want the third one to join; they will exclude her. The two boys, "Sure, come on in and join it."
- 6.108** It's not that there's something wrong with the girls. They just want the one-to-one connection. That's what they're oriented toward. The third girl spoils it for it. Whereas

the men, the larger group is fine. Sure, let the third boy come and join the fun. So, males have larger social networks. The intimate one-to-one relationship is probably more important for the species because the mother-child bond, there has to be a lot of caring there. That's really good.

- 6.109** But it's not as fertile a basis for culture to emerge. So if culture is sharing information, 20 people can share a lot more information than two, obviously—and so, another reason that culture may have more emerged from the men's sphere. So, anyway, the men's sphere is producing more resources, the women's sphere remaining fairly stable.
- 6.110** Each new baby girl, well, she could try to organize the women to set up a physics lab or something like that and do research; but probably, the most adaptive and useful path is to find a way to be attractive to men, to get a share of what the men are producing. The baby boy is a more complicated situation. He has to work out the men's sphere and the hierarchy and the competition, and figure out how to gain respect there, so that he will be attractive to women.
- 6.111** And human intrasexual competition, competition for mates, is much higher in humans than in any of the other apes. I think, for obvious reasons, since you're trying to form a 10- or 15- or 20-year partnership, rather than just an afternoon boink, which is about what the apes do. And again, recruiting the men into a provider role was a key to the evolution of our species.
- 6.112** This came across my desk a while ago. There's a lot of nonsense written about the gender pay gap, and there's some very good science, too, but you have to look for it. But it's not really a gender pay gap, because the single men have the same lifetime career trajectory of earnings as the married women and the single women. Those are all the same. It's the married men who earn more, and part of that may be embracing the provider role. You have responsibility.
- 6.113** Somebody's given the example of a guy who goes to college and does some creative writing. He said, "I want to be a poet. I think that's what I'll do in my life." And then a few years later, he's got a wife and a baby, and he, "Well, I'd better get a job that actually could pay money," and so on. Or there's the musician thing. I used to be an amateur musician. Musicians have a joke. "What's the difference between a professional musician and a large pizza? Well, a large pizza can feed a family of four."
- 6.114** Anyway, part of that may be the man going into the provider role, but also, he has a woman to take care of him, and that's probably crucial, too, and underappreciated. The single man is on his own for emotional care. Competing in the men's sphere is often difficult and stressful and all that; and so, having a woman who will take care of you and make you feel better, that may be a big part of why married men have better, more successful careers than single men.
- 6.115** So, women—we've talked about the men a lot because the men obviously need more rehabilitation than the women. But let's say some of the positive things about the

women. Again, they're equal partners in the biological flourishing of our species. Taking care of the children, the women still do the majority of that. Taking care of the husbands, as I said, may be a crucial aspect as well. Women do make men's lives better. Married men do live longer. That's one pretty strong sign.

- 6.116** There's a recent paper, another one in behavioral and brain sciences, saying that romantic relationships are more important to men than to women. It's because men need the emotional support from women. They can't get it from other men, whereas women can get emotional support from each other. It was an interesting argument, and I think there may be multiple things going on, but that's no doubt an aspect there. As I said, other things the women contributed—they selectively bred the men to accept the provider role without complaint, or maybe with some complaining.
- 6.117** They did make some contributions to cultural progress, just, it wasn't really their thing as much as the men did. The male is the provider for the woman, and she helps pass it on to the children, so perpetuating the species. And marriage and co-parenthood benefit both sexes, both men and women, and the children as well, obviously. So, ultimately, women are not really victims, but partners in the success of this.
- 6.118** And yes, men made most of the social innovations to produce progress in civilization, but women made this possible, taking care of men. Women make the men better in multiple ways. As I said, married men are better off, happier, longer-living, healthier, better behaved even, and things like that. Taking responsibility for the biological and emotional tasks enables men to focus more on the civilization tasks.
- 6.119** All this casts the women's movement in a slightly different light. They'd like to say, "Oh, the men were oppressing us, and we rose up and defeated them." That's a stylized argument. Obviously, women got the vote in an election in which only men voted, and a majority of men voted that women should be allowed to vote henceforth as well, and that happened all over the world.
- 6.120** But really, I think you can really look at it more that the women in their sphere did not really create that much of a society. The men had a long, difficult struggle to do it. Creating social systems is not easy. Lots of bad things happen, lots of victims, lots of violence, financial disaster, and so on. But they finally got it going pretty good; and then the women, at that point, petitioned that they wanted to enter into it and take part of it, in what the men had created, because they hadn't really created that desirable social sphere for themselves.
- 6.121** At most, we can blame men in saying, "Well, they should have included women from the start." But I think that's unrealistic in a variety of ways. It's certainly much more complicated to include women. As we see, women have been in the workplace for over half a century at this point, and we're still having to make adjustments and so on. It's hard enough to build a working social system in the first place, and to accommodate all the special needs of women would be difficult.

- 6.122** Go back in history, there was the men's sphere and the women's sphere, and neither of them really included the other in what they were doing. So I think we shouldn't be surprised by that. So how did society flourish? The keys of success, biological success means increasing the population. Women bore and nurtured the infants, ideally with help. Men worked together with other men to produce the resources to support the children and their mothers. They also developed the systems to protect them from the various predators and from nature and from other dangerous groups of men.
- 6.123** This is not what had to be or anything, but it appears to be what's worked everywhere. Again, this is descriptive, not prescriptive. It's not the way it should be or anything like that. In general, society transfers resources from men to women. I think that's the fact of life, just about everywhere. But that's what needs to be done, because women aren't likely to produce resources on that scale. Ideally, a society helps the men work together to create more such resources.
- 6.124** Just to summarize, one key to human evolution was, first women recruited men into the provider role; that made culture possible. Again, I've produced this theory without any reference to oppression or patriarchy, exploitation, misogyny. I don't think we need to postulate any of those, and that was one of the tasks of trying to produce an alternative explanation. Man's role has been to produce the resources.
- 6.125** In fact, the first time I came across the idea that, to be a man, you have to produce more than you consume, this was not an evolutionary book at all. This was a sociological book, a guy who'd studied men's roles in many different societies. Sociologists don't tend to even pay much attention to evolutionary theories, but he said that seems to be the theme everywhere on Earth, every society. To be a man, you have to produce more than you consume. That fits in what being the provider role is going to be all about.
- 6.126** Resources, not power, are the key. This modern fascination, to analyze everything in terms of power, I think is fundamentally misguided. It's not completely wrong, but resources are what matter. Resources without power are fine. Power without resources isn't really worth that much. People want power because you can get more resources. Power without resources is not any good. Resources without power was most of human prehistory. The hunter-gatherers didn't have power. They were all equal, but they got resources and shared them.
- 6.127** Women's role has included taking care of the men and children, including extracting the resources for herself and for the children. Both genders are fundamentally altruistic. Humans have to put in a lot more care for their offspring than any of the other great apes. And because of that, the species thrives. There's no value judgment, as far as I can take, in terms of which one is more important or better; both roles were essential.

- 6.128** And so, all I'm asking is, should we be allowed to consider this theory? There are people who say no. Even if this theory is true, nobody should be allowed to say these things. And that, to me, is fundamentally anti-scientific. And soon this will be a much more healthy way for men and women to understand each other, than that the women are helpless victims and the men are wicked oppressors.
- 6.129** I mean, you want to challenge them, here's some of the basic points. Are any of these wrong? Men evolved group cooperation earlier than females. That goes back, as I said, even into the other great apes. Turning males into long-term providers was one key to the human evolution. I think that's obviously true. The provider role imposed extra duties, extra burdens, on the males. Well, obviously it has to. You have to spend more time getting food.
- 6.130** Men got a headstart in developing social systems. That's clear as well. Neither the men nor the women typically included the other in their private rituals and whatnot in prehistory and early civilization. Once the men created the systems, there was less need for the women to do that. The reason men moved ahead of women is because they created most of the wealth, knowledge, and power.
- 6.131** So, men and women have cooperated to produce one of the most successful biological flourishings of any mammals. I restrict to mammals. I think some of the bugs are doing pretty well, but talking about mammals, we are doing well. Most of the other mammal populations are declining. And again, men and women get along well. Both men and women live longer, happier lives when they are married to each other on average. And so, we should start our theory with the fact that this is a big success story.
- 6.132** We could do good for society if this theory is correct by at least putting it out there and giving young men and women a more positive way to understand themselves and each other. There's a lot more good than bad in male-female relations. Recently, it's taken a turn for the worse, I think, under the influence of this negative oppressor-victim account. We should listen to the other theories, but don't let them silence us. Don't let them say we're not allowed to think these things or say these things.
- 6.133** A key point, human males are more fundamentally providers than oppressors. When a young man thinks, "How do I relate to the woman? What's my basic relationship to her? Is it that I can dominate her and boss her around and exploit her? Or is it, no, I want to take care of her and provide for her and make her happy and make her do well"? Women and men help each other and their children thrive.
- 6.134** Two last thoughts on a personal note. I'm old. My friends are old. Death is getting close. I had a couple conversations with some of my older friends. They talked about their wives, and I've known these guys for decades. We never talk about our wives. Talk about all kinds of things. But the approach of death, and one of them just died, made it imperative.

- 6.135** Do you think what they were talking about was, "As my strength fails, how can I keep my dominance over my wife? How can I keep her down in her place?" No. They were both thinking, "How can I provide for her after I'm dead?" They both said, "What if she lives 20 years after I do? Have I set things up to take care of her?" That was the concern. To me, that's more what a man is being about than oppression.
- 6.136** The other thing, a more positive note. This was a joke the adults had when I was a boy. I'm not sure I understood it then, but I think it's fitting, and it fits with all we've learned in the intervening half-century about men and women. You have to pardon my French, but it came from the French. The line was, "Well, ultimately, all things considered, there's very little difference between men and women. But vive la différence!" Thank you.
- 6.137** So, considering men and women have lived separately for, I guess, our entire civilization and human history, that doesn't seem to be happening so much anymore, obviously. So, what would you say the consequences of that are?
- 6.138** Potentially, it's very positive. As I said, if you choose by merit, then there's a bigger talent pool. The men who want to stay home and take care of the children who are better suited for that can do that, and the women can contribute in any field. So, again, this is descriptive, not prescriptive. People do sort of sort themselves habitually into that, but the upside is very strong.
- 6.139** Now, there are interesting arguments that, well, there are new kinds of problems that arise there, no doubt. No doubt there are. But ultimately, in terms of what's best for society, first of all, we have to believe in freedom, and it gives people more freedom to carve out the life that really suits them. Second, in terms of the flourishing and functioning of society as a whole, the more the best people can do the jobs that they're best suited for, the better the system will work.
- 6.140** So I think it's mainly positive, but there are difficulties, and romantic attachments happen at work. I guess they've now mostly cracked down on those. Finding the right policy is very difficult. I mean, for a long time, women went to work and would meet a man there and marry him, and now that's forbidden. One of my friends, I was at Harvard for a couple of semesters, and one of the women said there, "None of the happy marriages I know would happen today." They wouldn't be allowed because they're more, "Oh, you're not allowed to date any younger person in your department or even in the university."
- 6.141** And so, it makes it harder. Whether women are really better off having that excluded, I mean, it will prevent some sorts of abuses. But the difficulty—so, anyway, settling on the perfect policy for all these things continues to be difficult. Some years ago, I was a guest professor at a Saudi Arabian university, and I went there. There, the separation is much, much stricter. They had me give my talk in the medical school because they said,

"In the medical school, we're trying something different. We're having men and women work in the same place."

- 6.142** But the rest of the campus was divided. There were two chemistry departments and two physics departments and two everything, and a big wall down the middle. They're rich. They had modern technology, so occasionally a man could give a lecture and they would broadcast it over to the women's class, too, but there would be a female professor there to translate things for them as needed to be. Yeah, I'm not sure that's as productive. Who am I to judge? They all seemed quite happy with the way things were there.
- 6.143** Anyway, just looking at it from system aspects, the upside looks to me bigger than the downside, of mixing people into the same spaces. I will say, though, that the eradication of male spaces, a lot of people are saying that's doing harm to society. Some put it, "There's no place a man can go to be away from women anymore." And male-to-male interactions do have some positive aspects.
- 6.144** My view on that is the best thing for a system is to have all, even with the schools. Rather than saying all schools have to be co-educational, I'd say, why don't you have some co-ed schools and some boys' schools and some girls' schools, and different people will flourish in different ones, and we can see what works best? And again, probably different people have different needs and preferences, and they can transfer among them.
- 6.145** I would say that the eradication of male spaces and male clubs, I think, is partly what's producing some of the problems young men are experiencing today. And again, you might as well just retreat into the alternative worlds of online and video things. The men's clubs did serve valuable functions in the past, and I think we may have rushed too far. But other than that, it's more of a good, I'd say, to mix them.
- 6.146** I've seen a lot of content talking about the male loneliness epidemic. It's interesting to hear you say that men are more social, or at least more receptive to bigger groups. Have you read anything about that, the supposed male loneliness epidemic? Is it real or is it overblown?
- 6.147** I don't have a knowledge of that. I've heard about it, but I haven't looked at it systematically. But I can well believe it. Again, the male groups have mostly been abolished by the feminists who insisted somehow they were busy there promoting patriarchy and oppressing women. Well, they did good. There are women's groups and women's clubs, but I don't know that they're as productive as well.
- 6.148** I was at Florida State as a professor in other years, as was my wife, and they had a women's faculty meeting for the psychology department. Why they needed one, I don't know. Did they need to revisit the colloquium from an all-women perspective or whatever? Anyway, my wife went to a couple of them and said, "All they did is, they spent the whole time complaining about their husbands." And finally, she and another

woman just stopped going. They said they didn't want to complain about their husbands. But I told you, my wife's pretty nice.

- 6.149** That has become quite rampant, and women are encouraged to disrespect men, which I think is also a negative thing, weakening marriage, as men want to be respected. I've gotten to know Arthur Brooks a little bit, and he says, well, men and women want slightly different things from marriage. The man wants to be respected and admired. The woman wants to be cherished for who she is and what she does for him.
- 6.150** He doesn't want his wife necessarily to bring home a dead antelope and say, "Here, I've got our dinners for the coming week." He doesn't need her to do that. She isn't looking for that, but she wants to be appreciated as a special person, and he wants to be respected. And with this attitude of disrespecting men sweepingly, and women complaining about their partners and so on, that's very undermining. That's what my wife said. She thought it's really bad for marriages, to do that.
- 6.151** Since she joined the the Mormon Church, the LDS, they tell people not to do that. They have their women's groups, but they don't sit around complaining about their husbands and so on. Diane, my wife, was talking to them, and the ones who are lifelong members of the church, of course, they didn't know any difference. They just thought that's how it was. But all the other converts said, "Yeah, that's a surprising difference. I think the Mormons have stronger marriages because of things like that."
- 6.152** The men, likewise. I don't think men complain about their lives nearly as much as women complain about their husbands, at least certainly not in my experience. But the men also accept their role and embrace it.

Lecture 7: Unconscious and Conscious

At a glance — The unconscious and conscious.

- 7.1 All right. Today's lecture, we'll focus on one of the great problems of psychology, is trying to understand consciousness. This is something that's been discussed and debated, again, for centuries, by people in multiple fields. Multiple mysteries and problems and questions remain. Among other things, I sometimes say when I would lecture to a large group that I'm not religious, but some of my friends are, and I respect religion.
- 7.2 If I were religious, one place I would look for it is in the leaps between sciences. I think I spoke in another lecture about Anderson's hierarchy of sciences. How do we get from physics to chemistry, or how do the chemicals combine to create life, into biology? That remains a mystery. Lots of people mucking around in chemistry labs, but making living things from scratch is still beyond what we know how to do.
- 7.3 Anyway, a similar problem is, how does the brain create conscious experience? The brain is a physical thing, a bunch of nerve cells, electrical and chemical activity. Somehow, how does an electrical charge from the nerve cell combine with others to create our conscious feeling, our conscious experience, even in something as simple as pain?
- 7.4 And pain, by the way, my speculation, I can't prove it, but I would guess that that is probably, in evolutionary terms, the very first conscious experience. It would be a signal to the brain that damage to the organism is happening, and so you should do something about it. It could be, pleasure was the earliest one, but going by the principle that bad is stronger than good, I would guess getting away from damage would be a high priority and highly adaptive.
- 7.5 But still, from the simple feeling of pain in a simple animal, how do we get to the fancy conscious experience that human beings have? What is it useful for? It's one of the centrally defining traits of humans. I mean, it gives us a rich subjective experience of our life, so we have feelings and thoughts and all that. But does it do anything else? In particular, does it have causal influence on behavior?
- 7.6 Now, theories of consciousness pretty well sort. There are at least two levels, although people argue about everything. Are they two levels or two kinds? How should they be understood? But one, in simple terms, is the same as what we share in common with other animals. Lots of animals have enough consciousness that there's an agent there that sees things and guides behavior, responds to the environment.
- 7.7 So a lot of animals have that in common, but everybody pretty much agrees that humans have another level where you're going beyond that, where you can reflect on things and reason and simulate things away from the present, have self-awareness, all

kinds of more advanced things. So we'll call the one, the simple one, phenomenal awareness. That's the more basic one. Again, the one we have in common with lots of other animals, even bugs, perhaps.

7.8 Then conscious thought would be the distinctively human variety. I'm going to focus on the second one, conscious thought, because that's more, again, distinctively human. My project is to understand what makes us human, what are the human traits. I got started on this around the turn of the current century. At the time, there was a lot of disrespect for consciousness. The pendulum had shifted back and forth, that all human behavior is consciously determined and decided, or hardly any, and that the unconscious is doing most.

7.9 I'm not sure exactly where we are now. Social psychology got sidetracked with methodological concerns, and so it's been slower in terms of conceptual advances. Anyway, lots of skeptics and detractors at the time. Some doesn't do anything except allow us to entertain ourselves, and as I said, give a rich feeling experience of our own lives.

7.10 It wasn't a new idea. Thomas Huxley, in the 1800s, he was a contemporary of Darwin and an assiduous defender. I called him Darwin's bulldog. He said consciousness is just like the steam whistle on a train. We don't ride a lot of trains these days, but the steam whistle is that little thing on the side that basically goes toot-toot. So consciousness is like that, which means it's a product of what's going on inside the engine.

7.11 It might tell us something about what's happening inside, but it doesn't have any capacity to steer the train or power it or move it along. It's pure output, not at all input. And so, that has remained a view that goes, as I said, in and out of fashion. Some of the people writing about this, "The causal role of conscious thought has been vastly overrated."

7.12 "Most of a person's everyday life is determined not by their conscious intentions and deliberate choices, but by mental processes that are put into motion by features of the environment that operate outside of conscious awareness and guidance." John Bargh is one of the leading researchers in terms of looking at unconscious and automatic processes. And so, he was saying things like this.

7.13 And indeed, another statement by him, "Our psychological reactions from moment to moment are 99 and 44 /100% automatic." So pretty extreme there. He later came around a little bit, but he said this was in his feisty period, when he was quite assertive. But even notice the expression moment to moment, that he puts in there as a qualifier. I guess he goes, maybe this is close to right, because maybe the moment-to-moment reactions are really guided by the unconscious.

7.14 As I said, you're never aware of the nerve cells firing to move your arms or your legs, but consciousness might still have a really powerful and decisive role elsewhere. In my career, I've studied lots of different fields and so on and come with controversies that

go on and on for decades, like a number of the ones I've put up here. My sense is, why can't we resolve some of these things? Usually, either the problem is phrased wrongly or the two sides are talking past each other, in which case both sides have a valid point, but they usually overstate it.

- 7.15** To make sense of it, you have to strip away the overstatement and see what the valid point is on both sides, and they have to be able to fit together. Reality has to be plausible, after all. So let's look at some of the criticisms of consciousness. I came up with five of what people were saying that, look, consciousness is not able to do this. One, going back to the '70s, they showed that, when people try to explain why they did something, so explain their reasons, they're often wrong. They don't really know.
- 7.16** Some of the experiments that Nisbett and Wilson did, in one of them, they had people choose which was the best of four stockings, and they shuffled them around. They kind of made it difficult for them; the stockings were all really quite similar. But anyway, the data show that people tended to pick the last one, whatever one they saw last of the four. But when they asked them why they did it, they said, "Oh, it's the color, the texture." They never said, "I picked it because it would be last," because that would be stupid.
- 7.17** So they concluded, people, when they give explanations, they're just borrowing from a stock of accepted explanations in the culture. They're not really understanding what's going on inside their mind. I think people can report certainly accurately how they feel, but why they feel that way, some of those causes are unconscious.
- 7.18** Another criticism is that sometimes consciousness makes up false explanations. There's this inner narrator that goes along analyzing what you do and coming up with explanations. Gazzaniga's research on the split brain would show that, if you could present information to one side of the brain, the other side would devise some explanation that wasn't really true. It didn't really know why it got that, but it was manifestly wrong. So, consciousness doesn't really know entirely what's happening in the world, either; and so, explanations are at least unreliable and sometimes useless.
- 7.19** Third, consciousness may be too slow to guide behavior. I think I mentioned in another lecture, the Libet studies, where people were told to make an arbitrary movement of their finger and to record on a very detailed clock exactly when they made the decision to act. The brain data showed that the brain was starting to respond before that. Some people said, "Oh, look, the brain is doing it." Their conscious decision is too late to make a decision. That's not what Libet himself said. He said it gives you a chance to veto it and change it.
- 7.20** As I said before, to me, conscious experience has to be created by the brain, so it makes sense that the brain started beforehand. Also in that situation, it was a totally meaningless, arbitrary action, which is not what consciousness is for. They even told people not to plan when they were going to make it, so you basically removed

everything that we'll see consciousness is good for. Still, some people said, "Look, by the time consciousness gets involved, things are already underway."

- 7.21** Fourth, you can duplicate variety of behaviors that people had thought were required for consciousness, like a goal-directed pursuit. A lot of actions can be caused without consciousness; that at least unconscious things can activate, set a goal in motion, and the person doesn't know why. In one of his famous and much-debated experiments, he primed students with ideas about old people. He didn't even use the word old. He used things like Florida, retirement, sunshine, and so on, and then found that they walked more slowly going down to the elevator.
- 7.22** There was actually a follow-up. Young people who don't like old people walk faster to the elevator, but somehow they were adjusting their behavior based on their preference, and they had no idea that they were doing all this. So again, some behavior can be duplicated without consciousness.
- 7.23** And last, it could be mistaken about whether it causes action. Dan Wegner had a variety of clever experiments showing that, if you have a thought just before something happens, and in fact, the computer makes it happen, but you feel like you caused it, and so there's what he called the illusion of conscious will, which is the name of his book. He was one of my friends. He died a few years ago. I talked to him during this and followed it closely because I was a big admirer of his work.
- 7.24** He sort of went back and forth between, well, is all conscious deciding illusory, or is it just there's a system that can be fooled, which is what his data showed? So again, people tend to take one point and then overstate the implication. He showed clearly you can be fooled; you can think you did something when you didn't. But does that mean you're always mistaken or just you're mostly correct?
- 7.25** So, putting these together, consciousness is not so good at perceiving the self and the world, and it also isn't essential for the direct control of behavior. It can be dispensed with. So, first question, why? Why then do we have consciousness? Most theories assume that it's either there for input or output; in other words, that consciousness is there to help us perceive the world or to guide behavior. That's the output, obviously.
- 7.26** But the five criticisms reveal flaws or shortcomings in both of those. So, well, if it's not input or output, what else could it be? Plausibly, it's for some kind of internal processing. It may be something that goes on several steps removed from either immediate sensory input or immediate motor movement output. Essentially, the idea is that conscious thoughts are when the brain is making input into itself, taking information it has, integrating it and combining it.
- 7.27** As we'll see down the road, the unconscious is not so good at extensive integrative processing. It's really good at fast and simple processing, but to put things together into a bigger picture is somewhat hard for it. So, in a sense, the steam whistle theory is that conscious thought is just output, so unconscious processes are doing everything. They

create the conscious experience, they create the behavior, and consciousness has no causal role itself.

- 7.28** But maybe conscious thoughts do have an influence. They could influence back, on what goes back into the unconscious. You think about something and you think about how to do it, or you think about whether it's against the law, and decide not to do it or change your actions. And so, behavior could still be the direct result of unconscious processes, like Bargh said, but conscious has a big upstream input that could be very useful and adaptive.
- 7.29** Another point, too, is consciousness might help us relate to the cultural environment, for complex information processing. So again, even if behavior is always the immediate proximal result of unconscious processes, conscious thoughts can help us communicate with the world, get social input, and so on, feed back into the unconscious, and thus indirectly but profoundly affect behavior.
- 7.30** There's this question, what is the purpose of the brain? Why do we have brains? And the human brain, in particular, is for accessing information and social groups. Again, the point I've made in a number of the lectures is, my thinking starts from, we are cultural animals. Our biological strategy is to work together with others, shared information, interlocking roles, to produce more resources.
- 7.31** I pondered, thinking about consciousness, things like arithmetic. We learn arithmetic from our culture, from schools and so on. It's clearly learned, but all cultures get the same answer. 3 times 5 equals 15 everywhere. It's not like there's cultural variation. I'm pretty sure if we ever did space travel to another the galaxy and found intelligent life there, they would agree that 3 times 5 is 15. So, is that natural or cultural?
- 7.32** Well, arithmetic is a body of objective truth, but that we learn culturally that knowledge is built up through many mathematicians figuring these things out. If you started from scratch, you wouldn't get very far. It's not built into the brain, but you could gradually figure these things out. But collectively, we master this, and then pass that information along to the new generation.
- 7.33** Theories of human evolution, the early ones—and we named ourselves *Homo sapiens*, based on the big brain, they were intelligent people—and so, the early theories, when biologists started applying evolutionary theory to humans as well, was, OK, what set us apart is the big brain. Because of that, we started to walk upright, and that freed up the hands for making tools and led to the modern world.
- 7.34** But then they discovered the remains of *Australopithecus*, which was an upright-walking early human, hominid, but still had a very tiny brain, so the upright posture came first. Well, that raised a couple puzzles. Why then did the upright posture start compared to other animals, and then what did cause the brain to expand?

- 7.35 Well, I think a good answer to that starts with one of the exciting neuroscience discoveries of the 1990s, which was mirror neurons. These are brain cells that fire when you do something, but they also fire when you see somebody else do the same thing. These are in a variety of animals. Monkeys have them even.
- 7.36 A couple of facts they noticed about the mirror neurons—one is they're especially sensitive to the arm and hands, and the other is that they will mentally complete the action. I'm not sure how they found this out or figured this out, but even if it's not completed in reality, they can sort of infer what the intention is. So a monkey sees another monkey reach for a banana but not get it, it sort of knows, "Oh, it was reaching for that banana," which is probably then useful, because then, "I could go, if I have a longer arm. Maybe I could get the banana."
- 7.37 It's also obviously very useful for observational learning to have the same brain cells guard your own action as seeing somebody else do that. Well, the monkeys never did a whole lot with the mirror neurons and so on, but apparently, some of our ancestors began to understand that, if you're paying attention to my arms and inferring my inner mental states from that, then I can use them to signal you. There's pretty good evidence that human language started with gesture, that that was there before speech.
- 7.38 One sign of it is, we are able to teach chimpanzees and gorillas and so on sign language, but we're not able to give them speech. In fact, they were trying to teach them to talk, and they almost gave up because their vocal apparatus just isn't competent in those other apes. But then they hit on the sign language, and they could get them to master that. Probably the early humans did that.
- 7.39 It turns out, to give us the capacity for speech, there had to be a whole bunch of changes to the physiology, the anatomy, some of which are not advantageous. There's a higher risk of choking on a chicken bone or something like that. But that would be worth it, to be able to communicate better. But that means the communication was already there in the environment. In this way, culture was actually shaping physical evolution. People were evolving to be able to talk and communicate better. There was already communication going on, probably gesturally.
- 7.40 And so, the advantage for being able to be more effective at speaking, a young man, for example, born with, say, a gift of gab, who's very fluent, a good talker, well, if he's in a group that doesn't have language yet, that's not going to be any advantage. But if there's language in there, then he can probably charm the ladies better, and it will improve his reproductive prospects and so forth. In that way, again, physical and cultural evolution may have influenced each other, rather than one happening all at first and then the other starting.
- 7.41 So, human evolution, I think then maybe the original human trait was communication, although since then cooperation—we clearly cooperate much more than the other apes do. But communication helps with that, because you can make joint plans and

make agreements to share the loot and so on. So, other animals communicate some. It's not clear they intentionally communicate. They don't really understand that "I know something that others don't." But humankind took this to a big new level.

- 7.42** Perhaps the social sharing of information is the first distinctively human trait, the one that set us off on our unique path. As we moved out of the forests and into the grasslands, instead of competing with the other apes, we were competing with the big cats. We could have evolved to have more catlike traits and be more like them.
- 7.43** But instead, that's where we started to go our own way and have what they call the social cognitive niche, that we are working together, sharing information and relying on that to make ourselves ultimately the best hunters in nature, even though we don't have the fastest legs or the biggest teeth or anything like that. But it's by cooperating and communicating, we're able to become such effective hunters.
- 7.44** So what is the purpose of conscious thought? Well, William James had this pragmatic, pragmatism theory that it's always for doing, that thinking is for guiding behavior—and this, many people have repeated this. Susan Fiske, a leading scholar, writing in the Handbook of Social Psychology, says the same thing, "Thinking is for doing." In a trivial sense, that has to be true. But can we at least consider alternative versions at some point, to some degree?
- 7.45** One theme of my career has been that inner processes serve interpersonal functions. So, thinking might possibly begin to have interpersonal aspects, and so, it's somehow there to get along with others. And so, let me raise at least the possibility that conscious thinking is partly for talking. Going back to the research by John Bargh and the other guys, who show that, well, things that we thought we need consciousness for, somehow they can be done, or at least some version of them can be produced without consciousness, if you bypass conscious and just activate the unconscious automatic processes.
- 7.46** It's a very impressive program of research, but none of them has ever been able to show that you can have a conversation unconsciously. It's basically impossible to talk to someone while you're thinking about something else. You can't bypass consciousness there, so that's one of perhaps the key reasons for it. Talking is always conscious.
- 7.47** When you think silently to yourself, that's often imaginary talking. A lot of thoughts people have when they're alone, they imagine having a conversation with someone, replaying one they had or thinking about one that had come up or even just imagining a possible one. The production of speech and thought is very much overlapping a lot of the same processes until you say it. When people learn to read, for example, they learn to read out loud, and then at some point, they learn to inhibit the saying it.
- 7.48** But even adult readers, there are sort of incipient movements, as if you're about to say it out loud, so you've stopped that. So the saying and the reading are really closely tied together. In fact, there's an interesting paper on logical reasoning by Mercier and

Sperber. It was one of the most provocative, interesting ones. They looked at all the research on reasoning, and people make a whole bunch of mistakes and so on. So they said, well, the reasoning faculty is pretty good, but it has all these big flaws.

- 7.49** But they argued, well, they are only flaws if you think, like scientists do, that the purpose of thinking is to discover the truth. But maybe it's for arguing. For example, one of the standard flaws is that, when people have a theory they like, they look for evidence that confirms it, not so much evidence that disconfirms it. Well, any scientist will tell you, "That's not sound, that's a flaw." But if you're having an argument, of course, you want to have arguments that support your side. You're not looking for counterarguments.
- 7.50** As they pointed out, when people are arguing with someone, they have a disconfirmation bias when they're listening to what the other person is saying. They're looking for the flaws in what they say. It's not that they're unable to do that. And so, a very impressive argument, and what that means is that we evolved to discover the truth together, that people can argue an issue—one argues one side, one argues another.
- 7.51** And even if they don't persuade each other, often the third people, the third person, others who are listening, they are persuaded. They can see she has a better argument than he does, and so they go along with the one side rather than the other. The cognitive scientists who study thought processes and so on, one of their tests for whether something is conscious is, "Can the person tell it to you? Can they tell you that thought?"
- 7.52** And they use it, but they somehow never make the leap to say, "Maybe that's the point. Maybe that's why we have consciousnesses, so we can tell other people about it." It could be highly useful and adaptive, so there's a puzzler. If thoughts can cause behavior, why do they need to be conscious? I mean, yes, if I think I shouldn't put my hand in the fire, then that's probably good, and I won't burn my hand. But why does that need to be conscious? Why can't there just be an unconscious, "Don't put hand in fire"? That would work just as well.
- 7.53** Nothing is gained by consciousness. But if I can talk, if I can tell my children not to put their hands in the fire, then that will improve their fitness, and my evolutionary success will be better at passing along. So sharing information, making it conscious, enables me to communicate it to others and is useful in that sense. All right, so that gets us through the first part. Now we go to the what.
- 7.54** What does consciousness do? It's a more basic question. Why are there brains in the first place? Part of it is stimulus and response. A lot of early psychology, early scientific psychology, analyzed behavior like that, the stimulus in the environment and the behavioral response. The brain, of course, is where the stimulus meets the response.

- 7.55** Now, if you have a simple brain, a basic lizard brain or whatever, where you see something and all you have to decide is whether to eat it or run away from it or maybe have sex with it or ignore it, well, you don't need a very elaborate brain design to do that. But when you get a complicated environment, then it's much harder to match up the stimulus with the response. We have lots of possible stimuli and lots of possible behaviors.
- 7.56** As the environment becomes more and more complex, getting the stimulus to find the right spot in the brain with the requisite response is more difficult. So, what people argue is that consciousness is kind of the big board, where it can be put up there and broadcast out to the whole brain, and then all the little pieces of information stored at various places in the brain can chime in and say, "Oh, yeah, this is a better way to do that, or, I promised my wife I wasn't going to do that. That's illegal," or whatever.
- 7.57** So putting it in consciousness activates unconscious responses from different things. Example I like to use for that is, when you're packing to go on a trip, you think, "Well, I'm going to go on a ski trip." So I imagine myself there. "Well, what am I going to need? I'm going to need my skis. I'm going to need long underwear. I'm going to need all these other things." So you hold up in your mind this imagination of what you're going to do, and then the unconscious chimes in with, "Don't forget your toothbrush and don't forget this or that."
- 7.58** It's called the blackboard or the theater, depending on which theory, theorist, you read, but it's the same kind of idea. It helps communication across different parts of the brain. And so, the social brain would be the extreme case of that. This is a highly complex one. You have a lot of different rules or contingencies that you might need to think about.
- 7.59** Consciousness' inner process might still affect behavior, but mainly indirectly, mainly working together with unconscious processes—again, several steps removed from both the sensory input and the behavioral motor output. That brings the question, does consciousness cause behavior? The steam whistle theory says no. It has no causal role whatsoever.
- 7.60** I think I mentioned, when the Libet studies come out, the skeptics said, "Aha, look, the consciousness is too late. It doesn't have any causal role. The brain is already doing it automatically or unconsciously." But that's been debunked and that hasn't held out. I got a couple of colleagues and I set out to review the literature and see, does consciousness indeed cause behavior? We looked for studies that showed an independent variable that was some conscious thought or feeling. So they randomly assigned some people to have it and some people not to have it, or to have a different one.
- 7.61** Then the dependent measure was the behavior, which had to be some actual behavior of actually doing something. It's not enough to just go from one conscious thought to

another conscious thought, which lots and lots of psychology studies do. Thinking about this makes you think that. That wasn't enough. It had to show some causal effect on actually doing something. Could we find evidence for that? Well, we found a ton. Mental simulation, if you imagine doing something, you become more likely to do it.

- 7.62 One of the early studies done, I think here in Arizona, was when cable television was first spreading through society. And so, they connected with the cable TV company. They called people up, and some made their usual sales pitch. But in other cases, they said, "Well, imagine yourself having it and enjoying the cable and what it would be like to do it." And so, those people were more likely to take a trial offer and more likely to end up actually doing it.
- 7.63 There are studies of voting, too, that imagining yourself actually going to vote makes you more likely to actually do it. So imagining to do something makes it more likely to happen. A mental practice in sports, there are a lot of studies showing that that works, that you imagine doing something. One of the most elegant had people practice a golf put. So before doing it, they imagined themselves making the put and the ball going into the hole.
- 7.64 Others imagined themselves making the put and missing. And then there was a third group that didn't imagine either way. And so then they actually made it, and sure enough, it worked in both directions. The one who imagined making it were more likely than the control condition to make it. And the ones who had imagined missing were more likely than the control group to miss. And again, this has been shown with lots and lots of different sports.
- 7.65 Planning, obviously very powerful for producing results, even just specifically intending something. I've talked before about the implementation intentions, when you have a very specific idea of what you want to do. You translate the grand intention into a specific plan. "I should shop for Christmas presents. OK, tomorrow morning, I'll get up and I'll go to this department store and look for presents for my younger brothers and sisters or whatever." That works a lot better. So again, the conscious thought makes the behavior more likely.
- 7.66 Anticipated emotion works as well. I think I've talked about emotion, maybe in the first set of lectures. And it has a more reliable impact on behavior than the actual currently-felt emotion. So again, thinking about how you will feel or how you won't feel or how you want to feel, if you do this or if you do that, has significant effect on changing behavior. Anyway, there are a variety of others, replaying things in the past, reinterpreting.
- 7.67 There's a licensing effect. You remember yourself doing something virtuous, that makes you more willing to indulge in something not-so-virtuous in the present because you earned credits, you proved your goodness or whatever. False memories can be planted.

Well, it will avoid behavior. One study, they had people, they implanted the memory that, oh, at one time, you had food poisoning for this certain kind of food.

7.68 And then a month later, they had people back, where that food was served among others, and people wouldn't eat it. And it was made up; they had not had food poisoning from that thing, but somehow planting that false memory was enough to guide behavior. Likewise, counterfactual replay, you replay something that happened, only you said something different or did something different. Very adaptive. It's the way people learn to be better people and more successful and so on. Again, it improves performance down the road.

7.69 Reasoning. Now, one of the researchers had argued, look, unconscious thought is superior to conscious thought. You have some evidence for that, but most of the findings favor conscious thought is better. You can improve logical reasoning, such as by having people think aloud, or tell them they're going to have to explain why they came to this conclusion. Well, then they're more likely to come to the correct conclusion.

7.70 There are more things we found, but evidence is clear, in terms of the basic logic of research design. Conscious thoughts, randomly assigned or manipulated, will produce significant effects on behavior. Indeed, with studies on logical reasoning, we did a whole bunch in our lab, if you occupy the conscious mind with thinking about other things, suddenly people fail at logical reasoning. They're not able to do it without consciousness.

7.71 If you encumber the unconscious by putting a distraction in there, what you usually do is, you tell people to think about something, like a previous lover that you broke up with, and then say, "OK, try not to think about that." So you shut it out of your mind, but the unconscious is still remembering the previous relationship. But that doesn't interfere with logical reasoning at all.

7.72 And by the same token, if you stimulate the conscious mind, such as by these instructions, you're going to have to explain what you have to do, then people get better. If you stimulate the unconscious by planting the idea of logical reasoning, people don't get any more logical; they don't get any more effective.

7.73 Another important thing in terms of consciousness is other people's perspectives. This is something we do. So research on negotiation, for example. If you think of the other person's position, understand what they want and what they need, you're more able to come to a better outcome in the negotiation.

7.74 All right, well, I do want to spend a moment or two on the issue of creativity because this one has been particularly contentious. So Wegner, in his book on illusion of conscious will, he said, "Although most of us are quite willing to take credit for our good ideas, it's true that we do not experience them as voluntary. Rather, they happen to us, jumping into the mind." And so, "Losing oneself and one's will is a goal in art."

Basically, get the conscious mind out of the way. Creativity comes from the unconscious.

- 7.75** This is a new version of the more traditional argument, which creative people had a muse, some kind of a supernatural figure who would visit them and give them their creative inspirations. I don't know that these muses were actually cited, but people somehow believed, or they reflect the idea, that creativity doesn't originate in consciousness, that it relies on the unconscious to, or something outside of consciousness at least, to create it.
- 7.76** But there's a problem with this. Think about it. If consciousness doesn't really help creativity, especially if it's detrimental, why are humans the most creative species on the Earth? We're the only ones who do creative art and creative music and technology and all these things. So it seems like there must be some role for consciousness in there, even if it's true that the inspiration doesn't originate in consciousness. Maybe consciousness has to process it, in order to be something.
- 7.77** We did a number of studies like this. I used to play the guitar a lot. For one, we got musicians from the graduate students in music with guitar specialties. They're really good guitarists, making a career out of it. We brought them in and we had them jam. We recorded some chord progressions, and they had to improvise a solo over it. They were doing this for the first time, and we had them do several, one in which they just did it normally, one in which they started counting in their mind from 1, starting from 15, 16, 17, 18, and so on, and a more difficult one where they started counting backward by 6 from 917.
- 7.78** Initially, I had them counting backwards by 7, which is what I did for myself, but counting backwards by 7 was too hard for the musicians. Anyway, a lot of them even thought, "OK, the best one is going to be the one where my conscious mind is totally distracted by the backward counting." We recorded everything. We played it for judges to rate the creativity. It was clear that the creativity was less when the conscious mind was preoccupied.
- 7.79** They were able to do some things. They stayed on the beat perfectly well, so rhythm is pretty much automatic. They stayed in key. If you know music, there are 12 different notes that are repeated over and over, up and down the piano keyboard. And a key is usually a set of about 7 of them that you can use, and those ones will sound good; the others will sound wrong to varying degrees.
- 7.80** Even in each of the chord progressions, we slipped in a change of key, so that which 7 notes were right changed. They had no problem with that. Again, that was all automatic. They did that all perfectly. But the creativity, the judges said, yuck, it was mostly gone when the conscious mind was there. And my sense is that the unconscious is perfectly happy with [singing]. It's following the rules. It's making up a solo that's with the right notes and rhythm and stuff.

- 7.81** But it takes the conscious mind to say, "Wait, that's kind of boring. After about two "da-da-das," you'd better have something else and put in a variation." So conscious and unconscious work together, but consciousness plays a key role. We had several other studies with drawings and other things—again, distracting people, and had the same result. They could follow the rules, even when the conscious mind was preoccupied, and produce something, but it wasn't as creative as when their consciousness was involved.
- 7.82** So, consciousness causes behavior. We also looked for, what are the general patterns then, when conscious thoughts have this causal role? So first, there are a lot of them. So in my mind, the question is resolved. Does a conscious thought cause behavior? Yes. Scientifically solid evidence. Most of them are useful, some of them are not. I think I mentioned, like choking under pressure, and it's an advanced skill that you've learned to do automatically. Putting more consciousness into that messes it up.
- 7.83** But notice, that's trying to get consciousness really close to the direct causation of behavior. Sort of fits the idea that consciousness is most effective behind the scenes. It's several steps removed from input or output. Again, holding conscious thoughts triggers other mental processes. It's useful for translating broad ideas, abstract intentions, and whatnot into specific actions, like with the implementation intentions. "I'm going to go shopping tomorrow morning and go to the store," and so on. That's more effective.
- 7.84** The effects were indirect. Yes, consciousness causes behavior, but none showed direct guiding of muscle movements. It's much more useful for planning and strategizing and things like that. Simulating events away from the here and now, consciousness is really good for that. Again, we forget most animals live basically in the immediate present, responding to what's there in their environment at that moment; whereas we can think of, well, what will happen next Christmas or next summer, and what do I want to do, and what are other people doing that I should prepare for, to respond to as well?
- 7.85** Again, not direct control. No proof that decisions are made inherently consciously. It could be, I'm a little skeptical, but it could be that consciousness reviews all the information, and then the unconscious chooses, and then puts the decision into consciousness, so that you can act on it. Again, putting it up on the big board, so all the other parts of the brain can cooperate and respond to that.
- 7.86** And no proof that the impulses to act originate in consciousness. As with the creativity, the unconscious is tossing out ideas. It takes the conscious mind to fashion them into a beautiful solo, but it doesn't originate there. Still, be very adaptive. If the impulse is to do something, pass through consciousness so you can reflect on them. Again, put them up on the big board, so other parts can say, "Wait a minute, that's not a good idea. We tried that," or, "You'll be sorry tomorrow."

- 7.87 Again, fitting the view that conscious thoughts do feed back into the unconscious, even if behavior is always a direct result of unconscious processes. General patterns. When, what were the kinds of situations in which consciousness did have effect on behavior? First, it integrates across time. Animals respond to the immediate present, but we can act based on things even in the distant past and future.
- 7.88 It enables the culture, the social environment, to influence behavior. You get information from talking, of course. So when people tell you things, then you can use that to influence your behavior. It came up, a lot of the situations, where there were multiple possible alternatives, so making choices. Consciousness can simulate each of the options, and not just if it seems appealing in the immediate behavior, in the immediate present, but down the road. Will the downstream effects be something I can live with and be glad to have?
- 7.89 And again, there's this tradition in psychology, the conscious and unconscious are competing processes. People want to say, "Is this conscious or unconscious?", as if they're working against each other. But really, they seem to work together. I think moving forward in research on understanding consciousness and understanding the mind generally, we should focus on how conscious and unconscious processes complement each other and work together to promote behavior.
- 7.90 In fact, I'd go so far as to say, almost no human behavior is entirely the result of one or the other. Almost all of them are the result of combination of conscious and unconscious.
- 7.91 So third and final question then, how does consciousness work? One thing really important about it that became clear as we were reading all the research literature is sequences of thought. And so I came to think of it this way. You can think of conscious thought, or the conscious realm of your mind, as the place where the unconscious builds meaningful sequences of thought. The unconscious is pretty good at quick reactions, simple thoughts, and so on, but it's not so good at complex thoughts.
- 7.92 Well, I mean, one thing I struggled with for a long time is, well, what about language? We know the unconscious has language. People can prime individual words, and they will affect people's behavior, with them not realizing that they were primed. It flashed on the screen so fast, they didn't even see it, but it affected and changed their behavior. So yeah, the unconscious does have language.
- 7.93 But I talked to one of the experts on this, and he says, "Well, yeah, that's true, but only single words." The unconscious is a bit like a dog with a better vocabulary. He said there's not a single one of these priming studies that passed what he called the two-word test. If you put two words together, the unconscious is already beyond what it can do meaningfully. It can add them together, but it's just a simple combination.
- 7.94 For example, you flash unconsciously the word green. OK, green is pretty good. It's a nice color in the environment, so on. Then bread. OK, well, bread is good. That's food

and so on. So green bread is like two good things to the unconscious. It takes the conscious mind to say, "I don't want to eat any green bread. There's something wrong with that." Or on the negative side, somebody used the example of dead enemy. Being dead is bad, the enemy is bad, so that's two bad things. But wait a minute, the enemy is dead. No, that should be a good thing.

- 7.95 So it takes consciousness to put those together to the more integrative meaning. The unconscious can use words, but it needs to put things into consciousness to do sentences. You can communicate an infinitely greater amount of information with sentences than with words, let alone books and articles and speeches and paragraphs and all these other larger chunks.
- 7.96 Logical reasoning, as I said, the conscious mind is needed to do that. That's a sequence of steps. I notice this goes against what the Nesbitt and Wilson point, that when we explain why we do something, we explain our reasons, we don't really know. We just sort of make something up. Well, logical reason, you can inspect those. You can explain exactly why you think this falls from that, and somebody else can look at that and say, "Well, no, there's a mistake in your reasoning. This could be otherwise." So, that's open to conscious inspection and debate, again, though, requiring consciousness.
- 7.97 Counting numbers and so on, that generally requires the conscious mind to do it. You don't really do unconscious calculations, not very much. Causal understanding, what causes something else, that appears to require causal influence. Very simple things perhaps the unconscious can pick up, but a real analysis. All these things are objectively valid, but are learned from the social environment. Like math, it's really true, but we learned it from groups. Only groups can get certain kinds of knowledge. It requires to be accumulated collectively and then shared through the group.
- 7.98 That leads to two more things, that important sequence thoughts. One is stories. You don't have stories in the unconscious, but stories are a very important form of human communication. Stories have to follow the other things. They're usually told in sentences and paragraphs. Stories have to be logical, otherwise they don't make sense. If there are time or other quantitative limits, the numbers have to add up. I can't do something that takes five minutes, and somehow the story is three hours long or vice versa.
- 7.99 And stories are causal. This leads to this, and that leads to that, and so forth. So stories build on all these features, again requiring consciousness. Likewise, simulations, mentally imagining things. It turns out, even our view of the present is the simulation, our experience of the immediate surroundings. One example of that is, we know there's a blind spot in the middle of the visual field, and yet you don't see it. There's a place where there's no information coming in. If you move your finger back and forth, it sort of disappears at one point.

- 7.100** But it's not just that there's a small correction that is made. The implication is, your whole construction of the visual field is put together by the unconscious. There's wonderful research showing these kind of things. Sensory information, light waves hit the eye. The optical nerve transmits that to the brain, but then the information is split apart and processed in different places. What it is and where it is is processed separately, and then it's combined again, or various features, shape and color.
- 7.101** People make these mistakes sometimes. If you see red triangles and blue circles, and you see a lot of them coming through, your mind may remember that there were some blue triangles or red circles, that will make some mistakes like that, showing again that it was processed separately and combined. The mind is putting all this stuff—the unconscious, basically, is doing a lot of work to create the conscious experience.
- 7.102** Again, I'm not all that impressed by the Libet studies, that the brain has to be doing some work before you have a conscious thought. Essentially, in terms of the brain making a movie for itself, it's a useful way to think of our consciousness. It's true with the present, but it's easier to understand with thinking of future or past things. We can replay events that occurred to us last week or last year or even long ago, or think about future things, make things up.
- 7.103** We can replay things that happened counterfactually, imagining what would have happened if I only had said what was in my heart. It's like a movie. It's not a static image, but more making a movie. There are two ways of making a movie, too. I don't know how important this is, but I think it's interesting. There are two styles, animation and camera, at least traditionally; now things are done electronically.
- 7.104** The consciousness is more like animation, and the difference being, with the old film, each frame in the film is what the camera saw at that moment. Whereas with animation, each frame starts with the previous frame copied, and it just makes any small adjustments to it. And it seems that consciousness is much more like the latter.
- 7.105** So you see your friend Ben walk into the room, it's not at each moment that the unconscious is saying, "That's Ben, that's Ben, that's Ben." So he comes into the room, and it pegs him, "OK, that's Ben." And then it just leaves that and updates that. And there might be, at some point you think, "Wait a minute, that's not Ben. That's somebody who looks like Ben." So you can make a change; but you figure it out and leave it, and you only gradually update the image that's coming in.
- 7.106** Consciousness does one thing at a time, so it's not parallel processing. The unconscious can do multiple things at once. The conscious mind can just do one thing at a time, and so sequencing is important as well. So, these simulated sequences of thought are useful in lots of different contexts, useful for understanding other people. We imagine, "What is she feeling? What made her say that? What is going on in so-and-so's life?"
- 7.107** Or extracting knowledge from past events, replaying them, and "Why did this happen? And what would have happened if we had responded differently?" Decision-making,

you can imagine this will lead to this and lead to this. And, "Oh, I don't like that. Well, how about if I say this and do this?" It will lead to a better outcome. Understanding yourself. Self-control in particular is bolstered by imagining you'll be sorry if you relapse and go back to smoking cigarettes down the road. You'll regret it, you might get sick, or your parents, or your friends, or your spouse will complain and so forth.

- 7.108** Automatization, you work out how to do things, and gradually the unconscious takes over to do it more automatically. Anyway, so, to conclude, consciousness is one of the remarkable powers of the human mind. I would speculate that all human behavior is a product of combination of conscious or unconscious behavior. There might be a few things that are purely unconscious. If you want to include things like your body keeping a stable temperature, OK, you're not conscious of how that happens.
- 7.109** But anything you actually do, and certainly anything you do in the social-cultural environment, is a product of both conscious and unconscious processes working together. In terms of the three questions I started with, the how, what, and why, how it works, it constructs these simulations. It constructs these sequences of thought, that the unconscious again, very powerful in terms of doing a lot of simple things, but integrating into more complicated sequences. You need consciousness for that.
- 7.110** What it does is simulate action sequences, simulate events away from the here and now, other people's mental states and feelings, things like that. Talking, as I said, extremely important. You can't talk without being conscious of what you're talking about. And that indeed might be enough to explain why it evolved in the first place and why it helps you relate better to the social environment and culture. So again, it's a key part of the human mind, something we can't dismiss. Thank you.
- 7.111** Can consciousness be related to ego depletion and willpower, in the sense that you use your conscious thought to make your decisions until you get depleted and slips into unconscious thought, but you don't exactly then have an unconscious experience? You're still consciously experiencing.
- 7.112** Yes, you're certainly still having conscious experiences. So willpower is about changing your behavior. You are conscious in either case, and it often takes effort to implement something, especially if it goes against your unconscious impulse or your unconscious tendency to react a certain way. What happens when you're depleted is, the energy to do the override is less there. And so the unconscious becomes more influential and more decisive.
- 7.113** But you are still conscious. It doesn't eliminate it. It just somehow reduces the top-down conscious control of behavior. It makes it somehow weaker, so that you let the automatic and unconscious reactions occur more. I've even had people propose that you should be less likely to choke under pressure if you are depleted, since that's consciousness intruding into the automatic skill. I haven't seen anybody quite get that

to work. Some said they had preliminary data supporting it. I haven't seen that of that. It is an intriguing and plausible idea.

- 7.114** Because of our understanding that the unconscious is such a powerful force in causing our action, is there any way to leverage the subconscious towards goals and aspirations that we might have for ourselves?
- 7.115** Well, that's what the implementation intentions do. It tells the unconscious, "This is the goal, this should get priority." So that's certainly one. Impulses to behave mostly originate in the unconscious. We evolved from animals who behaved all the time without our advanced consciousness. The human conscious thought capacity is a new advance on top of the others, to work with the unconscious to improve behavior. But it does go back and forth.
- 7.116** Even the moral reasoning thing. Jonathan Haidt, in his career, he rebelled against the tradition that people engage in moral reasoning whenever they have a moral dilemma. He said, "No, they mostly do what feels right." They have an intuition that this is wrong and they shouldn't do it, even if they can't explain why. But then when things go badly, in particular, they will engage in moral reasoning and think about it.
- 7.117** And so, the next time they're in a similar situation, they might have a different intuition. So the unconscious will remember, "Yeah, that turned out badly the last time." At least it will give a twinge of unpleasant anticipatory guilt. And in that way, the consciousness shapes the unconscious to behave better in the future.

Lecture 8: Sexual Exchange

At a glance — Sexual exchange.

- 8.1 All right. For this lecture, we'll take a fun and different way of approaching analyzing human sexual behavior. As I've said before, sex is an important aspect of social psychology. It's a way that people interact with others and obviously has some value for perpetuating the species and other things. So let's take an economic analysis of it.
- 8.2 In general, when people have thought about sex, theories of sex, we've borrowed mainly from two other fields. One is biology. The evolutionists say patterns of sexual desire are based on what our ancestors produced the most offspring. So that's one, and there's certainly something to be gained from that. The other is politics, particularly the feminist approaches, understood sex in terms of intergroup relations and oppression and other aspects as well.
- 8.3 So we've gotten lots from both of those analyses, but economics might have something to offer, too. Now, I have to caution, economics is a decidedly unromantic discipline. Those of us who have a romantic streak don't really necessarily like this kind of theory very much. First time I heard anything like it, I had a real negative reaction. But you want to be a generalist, as I said, and get to the truth, you've got to accept some things that you don't really like.
- 8.4 That comment was in a, actually, a feminist historian, Nancy Cott, talking about why women had such low sexual desire during the Victorian period, and they were famously prudish. And the way she put it, well, it had gotten to where that was the main thing women had to offer. As you know, the Industrial Revolution started by replacing most of women's work.
- 8.5 So women had been essential for making clothes, but the textile industry was the beginning of the Industrial Revolution; and all those hours women spent spinning and weaving and all those things, suddenly that wasn't needed anymore. And canning and storing food was another one. So it got to a while where women were economically not having any function. So Cott said, well, sex was the woman's main ticket to the good life, so she needed to have a high price. And so, restricting the supply was a classic tactic for raising the price.
- 8.6 It made intuitive sense. But as I said, I didn't like that at all. But after I started looking into the evidence later, and a couple of us started reviewing the research on sex, it was just clear that this analysis really does have a lot to offer. So, psychology has used what's called social exchange theory to analyze certain kinds of interactions, basically looking at what the costs and benefits of particular behaviors are. It's one way of analyzing people's behavior.

- 8.7** Certainly with sex, it's been applied to sex. So there are benefits in terms of pleasure and maybe bonding, forming a relationship. There are costs. Traditionally, the risk of unwanted pregnancy was a huge negative that had to be considered and regulated. Disease, another possible one, reputational damage.
- 8.8** So, there have been social psychologists who used exchange theory to apply to sex, but I think they missed the key point, which was, if you saw the opening slide with the quotations from Don Symons and Andrea Dworkin, that's something that men want from women. So sex functions like a resource that may be exchangeable, but the men want it from the woman more than the reverse.
- 8.9** Social exchange theory does have something they call the principle of least interest, which is, if two people are thinking of doing something together, the ones who want it more is at a disadvantage. There's a power advantage to the one who's less interested. So if you and your buddy or whatever are thinking of going to a movie and you really want to go, and they're just sort of so-so, often you have to sweeten the deal. They're, "Well, I'll drive and I'll buy the popcorn," or something like that.
- 8.10** Again, wanting something less confers an advantage. As we saw in another lecture, evidence is pretty overwhelming that men are hornier than women, when men want sex more than women do. And so even if women want it, too, as long as the man wants it more, he's at a disadvantage and needing to do something to persuade her. This theory builds on both nature and culture, the nature part, the need for resources is key.
- 8.11** Of course, sex itself is a vital aspect of reproduction and evolution and everything else, but the cultural aspect is that there's a marketplace in a community in which these transactions happen. Think of it as an individual man and woman, deciding whether to have sex or not. But it's usually in a social context where there are expectations as to what level of commitment or relationship or what else should be happening, in order for, it's OK for them to do various kinds of sexual behavior.
- 8.12** And one more aspect of the context, in most societies, in the history of the world, the material resources were mainly created and controlled by men, and so women need access to them. So, it sets up the basis for a marketplace. The man wants sex from women. He has a lot of other things that she wants. She has something that he wants.
- 8.13** Furthermore, key point, sex is basically a female resource. Sorry, guys, male sexuality has no exchange value. I mean, if you're a young man with a Mafia debt or whatever and hoping to pay it off by getting women to pay you to have sex with them, you'd better have plan B. You're not going to get very far with that. But women in that situation, they can get paid for sex. So, men give women resources for sex. It's not just money; it can be all kinds of other things. It can be love, commitment, respect, attention. It can be money.
- 8.14** Sex can be traded for opportunities, freedom from punishment, all sorts of other things. And the local sexual marketplace interlinks the sexual decisions of many

different couples. So if a particular woman is saying, "Well, before I go all the way, I want to have at least an engagement ring," but the other women in the community are going to bed on the first date, after a medium-quality restaurant meal or something, she's going to have a hard time getting what she wants for it.

- 8.15** But if all the other women are also holding out for marriage, then she has a better chance. This is a side implication. People are always interested in other people's sex lives, and there's a lot of talk on television and elsewhere, and why is it? Is it just that they're so fascinated by sex? But this gives a reason. People want to know what is the appropriate amount? What it should, the price of sex be? How much should the man do for the woman, in order for sex to happen?
- 8.16** Sexual interactions are not equal. We like to think of it as equal, two people in bed together and so on, but essentially she is giving it and he is receiving it. Or in economic terms, women are the supply and men are the demand. And so we can see economic operations of supply and demand processes, when they are out of kilter, when there's more supply than demand or vice versa.
- 8.17** So, I mean, sex will have a local market price, not necessarily in money, but again often in commitment. How far along should they be? Because of the danger of unwanted pregnancy, in many traditional societies, sex was not permitted until marriage, although I read a number of historians who spent time counting how much time elapsed between the wedding and the birth of the first child and so on. And they inferred that, in many, it was OK to start having sex when you were engaged. Once the commitment was there, the first baby was often born in less than nine months after the wedding day.
- 8.18** But still, something like that, the norm, and that was understood by everyone and not necessarily followed by everyone. But if you departed from the norm, you knew it and you were taking a chance there. The price, how much the man gives the woman to commence a sexual relationship, will fluctuate with supply and demand and other factors and local norms and so forth. Basically, a higher price favors women, a lower price favors men. But they can find a mutually satisfactory one.
- 8.19** The last point I got from a primer on economics by Gary Becker, who won a Nobel Prize in there, that the sellers tend to work together more than buyers, which would say that women would probably work together to cooperate in regulating sex norms more than men will. I was at a conference with economists who said, well, he was a brilliant guy, but that was a while ago, and things have become more complicated, as is usually the case in all sciences and social sciences.
- 8.20** So, they don't necessarily believe that anymore. I'm not expert enough to say, but it does seem to apply in the field of sex. As I said, I went to this economics conference and talked to a bunch of them, and they gave me a lot of interesting ideas to update

and upgrade the theory. So the man is offering resources, but there's some ambiguity as to exactly what it is that the woman offers. There are two possible answers.

- 8.21** She's offering him lots of great sex, or she's offering him exclusivity, just that she's not going to have sex with anyone else. How much she has with him is not necessarily part of the deal. So this can lead to marital misunderstandings. The paper by McNulty et al, and I was involved in this, too, a couple of years ago, tracking married couples over the first five years of marriage to see what happens. Now, everybody knows the frequency of sex does go down over the early years. The newlyweds are doing it quite regularly, and that drops.
- 8.22** But does the desire change? That's not nearly as well known. But these guys, they had a couple studies where they had tracked people over five years. The basic pattern was, five years into the marriage, the husband's sexual desire is about the same as it was when he walked down the aisle. There's really no change. But on average, the woman's sexual desire has dropped substantially. And so, any decline in it is more attributed to her changes.
- 8.23** And it's important, particularly because marital satisfaction goes down for both of them, in step with her loss of sexual desire. And I like people to understand this because this damages marital satisfaction, because they're probably inclined to blame each other. "What's wrong with you? You used to love this. And why are you bothering me all the time? I don't feel like it today. I have other duties."
- 8.24** And so, they blame each other for it, when they should understand, it's not a sign that there's a problem in the marriage. It's just a typical problem that occurs generally, and it's a problem you should solve as a married couple. If people could understand this and anticipate it and plan for it, then it might not damage marriages as much as it often does.
- 8.25** Anyway, translating that in terms of sexual economics, you could understand it this way. Who owes what to whom, when the wife doesn't want to do it? Man thinks he's going to have this great sex every day for the rest of his life, and then after a year or two, she doesn't really want to do it that often anymore. So he thinks she's going back on the deal.
- 8.26** But she may think, "Well, I said I just wouldn't have sex with anyone else, so I'm keeping up my bargain. And when we don't have sex, you're looking at pornography and other women. Maybe you're not keeping your end of the deal. You're supposed to be exclusive to me." So they both may feel betrayed by the other, even just because they have a different understanding of what exactly the deal was that made satisfaction.
- 8.27** She said, "I never—nothing in the wedding vows promised that I'd go to bed with you four times a week or anything like that." So, conflict may arise because of that, and hence, damage the marital satisfaction. Let's go start looking at some of the research

findings that really persuaded me to come around and think this economic analysis really had something to offer in terms of understanding how people negotiate the start of a sexual relationship.

- 8.28** The first and most obviously, paid sex, prostitution, sex work. It's extremely asymmetrical. Women don't pay for sex, and why not? Well, they don't have to. A woman can get sex. There was a famous study, actually replicated a couple times at Florida State, where the research assistants went out on the campus and just approached attractive people, someone out of the blue, and said, among other things, that, "Would you like to go to bed with me tonight? I've been noticing you around campus. I think you're really attractive."
- 8.29** The men usually said yes. Three-quarters of them in both studies said, "Sure, I'll go to bed with you tonight." Even the ones who declined said, "I'm sorry, I've got to do something tonight, but how about tomorrow?" Or, "I can't, I just got a girlfriend," or whatever. The women said no. Not a single woman said yes in either of the studies; they ran it a couple of times.
- 8.30** Basically, a woman who wants to have sex can get it. She can go out and ask, basically ask two or three people; you're almost sure to get a yes, whereas a man can talk till he's blue in the face, and that's not really the way it happens. In terms of paying for sex, empirically clear, there's a widespread asymmetry. Men paid women for sex all over the world throughout history and so on. Women hardly ever doing the reverse.
- 8.31** The research has found a couple cases, but even when that happens, it has to be camouflaged. There are these studies of so-called beach boys in the Caribbean. So, like, middle-aged Canadian women will go down to the Caribbean for their vacation. Then, these are usually black men in their 20s will come and hang around them and talk to them and form friendship and pay attention.
- 8.32** And then they start having sex, but she doesn't pay them for it. If a Canadian man goes down there and has sex with a local woman, it's, put the cash right on the dresser before you get into bed. But in this case, at one point to say, "Oh, I'm bad. My father is really sick. He needs an operation. I don't have enough money." She says, "Oh, let me help you." And she gives him something that way. So it's camouflaged, even though arguably it is basically just money for sex. So that's one sign, obviously, of transfer of resources for sex, and it definitely goes mainly one way.
- 8.33** Infidelity and divorce. In a lot of cultures, if you're married, if either party has sex with someone else, that's grounds for divorce. But when it's only one, it's almost always, if the woman has sex with someone else, the man can divorce her, but the man has sex with someone else, no, she can't divorce him just because of that. In many cultures, that's not there.
- 8.34** Interestingly, providing goes the other way. A woman can divorce her husband because he's not providing for her, he's not earning enough money, not taking care of her.

That's a legitimate complaint in many societies, whereas a man can't divorce his woman because she doesn't produce or bring home enough money. Again, it seems to be that the assumption in these unequal cultures is the man's providing money and resources, the woman's providing sex.

- 8.35** There's some interesting studies. I've seen two different ones that they seemed unaware of each other because the later one didn't mention the earlier one. But the question of, what would bother you more— if your spouse has sex homosexually or heterosexually with someone else? And so they asked both men and women. "Suppose your partner had sex with a man, suppose your partner had sex with a woman."
- 8.36** Only, it didn't break down as homosexual or heterosexual. It broke down that both parties, both men and women, are more bothered if their partner has sex with a man than a woman. And exactly why this is, is not entirely clear, but it certainly fits the idea that the woman who has sex with somebody, well, the man is a taker and the woman is a giver, in terms of sex. If your partner, whether it's a man or a woman, has sex with a man, well, the man is taking away something from us, whereas the woman is giving something, and nobody likes it in either case, but it's more bothersome if the interloper is a male.
- 8.37** Then there are studies how people steal other people's mates. You want, somebody else has a husband or a wife or boyfriend, girlfriend, and you want to have it, how can you do it? How can you lure that person away from their current partner and make them your partner? Offering to have sex is one strategy. That only works one side. Women offering to have sex with a man, that often works. Man having sex, offering to have sex with somebody else's wife, will generally get you nowhere. It's not an effective way to start the process of trying to poach, as they call it, to steal away somebody else's romantic partner.
- 8.38** Courtship. Generally, the man pays for dates. That's my subtitle and so on. In modern life, even when women earn as much or more than men, they still expect the man to pay most of the time. Sometimes the woman will pay for the third date or something like that and show it. But there's a general pattern that it's an expectation for the men to do it. I remember Maureen Dowd's book, *Are Men Necessary?*, she interviewed a lot of women, and she said, "Well, when you earn your salary, too, why don't you pay for the dates? Why do you wait for the men to pay for it?"
- 8.39** But they're all, "Oh, you can't mess with the natural order of dating. That's nature," which is an oddly hypocritical thing, because the feminists are mostly against nature explanations. But, oh, when it's in their favor, that's natural. You don't want to mess with that or change that. Females condemn other women who give away sex too easily, who do it without love or basically demand a lower price. They are equivalent to the rate-busters in the factories or something. They're affecting the community.

- 8.40** I mean, why should you care if another woman has sex with a man on the second date or something, if you still want to have an engagement ring before you do it? Well, again, that's changing the marketplace and the going rate. And so, if everybody else does it, it weakens your chance to get a higher price for yourself.
- 8.41** It's clear women control the onset of sex. There are studies looking at, when do you think it's appropriate, how many dates into a relationship before you have sex? And men had certain responses and women had certain responses. Then they ask, "Well, in your current relationship, how many was it?" It's almost exactly whatever the woman thought was appropriate. The man's preferences or ideas weren't even significantly correlated with when they have sex. So it's really up to the woman.
- 8.42** Now, some of these things have changed. There are certain modern patterns where people just have sex right away. There's an intriguing book applying sexual economics by Mark Regnerus, called *Cheap Sex*, talking about how, in the modern social environment, the price of sex has gone so down that sometimes people have sex—one sign of it is that researchers asked people, "OK, in your current relationship, how many dates did you have or how long were you together before you started having sex?"
- 8.43** And now there's a sizable majority of people who say, "Well, we had sex before there was a relationship. We went to bed just as a hookup and then decided we liked each other, and so that led to it." So that, in some ways, is changing the dynamic. But at least in the traditional, it was usually, the relationship had to be somewhat established for sex to begin, and it was up to the woman to decide. It's partly because the man was usually ready earlier, but he just waited until she was there.
- 8.44** In terms of men and women's complaints about each other, women complain about the men they had sex with who didn't treat them properly, and so on. Men complain about the women that they didn't have sex with. In particular, it's men's complaints about women are, "Oh, she got me to spend all this money on her and take her out to all these things, but then she wouldn't have sex with me." The woman's complaint is, "Well, I went to bed with him because he said he loved me and so on, and then he ran off. He didn't really love me. He just faked it, so he could get into bed."
- 8.45** Virginity. Traditionally, it's regarded as a gift, as a very positive thing that the woman saves for someone special; whereas the man, if he's a virgin, it's almost as a negative thing. It's like, what's the matter? Couldn't you get any woman to have sex with you? So being a virgin is a negative thing.
- 8.46** Shotgun marriage is an interesting phenomenon. There's an excellent paper by Akerlof, an economist who got a Nobel Prize for economics. One of his themes is, when there are big social changes, there are unintended consequences, winners and losers. He said, well, one of the big paradoxes in modern society, starting in the '50s, '60s, '70s, is that effective birth control became better than it ever was before. People have been trying to produce birth control for centuries. Women used to put camel dung inside

their vaginas and all sorts of things that supposedly work. None of that worked very well.

- 8.47** But now we have IUDs and condoms and pills and all these things. It's really possible to prevent unwanted pregnancies. So what happened? Have unwanted pregnancies disappeared? No. We have more than we ever had before. And so he said, well, this is a paradox. Why would this happen? But as he said, well, each change in society brings winners and losers.
- 8.48** When the machines started taking over the textile industry, it made a lot more benefits for many people who made the machines and who got the cheaper clothes and so on. But the people whose job was weaving clothes, they lost their livelihood and so on. So he said now it is possible to have sex before marriage without getting pregnant, and that becomes normal. But there are some women who can't use, who can't take the pill, either for religious reasons or for medical reasons, that it doesn't agree with them in some way.
- 8.49** But the problem is, these women have to compete in the same sexual marketplace with the women who are taking the pill. And so it becomes expected that you'll have sex before marriage. And so the woman has to go along and do this because that's common. Otherwise, you might lose the man who'd go to somebody else who is able and willing to have sex. One side effect of this is, you'll have more women taking chances, having unprotected sex, and getting pregnant.
- 8.50** In the past, this was enforced with the shotgun marriage. If a couple was having sex, the community would gather around, as in this image, to pressure the young man to make an honest woman of her, as they said, to get married, and in that way, prevent out-of-wedlock births, but that's no longer the norm. Society doesn't do that anymore, and it's considered to be her problem and her failure that she got pregnant, and he can decide whether it's his responsibility or not.
- 8.51** How do people mate? Well, unequal status is well established. Sociologists have what they call the marriage gradient, which, the average marriage, the man is a couple years older than the woman, has higher salary than the woman, maybe slightly higher social status, maybe half a degree more education. This last thing is going to be an intriguing thing in the future because, of course, now women are more university students than men. So there won't be enough high-status men for all of them to marry.
- 8.52** The men who do make it through and get a PhD, they'll probably be like studs on a farm or something. They'll have a large selection of female partners. But a lot of the women with PhDs are going to have to marry a man, if they're going to marry at all, with a lesser degree. But anyway, it has been traditional, the man has more status and resources than the woman, which again suggests that, for sex to start, he has to bring something else to the table, because she's contributing sex, and that has value. He's contributing sex, but it doesn't have any value, so he has to do other stuff.

- 8.53** It's shown also with dating. People tend to date by that pattern. Even short-term sexual encounters happen more in that context. An extreme example is groupies, and that's a really interesting one. I remember HBO, I think, had a special some years ago on athletes and sexiness, and they tried really hard to be sensitive and make it look like it's a totally different thing for the men and for the women. But it looked to me like it's the same thing.
- 8.54** Athletes have great bodies; lots of fans admire them and want to have sex with them. Only for the men, they love this, the male athletes; they love having women throw themselves at them and so on. For the women, they hated it. They wanted to be, "I'm just here to play basketball. You shouldn't be looking at my legs." Anyway, the point, though, is that both genders fantasize about having sex with celebrities, movie stars, athletes, and so on. But only women can do it.
- 8.55** And why is that? Well, again, female sex has value; the male sex doesn't. A nobody-woman can approach a celebrity and offer to go to bed with him, and in many cases, he'll say, "Sure, that it sounds like fun." A man in the same position approaches a famous woman and offers to have sex with her, she's going to call security and get him thrown out.
- 8.56** They said there's a joke in the National Basketball Association about what's the hardest thing about a road game in the NBA? Well, it's trying not to smile when you kiss your wife goodbye before you get on the airplane, because you're going to go to another city, and there are going to be women available to have sex with you. Anyway, again, fitting the economic analysis that female sexuality has value for exchange, that a low-status woman can give something to a high-status man, but not the reverse.
- 8.57** There's even evidence, I did some reading about sadomasochistic kinky sex stuff. And one reliable finding is do— so once one person ties up and dominates the other with some or whatever, and then there's a question, do they have genital intercourse? And if it's a male dominant and a female submissive, a lot of times they do. That's very typical. If it's a woman dominant and a male submissive, no, they don't usually do that, because they feel like he would be possessing her or something like that. So he just does oral sex to please her, but that doesn't usually translate into intercourse.
- 8.58** Again, all these reflect the asymmetry. Now, here's a really important set of evidence. Supply and demand should really come into play, as I said, when the supply exceeds the demand, the price goes down. If demand exceeds the supply, the price goes up. And certainly, there are places where the sex ratio is unequal. Often, same numbers of men and women are born, so you have the same number of adults, but they can be way out of whack.
- 8.59** For example, after a major war where a lot of the men have been killed, then there's a surplus of women. And by all accounts, sex becomes free and easy. A lot of premarital, extramarital sex and so on. Also in some low-income communities in the US today,

where there's a shortage of men, either because a lot of the men are in prison, or the men are marrying other groups and the women only marry within their groups, there's a shortage of men. Again, the woman can't really ask very much for the men; the price of sex becomes very low.

- 8.60** At other times, there's a surplus of men. The most famous recent one was in China, which had a one-child policy, which led to lots of couples finding out their baby in advance and deciding to have an abortion, particularly if it was a female. I thought this was, they must not like women or daughters or something like that. But it turns out, in Chinese law, they didn't have much of an old age pension. And in Chinese law, the parents can force their son to support them legally and take him to court, but they can't do this to their daughter. So, well, you want a son.
- 8.61** And so, it's something like 300 million more men than women, which is a huge asymmetry. Again, shortage of women. In that case, the demand far exceeds the supply. What happens? The price goes way up. And so, there's not much premarital or extramarital sex. The man basically has to sign his wife away before he gets into bed with her. Another place with a high sex ratio of men to women was on the Wild West. In the 1800s, it was "Go west, young man." So lots of men went out west to seek their fortune.
- 8.62** There were places where there were 100 men for every woman. So some of the women were prostitutes, to profit and make big money off it that way. But to get married, the woman could have her pick. The man, if he lost her, he was not going to get another one, so you had to give her whatever she wanted. And so, the price of sex fluctuates with the sex ratio.
- 8.63** This is shown in a classic book by Paul Secord, and it's called Too Many Women?. And they looked at sex ratios, unequal sex ratios all over the world, and found this pattern. They talk about it in sexual economics terms. They said that the norms clearly change, depending on which source. And notice it's not political, it's majority rule—it's not majority rule, it's minority rule, which is economic because you want to be, if you're the supply, you want the supply to be the minority and you want a huge demand. Then you're better off. So it's economic, rather than political causality.
- 8.64** Interesting studies of other things, one that really struck me, as they looked at rates of teen pregnancy. I started out in math, and you always look at the limiting case. So when there are no men, the woman can't get pregnant, right? And so, you'd think the more men around, since that's what causes the teenage girl to get pregnant, the more men around, the higher the rates of teen pregnancy. But it's the opposite. It's the shortage of men, when there aren't that many, that the rate of teen pregnancy goes up.
- 8.65** Why? Again, the women are competing for the men. They have to lower their price and be more willing to have sex. Otherwise, they'll move on to someone else. And this guy, I

think his name is Barber, looked at these things. Skirt lengths also change in sync with the sex ratio. The woman has to advertise more aggressively to compete for male favor.

- 8.66** Turning to rape, as Dworkin said, that's basically, the man can steal sex from the woman when he wants to, or take it illicitly. And again, gender asymmetry, it's not as extreme as you'd think from the media and all that. Some women do coerce men or pressure men into sex when they don't really want it, but it's still much more common to the extent it happens mainly, men forcing women or pressuring women into having sex when they don't want it to.
- 8.67** They often rationalize it as that, well, she owed it to him. He had done this for her or whatever. Or he was just such a great person in his own eyes that—rapists tend to be quite narcissistic. "I deserved it, and she had an obligation, and she didn't want to do it." That's the way they talk about it. It's kind of absurd and offensive, but that is the way they think. And again, it does seem to reflect this economic assumption that he wanted sex from her, and he had done enough, that she owed it to him.
- 8.68** Indeed, statutory rape laws are very asymmetrical in both laws. And even if they're written in egalitarian fashion, the enforcement is one-sided. Friedman has this lovely book, *History of Law in America*. If you're interested in that, I highly recommend it. But one of the lines from it is, when two teenagers have sex, the boy is by definition a rapist, the girl by definition a victim, even if they're both completely willing or even eager to do the deed, even if she started it. So again, it's not an equal thing. Even though they're both, say, 16 years old, the law says he has committed a crime and she is the victim of it.
- 8.69** A couple more things, some social exchange theory. What happens when you have more and more partners? Well, the value of a reward diminishes the more widely it's given. That's also, people who just looked at all social behavior in exchange terms talk about this. So praise, for example. Some people who just go around complimenting everybody, and it's nice, people like them, but it's not much of a compliment.
- 8.70** In contrast, I sometimes think of my own dissertation advisor. His theory of education was, if the criticism was really careful and thoughtful and thorough and detailed, and did I say thorough, then he didn't have to bother with praise. I think I remember all four positive things he ever said to me, but they had a huge impact because he didn't really praise anybody. He was just a critic. So the same reward, it does devalue by being given and used a lot. And there's some of that with sex.
- 8.71** So a woman has sex with lots and lots of guys. It's not considered as valuable, particularly a woman who gives you her virginity, as you're the first one. Wow, that's an implicit, very strong compliment, that she said no to everyone else up till now, but to you she says yes— that's a big testimony to something that's good about you. And in economic terms, it creates something of a dilemma for the female. To get any benefit from sex, in terms of getting resources, she has to have sex. But if she has too much,

then it reduces the value of her sexual favors, and then it might reduce what she can get going forward, so managing that.

- 8.72** There are these interesting paradoxes in the sex literature. Even when there are equal numbers of men and women, they ask you, "How many partners have you had?" Men always say they've had more than women, which is somewhat implausible. I mean, each time a new coupling between a man and a woman, that ups the number for both, but women count more restrictively. So men tend to round up, estimate and round up, and women count each one and forget some.
- 8.73** And then they don't count certain things, like "it doesn't really count as sex unless it was full genital intercourse, and we both had an orgasm." Whereas for the man, oral sex counts and other things. Anyway, there's a variety of things, but the man wants to think he has more, and the woman wants to think she has less and be able to report a smaller number, which again conserves more the value of her sexual favors.
- 8.74** Likewise, female virginity traditionally has had high value. In some cultures, they even inspect the bride to make sure she still has an intact hymen. Male virginity, well, there's no hymen anyway, but nobody cares, and it's not considered a particularly desirable trait. Indeed, for the male, having more partners indicates that you're wealthy or you're attractive or you're very persuasive, so are reflecting good things.
- 8.75** What about violent relationships? Well, here is a surprise. I read the literature, and then later another paper came to the conclusion. I mean, you would assume, if they're fighting each other or one is hitting the other or whatever, that it's not going very well. It's a bad relationship. And so sex, you'd think, would go with a good relationship. So you would assume less sex in violent relationships. But no, they actually have more frequent sex in the violent relationships.
- 8.76** But when you look closely, it's one-sided. If the man is violent and the woman is not, then they have more frequent sex. If the woman is violent and the man is not—and that's actually equally, if not more, frequent. The media always try to portray it as, domestic violence is a bad male thing, but the best data and review articles show it's equal, and if anything, slightly more common for women to do it, although the woman gets hurt more, because the man is more built for fighting and things like that.
- 8.77** But if the woman is violent and the man is not, no, they don't have increased sex. And so, why is it, that asymmetry? Well, it fits again the sexual economic aspect. If she sees him getting mad, she can give him sex to persuade him to forget his anger and have a good time. So, offering him sex is a way of basically buying off the aggression. For the man to offer sex with her, no, that has no exchange value; and if she's getting mad, he can't dissuade her by trying to get her into bed.
- 8.78** It's somewhat of a perilous—it's not a conclusive finding, but I moved to Australia. I read some histories of it. And initially, it was a penal colony, where a few guards were

keeping control of large number of prisoners, so the discipline was fairly severe. If you did something wrong, they would give you a whipping.

- 8.79** One of the practices that it developed is, if it was a woman who had to be whipped, if she would agree to take the whipping naked, then they would cut the penalty in half. So she only had to take half as many whippings. And for the man, there was no such offer. And you can have whatever moral spin on that that you want to, but it again indicates the woman has something sexual to offer that has value, and the man doesn't.
- 8.80** There's also some evidence about initiations into youth gangs. Now, most gangs are all one gender, all male or all female, but some of them have both. In those, typically in a lot of gangs, when you're introduced into the gang, you get beat up. There's a ritual. Everybody gets to beat you up. In the youth gangs, it was the same thing, except if it was a girl joining the gang, she could, instead of getting beaten up, she could offer just to have sex with everybody in the gang who wanted to. And again, this was not an option for the boys joining the gang; they had to take a beating. That was that.
- 8.81** All right, a couple more things, prostitution and pornography. Well, how do we understand these, translated into economic terms? These are low-cost alternatives sources of satisfaction for men. So, in an economic marketplace of any sort of good, if the price of something is high, somebody will come along with a cheap alternative version, and some people will be content with that.
- 8.82** And indeed, we're seeing today, as many young men are withdrawing from the romantic scene and contending themselves with pornography. Pornography has gotten better, and some say women have gotten more difficult. But anyway, it is a standard market response; and data very consistently show that women are much more opposed to these things than men are. Now, you can ask, why should a woman care if another man in her town watches pornography and masturbates? I mean, what does that affect her?
- 8.83** But arguably it does, because it's reducing the price of sex a little bit. It's slaking the demand in the community. If lots of men are doing it, then the demand for her sexual favors may be reduced. It does make sense for women to be more earnest to suppress these things, to keep men in the position of highly demanding and highly wanting sex with women.
- 8.84** That's not what they say when they oppose pornography. They say it's degrading to women; and yet, why is it degrading to women? It's two people having sex. It couldn't be depicted much more equally. There's nothing negative. The women in the pornography are doing the same thing and enjoying it and so on. The same with the man. In fact, I talked to a guy who worked in the porn industry for a while, and he said the women are paid three or four times as much as the men.

- 8.85** So if anything, it should be flattering to women. They should be rejoicing. It's one of the fields where, basically, doing the same thing, the woman gets a much higher reward for. But the women are against it. And again, it does give a message of low-cost sex. Pornography doesn't usually show the men making a long-term commitment or being in love or anything like that. It's just to show a character wake up and say, "Oh, well, let's go find somebody to have sex with." Everybody eagerly does it.
- 8.86** Recent flap over sex robots shows the same pattern. They started saying, "Oh, we can make sex robots that will be sort of realistic and mainly for men to have sex with." Immediately, the feminists were against it. Why? I mean, sex robots seem like a very positive thing, even for women. As we saw, lots of women don't want to have sex with their husbands as much as the husbands want to have sex with them. They even have survey data showing that the average woman thinks the amount of sex in their marriage is about right. The man, on average, is wishing for about double what they are having.
- 8.87** Well, a sex robot is certainly the best way to take care of that, as opposed to, he could go to prostitute, sex worker, himself; he could get a mistress and so on, but that risks him leaving her. Prostitute risks, getting a disease or something, costing money or creating a scandal. So it's unclear. The sex robot might be, logically, a very positive thing for women, but it might reduce men's dependence on women. So immediately the outcry was, "No, we can't allow sex robots."
- 8.88** A couple more things. Competition among females. So obviously there's competition, both males as well as females. The woman wants to advertise and make herself look more attractive than she is. I mean, use makeup, which basically simulates what your face looks after you had an orgasm, with the flushed cheeks and the redder lips and the bigger eyes and all that. It's useful and rational for the woman to stimulate more demand than she needs, so that she can have the pick of whatever's the best suitor or partner that she wants.
- 8.89** When women compete against each other, like in girl gangs, if two girls are going to get into a fight, they usually start by insulting the other as promiscuous. "Oh, you have sex with lots of different people." Two men getting in a fight, they never say that. They sometimes say, "Oh, your mother does that." Promiscuity is more of a negative reflection on women than men.
- 8.90** There's interesting new research on female competition coming out from a number of scholars, Tania Reynolds in particular. And women use gossip, often depicting the other woman as highly promiscuous or having low morals, which suggests again that her sex would have low value. Now, one of the classic questions is, many cultures have suppressed female sexuality, have imposed controls on this. So there's the question, why does this happen? And why do they focus on the female?

- 8.91** Obviously, it's in the interest of a society to restrict the number of unwanted babies because they end up being a problem. Individuals, often a burden on society; other people are going to have to come up with the money to support them, things like that. So unwanted babies are a problem, and it's not surprising societies try to control their sexuality, and they focus more on the female.
- 8.92** Now, the two big theories in sexuality, the two big theoretical camps, are the evolutionary ones and the sort of constructionist feminist camp. This is for once, they agreed that it's the men's fault—only I think this time they were both wrong, because sexual economics theory suggests, no, it would be in women's interests to control female sexuality, to restrict the supply, to drive up the price. This was kind of what Nancy Cott, the feminist historian, was saying about women in the 1800s, which was my first inkling of this idea.
- 8.93** So we looked at the research literature on that, and I thought this is going to be a whole complicated mess. I've done many literature reviews, and cultural control of female sexuality is going to be really all kinds of, data all over the—but no, it was very consistent, lined up. All the influences on the young woman to restrain her sexuality come, are female sources, hardly any from men.
- 8.94** So it's the mothers who tell the daughter, leave the pamphlet on her bed, that you should read about the dangers of it. The female peers regulate, and the norms vary, but they are from place to place, but they are enforced by the female group, and they ostracize anyone who goes too far and thus lowers the price of sex in the local marketplace.
- 8.95** The double standard that it's OK for men but not for women to have sex before marriage, historians have talked about this as being common. Actually, it's not that common, but to the extent it's been supported, it's supported much more by women than by men. There was a big meta-analysis about 20 years ago, looking over time and everything. In every study that found a sex difference, the women are more likely to say it's OK for men, but not for women to do this sort of thing.
- 8.96** At the more extreme and creepy level, the female genital surgery, the two kinds of things they do, they kind of sew the vagina up tight or they cut off the clitoris. This is something that's entirely in the women's sphere. There's a big study of men in the Sudan, before the country—it's a real mess now, but this was a while ago—of men who had multiple wives, one of whom would have had this surgery and one of whom had not. And it was a couple hundred men. And they said almost everyone preferred having sex with a woman who has intact genitals.
- 8.97** I mean, it makes sense. You're going to enjoy it more with a partner who enjoys it herself, rather than the one for whom it's painful. So the authors were puzzled with this, because they'd had the feminist ideology that this is something men make women do. But no, it's something that the women do. Often, the father doesn't want his

daughter to go through it, but the mother insists that it has to be done. Why is it? Well, again, it will restrict the supply of sex and drive up the price.

- 8.98** Modern Western scholars go into these countries and insist on stamping out the practice, but it is taking away one of the ways that women exert control over the situation. What really works is when women can get access to education and employment and start support themselves, then they're less dependent on men, and then those practices disappear automatically. Again, fitting the economic analysis that, once women have other ways of supporting themselves, they don't need to maximize the price of sex by restricting the supply artificially.
- 8.99** Reputational pressures and so on, I mean, they used to say that, "Oh, if a girl has too much sexual activity and so on, no one will want to marry her." But that's not true. The reputational pressures mainly come from the other girls or the other women from the peer group. The boys, if anything, seem to like the girl who is a little more sexual than the other.
- 8.100** Even prostitutes get married. You'd think, OK, there's somebody who's had sex with lots of people for money, if anybody's going to be undesirable as a mate. But a fair number of them do eventually find, often one of their customers who becomes a regular customer and starts to develop an emotional attachment. So they do get married. I don't know whether they're really happy marriages or not, but it's certainly not true that no one will marry you if you've had too much sex or with too many different people.
- 8.101** Why does this happen? Again, it's a standard cartel or monopoly tactic; restrict the supply, and the price goes up. Women who undermine the high price, the cheap women—and they actually use that term in many slangs, that the woman who has sex too easily or too freely or with too little commitment is cheap.
- 8.102** Why is that bad? Well, it's undermining the market price there and making it harder for other women to get the price that they want. It's an economically rational response to a situation of dependency. As I said, a lot of these things, including genital surgery, once women become more economically self-sufficient and independent, these things disappear.
- 8.103** Another interesting point is the sexual revolution. Why did this happen at this point in history? Well, there are multiple factors, including birth control being there, but oral sex was possible and doesn't make babies. You can do that, but still, suddenly a lot more happened. I think part of the explanation is the women's movement in the middle of the 20th century, suddenly women were getting jobs, were able to support themselves, so they were less dependent on a man, and less dependent on needing sex and virginity to attract the man.
- 8.104** And so, they could relax and let the price of sex go down. And some people have said, points out a paradox, that men wanted to keep women in an inferior position so they

could get more sex from them. But actually, it works the other way. If you let women have more equality, then men's sexual opportunities increase. If men were doing that, it was a self-defeating, short-sighted strategy.

- 8.105** I'll just describe quickly a couple of studies. The original paper on this was by my colleague, Kathleen Vohs, with myself. And she went on, with a number of her colleagues, to do a variety of tests of the theory. The first one was this Dutch guy I did. He went online in 2005, and there was a data set of 41 countries with various responses to questionnaires. The country is the unit of analysis. We got the gender equality rating from the UN index, which is flawed, but it still shows how much women are well-off compared to men.
- 8.106** And looked at some of their—how this correlated with the average responses to sex. It all fits the idea that more female equality leads to more sex. Do people think you should abstain from sex before marriage? Well, this is negatively correlated with inequality. In unequal cultures, people say, "Yeah, abstain until marriage;" in more equal cultures, "Yeah, let's go for it." Casual sex, much more casual sex in more egalitarian countries. People report more sex partners in those. The age of first sex is earlier in those countries.
- 8.107** So again, more gender equality means more sex. These are correlational, so you can ask what causes what. I don't know if having a lot of sex is the cause of equality. That doesn't seem nearly as plausible. More likely, equality leads to more sex, because then women have more opportunities and don't need to restrict access to sex to make it a rare and expensive good, to get a high price for it.
- 8.108** Two of Vohs's colleagues did a study somewhat with her help. They're in a marketing department, and so they had good budget. They were able to have professional ads made. And so they're interested in, do people like sexual explicit ads? And in general, the finding is, men like sexy ads more than women do. But they had two ads made up, two ads for a woman's watch. It was either gift-wrapped or not, and it was either a very expensive watch or a very cheap watch.
- 8.109** OK, so which ad do they like? There's one of mountains, "for that special moment in time," is the slogan they used. And then the other was a kind of a sexual, sexy ad, again for that special moment in time. Well, what they found is, in general, men liked this ad, women liked the mountains ad. But if the ad was for a very expensive watch, women's watch packaged as a gift, then they like the sexy ad, too. So, women don't like, again, cheap sex, but when it goes with something very impressive and expensive, then sex is OK in advertising.
- 8.110** Kathleen herself did a study on Valentine's Day, getting a large sample of men and women, asking them the day after Valentine's Day, "Did you get a gift and did you have sex?" Three variables, their own gender, did they have sex or not, and did they get a gift or not? Did you feel guilty afterwards? So there are eight different means, but only

one of them is different from all the others. So see if you can guess which, but I'll just tell you, it's the women who got a gift and did not have sex, they felt guiltier than anyone else.

- 8.111** So again, they're sort of violating the implicit economic bargain there. The men who got, in a relationship, who got a gift, but didn't have sex, they didn't feel guilty, than the men in any of the other conditions. But when men are more equal, they can support themselves, they can live fine without a man, no longer so necessary to keep the price of sex high.
- 8.112** All right, there are a lot of issues that are not settled here, or what happens in long-term relationships, where in marriage, you own all the same things, so these things don't really apply anymore. Some people talk about, well, she has sex to reward the husband for helping with housework, washing the clothes or something. But turns out, if the man does more housework, they seem to have less sex. So I'm not sure that works.
- 8.113** Same-gender sex, well, there isn't the same kind of marketplace, but that's interesting, too. Anyway, there's more work to be done on this. But still, here are the main conclusions. First of all, it's clear female sexuality has a kind of value that male sexuality does not, basically exchange value, and the price may fluctuate in different circumstances, and how much she can get in exchange for commencing a sexual relationship varies. But it's often not nothing, whereas for the men, it's nothing.
- 8.114** A sexual marketplace exists, subject to economic principles, such as women are the supply and men are the demand, and if one is much larger than the other, the local price of sex adjusts accordingly. Differences in product quality influence price, other sorts of implications that one can develop from this in further work.
- 8.115** Anyway, as I said, it's a decidedly unromantic way of thinking about one of the marvelous forms of human interaction. But I certainly did come around, even despite my initial dislike of the theory; and I think anyone else who takes an open-minded look at the evidence would have to come to pretty much the same conclusion. All right. Thank you.